

CSC-204 Software Engineering
Semester Project

Library Management System



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Table of Contents

Abstract:	5
Introduction:	5
Problem Statement:	7
Objectives:	7
Software Tools and Technologies:	7
Vision:	8
Scope:	8
Requirements Analysis:	8
Functional Requirements:	8
Admin:	8
Librarian:	8
Student:	9
Non-Functional Requirements	10
Business Requirement:	11
Physical Requirement:	12
Development Requirement:	12
Wireframes:	13
User Stories:	31
Administrator:	31
Librarian:	31
Student:	32
Story Board:	33
Student:	33
Librarian:	34
Admin:	34
Uses Cases:	35
Login	35
Add Book	36
Books Report	37
Book Details	38

Add department	39
Edit department.....	40
View department.....	41
Delete department.....	42
Add Student.....	43
Add Librarian	44
Remove Librarian	45
Edit Librarian Info	46
Students Report	47
Librarian Report	48
Student Details	49
Book Issue	50
Issued books report	51
Return Book.....	52
Return books report.....	53
Paid Penalty Fine	54
Waive Penalty Fine	55
Logout	57
View Account.....	58
Edit Account	59
Borrowed book list.....	60
Return book list	61
Penalty Report list.....	62
UML Diagrams	63
Use Case Diagram:	63
Deployment Diagram:	63
Class Diagram:	64
Object Diagram:	64
Component Diagram:	65
Package Diagram:	65

Sequence Diagram:	66
Activity Diagram:	82
State Machine Diagram:.....	99
Collaborative Diagram:.....	115
TESTING	132
Unit Testing	132
Integration Level Testing	159
System Level Testing	162
Performance Testing	173
Proposed Methodology/System:	175
Related Work:.....	176
Data Gathering Approach.....	179
Team Members Individual Tasks:	180
Timeline/Gantt Chart:	181
References:.....	181

Abstract:

Managing library resources is often challenging for schools and colleges because traditional systems can be slow and hard to use. Existing library solutions usually do not provide an easy way to track books and student accounts. Our Digital Library Management System aims to solve these problems by creating a simple and user-friendly platform for administrators, librarians, and students. Using C# and MongoDB, the system includes enhanced functionalities such as admin-level controls, real-time book tracking, and penalty management. This solution offers benefits such as easy access to library resources, real-time tracking of borrowed books, penalties, and less work for librarians, leading to a more organized library experience for everyone.

Introduction:

Libraries play a crucial role in schools and colleges, providing students and faculty access to a wealth of books and resources. However, managing a library effectively can be a daunting task, especially when traditional systems are slow and complicated. Many libraries still rely on manual processes or outdated software, leading to confusion and frustration. For librarians, this means a lot of extra work, while students often struggle to find and borrow the books they need.

Traditional library systems come with many challenges. First and foremost, they can be clunky and hard to use. Many libraries still track books with paper logs or simple spreadsheets, which can result in lost records and mistakes. When a student wants to check out a book, librarians might spend a lot of time looking for it or verifying its availability. This often leads to frustration for both librarians and students. Furthermore, many existing systems do not provide real-time tracking, meaning that students might discover a book is unavailable only after making a trip to the library. These inefficiencies can discourage students from using library resources, making it tough for librarians to foster a love for reading and learning.

Another significant problem with traditional systems is the high risk of human error. A book might be mistakenly marked as available when it has actually been checked out, or a student might receive an overdue notice for a book returned on time simply because the system wasn't updated correctly. These inaccuracies can create distrust in the library system and discourage students from using it. Additionally, many older library systems do not have strong security measures for protecting sensitive information, such as student data and borrowing history, raising concerns about privacy.

Our proposed **Digital Library Management System** addresses these challenges by offering a streamlined, easy-to-use platform for administrators, librarians, and students. With our system, librarians can log in and quickly add books, view all available titles, and even generate reports to track how many books are in the library or to monitor student activity and also track penalties. This feature simplifies the process of managing a library's collection, making it easier for librarians to keep everything organized. They can also add departments and students, allowing for a more structured way to manage resources.

When a librarian wants to issue a book to a student, they can do so easily within the system. They can also process returns with just a few clicks, automatically updating the book's status without the hassle of paperwork. This efficiency means librarians can focus more on helping students rather than getting bogged down in Librarian tasks. At the end of the day, when they're finished, they can simply log out of the system, ensuring that their session is secure.

For students, the experience is equally user-friendly. They can log in to their accounts and easily view their details, including books they have borrowed or returned. If they need to update their information or change their password, they can do so without any trouble. The ability to see all available books in the library encourages students to explore more resources, fostering a love for reading. They can even check which books they have borrowed and returned, keeping them informed about their library activity. When they're done, they can log out, ensuring their information remains secure.

The admin functionality further strengthens the system by allowing administrators to manage librarians comprehensively. Admins can add, update, or remove librarian accounts and view detailed reports on their activities. This layer of control ensures that library operations remain transparent and well-organized.

Security is a significant aspect of our Digital Library Management System. We have designed it with strong security measures to protect sensitive student information and borrowing records. Regular backups of data ensure that important information is always safeguarded and can be restored if needed. This provides peace of mind for both librarians and students, knowing that their data is safe from unauthorized access.

Flexibility is another key strength of our system. As libraries grow and change, so do their needs. Our Digital Library Management System can easily accommodate more books and users without losing performance. Whether a small school or a large university, this system is tailored to meet the specific demands of various institutions, making it a practical solution for modern library management.

Moreover, libraries are not just about books; they are spaces where students can come together, learn, and collaborate. Our system promotes a more interactive library experience, encouraging students to engage with the resources available to them. By making the library more accessible and easier to use, we aim to foster a culture of reading and learning that benefits everyone.

In conclusion, our Digital Library Management System effectively addresses the challenges faced by traditional library systems. By automating many manual tasks and providing a simple, user-friendly platform for administrator, librarians and students, we significantly improve the overall efficiency of library operations. Real-time tracking, easy search options, and secure account management make this system a valuable tool for modern libraries. Ultimately, we believe this solution creates a more organized and enjoyable library experience for everyone involved. In an increasingly digital world, it's important to embrace technology that not only simplifies processes but also enhances the educational journey for all. With our Digital Library Management System,

we are confident that libraries can continue to thrive as essential resources for learning and community engagement for years to come.

Problem Statement:

How can we overcome the challenges faced by traditional library systems to provide a more efficient and user-friendly digital library management system for both librarians and students?

Objectives:

- **User-Friendly Interface:**

To design an intuitive interface that allows both librarians and students to easily navigate the library system, reducing the learning curve for new users.

- **Automated Book Inventory Management:**

To implement an automated system for tracking book availability, including adding, updating, and removing titles from the library database with minimal manual input.

- **Efficient Student Account Management:**

To provide features for students to create and manage their accounts, enabling them to view personal borrowing history, update contact information, and reset passwords easily.

- **Seamless Book Issuing and Returning:**

To streamline the process of issuing and returning books, allowing students to complete transactions quickly and librarians to track all activities efficiently.

- **Enhanced Admin Control:**

To introduce administrative functionalities, allowing administrators to manage librarian accounts, update system settings, and generate librarian activity reports for comprehensive oversight of library operations.

- **Comprehensive Reporting and Penalty Management:**

To develop detailed reporting tools that enable librarians and administrators to generate insights on book loans, returns, penalties, and overall library usage statistics, improving decision-making and resource allocation.

Software Tools and Technologies:

- **Programming Language:** C#
- **Documentation:** MS Word
- **Framework:** .NET Windows Forms

- **Database Management System:** MongoDB
- **Integrated Development Environment (IDE):** Visual Studio 2022
- **Mendeley:** References
- **Pencil:** Wireframes, Story Board
- **StarUML:** Diagrams

Vision:

To create a digital library that makes it easy for everyone to manage and use books.

Scope:

The scope of the Library Management System includes functionalities for administrator, librarians and students. Administrators will have the ability to manage librarian accounts, update information, and view reports of librarians. Librarians will be able to log in and perform a range of library management tasks, such as managing books and departmental records, issuing and returning books, tracking penalties, generating detailed reports, and overseeing student transactions. Students will have the ability to log in, view and edit their account details, browse available books, penalty history, and check their borrowing history.

Requirements Analysis:

Functional Requirements:

Admin:

- Admin shall Login to the System.
- Admin shall add Librarian.
- Admin shall View Librarian Report.
- Admin shall Remove Librarian.
- Admin shall Update Librarian Information.
- Admin shall Update his Information.
- Admin shall logout from the System.

Librarian:

- Librarian shall login to the system.
- Librarian shall update his information.

- Librarian shall Add books.
- Librarian shall view all books report.
- Librarian shall see Book Details.
- Librarian shall add department.
- Librarian shall edit department.
- Librarian shall delete department
- Librarian shall view all departments.
- Librarian shall add students.
- Librarian shall view students report.
- Librarian shall view student Detail.
- Librarian shall issue book to student.
- Librarian Shall view Issued Books Report.
- Librarian shall return book.
- Librarian shall view Returned Books Report.
- Librarian shall view Penalty Report.
- Librarian shall Paid the penalty fine of student.
- Librarian shall waive the penalty fine of student.
- Librarian shall logout from the system.

Student:

- Student shall login to the system.
- Student shall view his account.
- Student shall edit his account detail.
- Student shall change his password.
- Student shall see all books
- Student shall see detail of specific Book.
- Student shall see books that he borrows.
- Student shall see his Penalty Reports.

- Student shall see books that he returns.
- Student shall logout from the system.

Non-Functional Requirements

1. Response:

The system must ensure quick responses, with login and search actions completing in 2-3 seconds.

2. Throughput:

The system must handle multiple transactions efficiently, supporting at least 10-20 book transactions per second during peak hours.

3. Security:

The system must ensure secure access with encrypted data transmission, multi-level authentication, and role-based access control.

4. Scalability:

The system must support growing datasets and user numbers, allowing for easy scaling of both database and infrastructure.

5. Availability:

Ensure 99.9% uptime with minimal downtime through failover mechanisms and redundancy.

6. Capacity:

The system must handle up to 10,000 books and 5,000 users initially, with room for expansion.

7. Usability:

The interface must be intuitive, requiring minimal training, and accessible to both Librarians and students.

8. Performance:

The system must maintain performance levels under various load conditions, ensuring responsiveness and stability even as user numbers increase.

9. Utilization:

The system should effectively utilize available resources (CPU, memory, and storage) to maximize performance without overloading any single component.

10. Reliability:

The system must consistently perform its intended functions without failure, providing accurate information and services to users.

11. Recoverability:

The system must have backup and recovery procedures in place, allowing for data restoration within an acceptable time frame in the event of data loss or corruption.

12. Serviceability:

The system should facilitate easy maintenance and updates, minimizing downtime during servicing while ensuring ongoing functionality.

13. Regulatory:

The system must comply with relevant data protection and privacy regulations to ensure the secure handling of user data.

14. Manageability:

The system should provide tools for monitoring and managing system performance and allowing Librarians to efficiently oversee operations.

15. Interoperability:

The system must be able to integrate and interact with other systems (e.g., external databases, library networks) using standard protocols and formats.

Business Requirement:

- **Efficient Library Management:**

The system must simplify and automate the management of books, departments, and students for the Librarian.

- **Secure User Access:**

The system must ensure that only authorized Librarians and students can log in securely and access their respective features.

- **Book Issuing and Returning:**

The system must track book issuing and returning for students and ensure an accurate record of transactions.

- **Student Self-Service:**

Students must have the ability to view and manage their own account details, borrow and return history, and password updates.

- **Reporting:**

The system must generate detailed reports for both books and students, allowing Librarians to make informed decisions.

Physical Requirement:

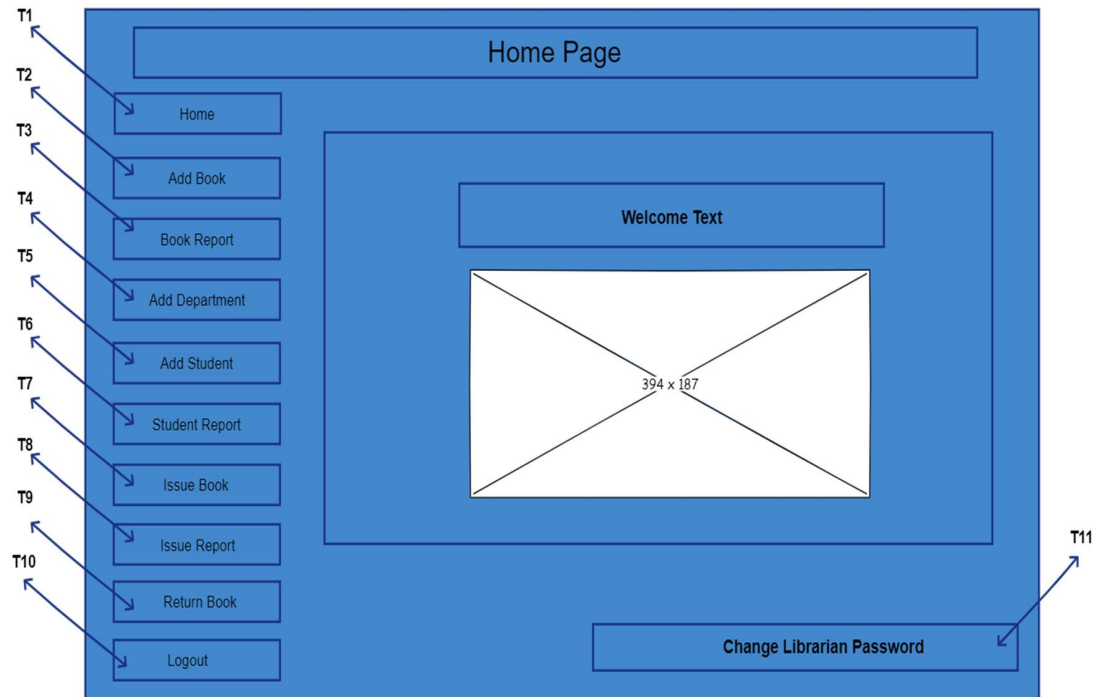
- Internet
- RAM

Development Requirement:

- **Programming Language:** C#
- **Documentation:** MS Word
- **Framework:** .NET Windows Forms
- **Database Management System:** MongoDB
- **Integrated Development Environment (IDE):** Visual Studio 2022
- **Mendeley:** References
- **Star UML:** For Diagrams
- **Pencil:** Wireframes, Story Board

Wireframes:

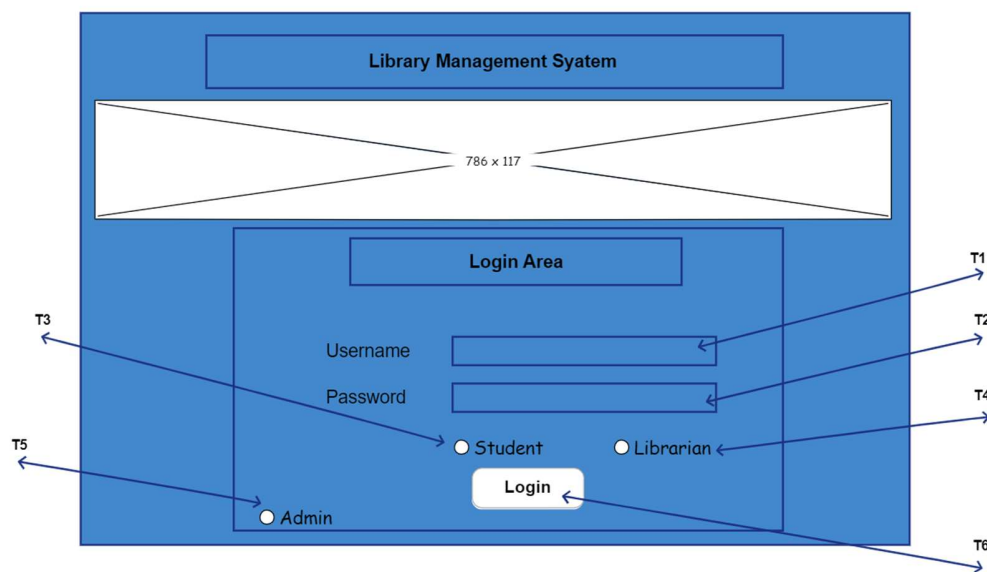
Librarian Home Page:



Requirements	Librarian
T1	As a Librarian, I shall view the Home page after logging in.
T2	As a Librarian, I shall add a new book to the system by click on Add Book button.
T3	As a Librarian, I shall view and generate reports for books by click on Book Report button.
T4	As a Librarian, I shall add a new department to the system by click on Add Department button.
T5	As a Librarian, I shall add a new student to the system by click on Add student button.
T6	As a Librarian, I shall view and generate reports for students.
T7	As a Librarian, I shall issue a book to a student.
T8	As a Librarian, I shall view and generate reports for issued books.

T9	As a Librarian, I shall return book from students.
T10	As a Librarian, I shall log out of the system securely.
T11	As a Librarian, I shall change my password.

Login Page:



Requirements	User
T1	As a User, I shall be able to enter username.
T2	As a User, I shall be able to enter password.
T3	As a User, I shall be able to select my role as a student.
T4	As a User, I shall be able to select my role as a Librarian.
T5	As a User, I shall be able to select my role as an Admin.
T6	As a User, I shall be able to click on login button to login to the system.

Add Book:

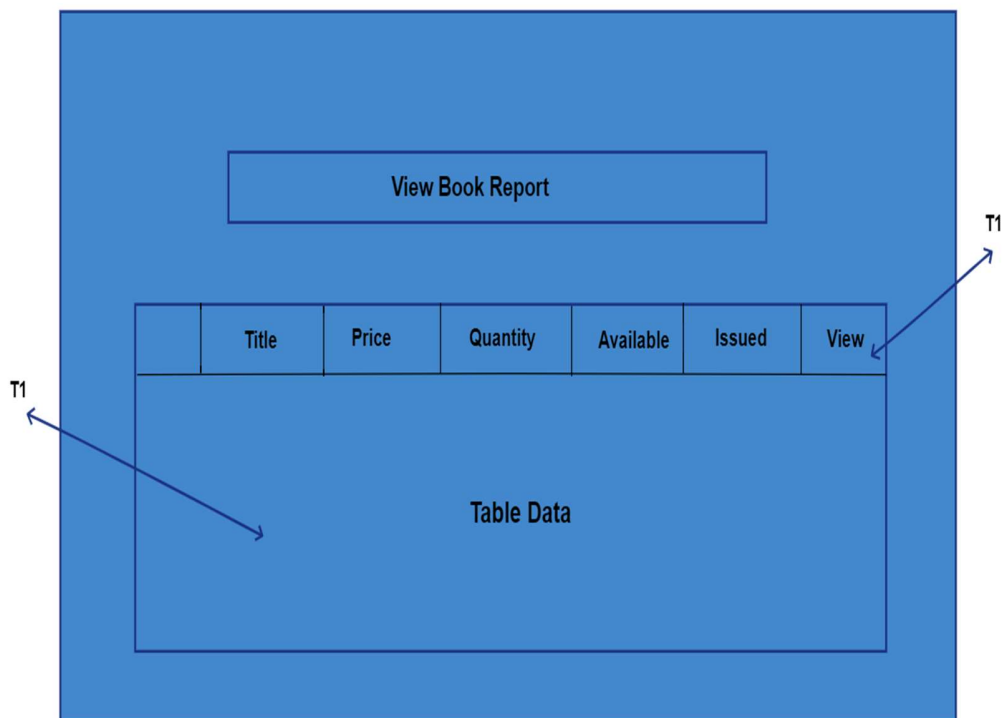
The image shows a web form titled "Add New Book". The form contains several input fields and a submit button. Arrows point from requirement labels (T1-T8) to specific parts of the form:

- T1 points to the "Book Name" input field.
- T2 points to the "Description" input field.
- T3 points to the "Auther" input field.
- T4 points to the "Edition" input field.
- T5 points to the "Price" input field.
- T6 points to the "Quantity" input field.
- T7 points to the "Upload Image" button within the "Book Image" field.
- T8 points to the "Add Book" submit button.

The form fields are: Book Name, Description, Auther, Edition, Price, Quantity, Book Image (with an "Upload Image" button), and an "Add Book" button.

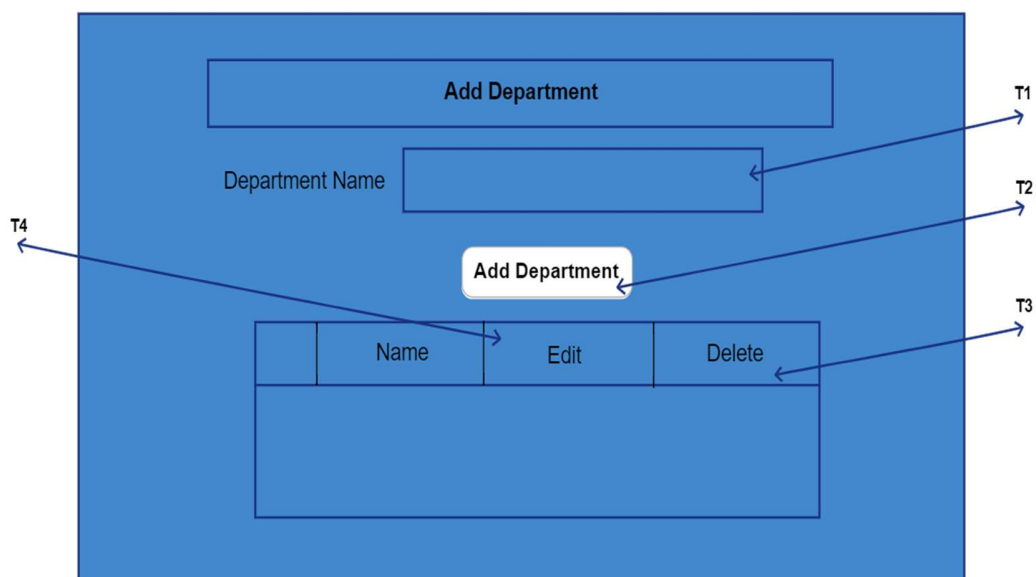
Requirements	Librarian
T1	As a Librarian, I shall be able to enter the book name.
T2	As a Librarian, I shall be able to provide a description for the book.
T3	As a Librarian, I shall be able to enter the author's name.
T4	As a Librarian, I shall be able to specify the edition of the book.
T5	As a Librarian, I shall be able to set the price of the book.
T6	As a Librarian, I shall be able to enter the quantity of books available.
T7	As a Librarian, I shall be able to upload an image of the book.
T8	As a Librarian, I shall add the book to the system by clicking the 'Add Book' button.

View Books Report:



Requirements	Librarian
T1	As a Librarian, I shall be able to view the book report.
T2	As a Librarian, I shall see a table displaying the Title, Price, Quantity, Available, and Issued status of books.

Department:



Requirements	Librarian
T1	As a Librarian, I shall enter new department in the system.
T2	As a Librarian, I shall add department by pressing Add button
T3	As a Librarian, I shall edit department name.
T4	As a Librarian, I shall delete department.

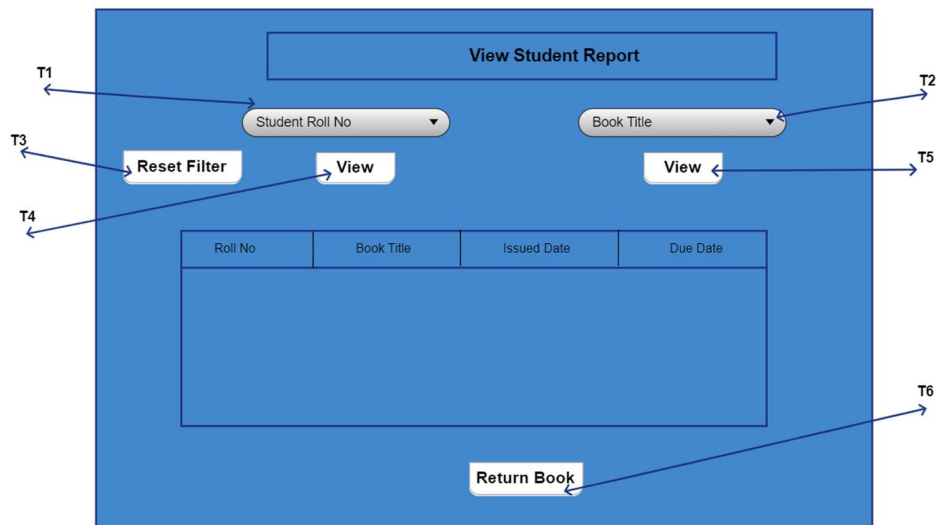
Add Student:

The screenshot shows a web form titled "Add New Student". The form contains the following fields and a button:

- Student Name (text input) - linked to T1
- Roll no (text input) - linked to T2
- Gender (text input) - linked to T3
- Department (dropdown menu) - linked to T4
- Mobile no (text input) - linked to T5
- Address (text input) - linked to T6
- Username (text input) - linked to T7
- Password (text input) - linked to T8
- Add Student (button) - linked to T9

Requirements	Librarian
T1	As a Librarian, I shall add student by enter student name.
T2	As a Librarian, I shall add student by enter student roll no.
T3	As a Librarian, I shall add student by enter student gender.
T4	As a Librarian, I shall have a drop down to respective for student.
T5	As a Librarian, I shall add enter student mobile number.
T6	As a Librarian, I shall add student by enter student address.
T7	As a Librarian, I shall add student by enter student username.
T8	As a Librarian, I shall add student by enter student password.
T9	As a Librarian, I click press Add button to add new student.

Student Report:



Requirements	Admin
T1	As a Librarian, I shall have a drop down to select a student.
T2	As a Librarian, I shall have a drop down to select a Book.
T3	As a Librarian, I shall click on reset filter button to reset selected page.
T4	As a Librarian, I shall click view button to see selected student.
T5	As a Librarian, I shall click view button to see selected student.
T6	As a Librarian, I shall click on return book button.

Admin Change Password:

The image shows a blue rectangular form titled "Change Admin Password". Inside the form, there is a white rectangular area containing four labels with corresponding input fields: "Username", "Old Password", "New Password", and "Confirm Password". Below these fields is a white button labeled "Change Password". Five arrows point from labels T1 through T5 to the following elements: T1 points to the Username input field, T2 points to the Old Password input field, T3 points to the New Password input field, T4 points to the Confirm Password input field, and T5 points to the Change Password button.

Requirements	Admin
T1	As an Admin, I shall be able to enter username.
T2	As an Admin, I shall be able to enter old password.
T3	As an Admin, I shall be able to enter new password.
T4	As an Admin, I shall be able to enter confirm password.
T5	As an Admin, I shall be able to click change password button.

Add Librarian:

The screenshot shows a web form titled "Add Librarian" with the following fields and requirements mapping:

- Librarian ID**: T1
- Name**: T2
- Gender**: T3 (Male) and T4 (Female)
- Mobile No**: T5
- Address**: T6
- Username**: T7
- Password**: T8
- Change Password**: T9

Requirements	Admin
T1	As an Admin, I shall be able to enter librarian id.
T2	As an Admin, I shall be able to provide my name.
T3/T4	As an Admin, I shall be able to Select my gender.
T5	As an Admin, I shall be able to enter my mobile no.
T6	As an Admin, I shall be able to enter my address.
T7	As an Admin, I shall be able to enter the username.
T8	As an Admin, I shall be able to enter password.
T9	As an Admin, I shall add Librarian to the system by clicking the 'Add Librarian'.

Book Issue Form:

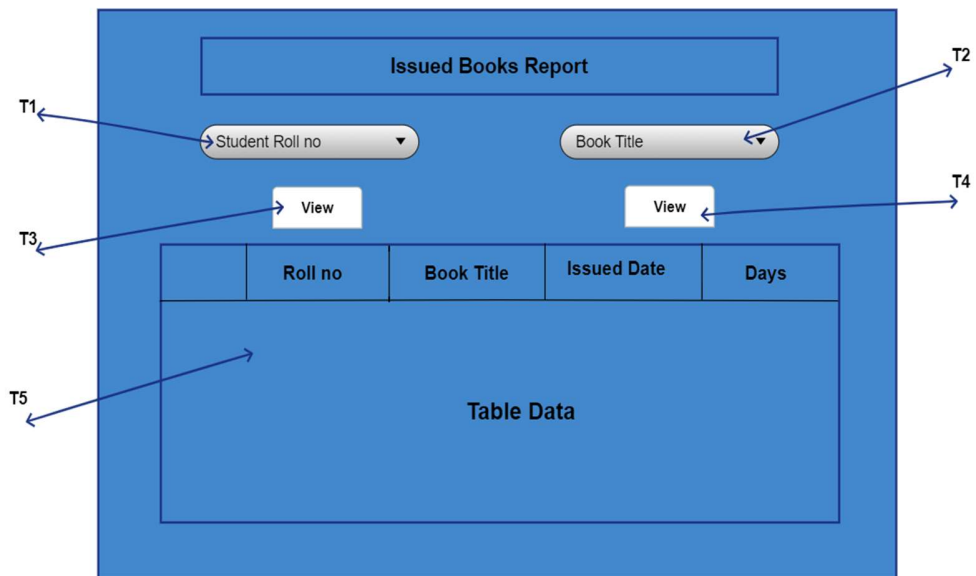
The diagram shows a 'Book Issue Form' with the following components and test points:

- T1**: Points to the 'Book Select' dropdown menu.
- T2**: Points to the 'Select' button.
- T3**: Points to the 'Select Roll no' dropdown menu.
- T4**: Points to the 'Days' input field.
- T5**: Points to the 'Issue Book' button.

The form layout includes a title bar 'Book Issue Form', a 'View Book Detail' section with a placeholder image (192 x 161) and a 'Book Detail' section, and a 'Select Student Detail For Issue Book' section.

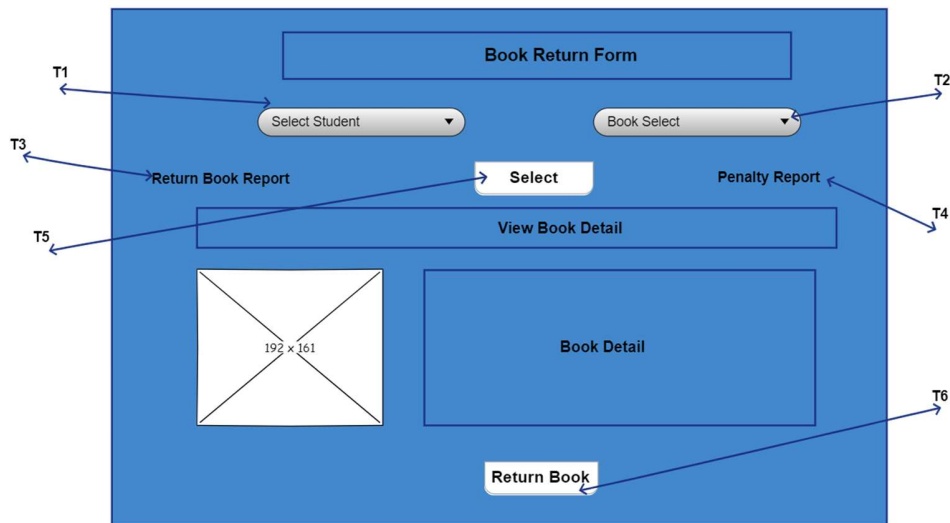
Requirements	Librarian
T1	As a Librarian, I shall have a drop down to select a book.
T2	As a Librarian, I shall click Select button to confirm selected book.
T3	As a Librarian, I shall have a drop down to select a Student.
T4	As a Librarian, I shall be able to enter the number of days for which the student requires the book.
T5	As a Librarian, I shall be able to issue the book to the student by clicking the 'Issue Book' button.

Issued Book Reports:



Requirements	Librarian
T1	As a Librarian, I shall have a drop down to select a student roll no.
T2	As a Librarian, I shall have a drop down to select a book.
T3	As a Librarian, I shall click view button to see selected student.
T4	As a Librarian, I shall click view button to see selected Book.
T5	As a Librarian, I shall see all details include Roll no, Book Title, Issued Date and Days.

Book Return:



Requirements	Admin
T1	As a Librarian, I shall have a drop down to select a student.
T2	As a Librarian, I shall have a drop down to select a book.
T3	As a Librarian, I shall click on return book to see Return Book report.
T4	As a Librarian, I shall click on Penalty report to check penalty.
T5	As a Librarian, I shall click select button.
T6	As a Librarian, I shall click return book button.

Return Book Report:

The mockup shows a web interface titled "Return Book Report". It features two dropdown menus: "Student Roll No" and "Book Title". Below these are two "View" buttons and one "Reset Filter" button. A table with columns "Roll No", "Book Title", "Issued Date", "Due Date", "Return Date", and "Penalty Status" is positioned below the buttons. Test points T1, T2, T3, and T4 are indicated with arrows pointing to the "Student Roll No" dropdown, the "Book Title" dropdown, the "Reset Filter" button, and the right "View" button respectively.

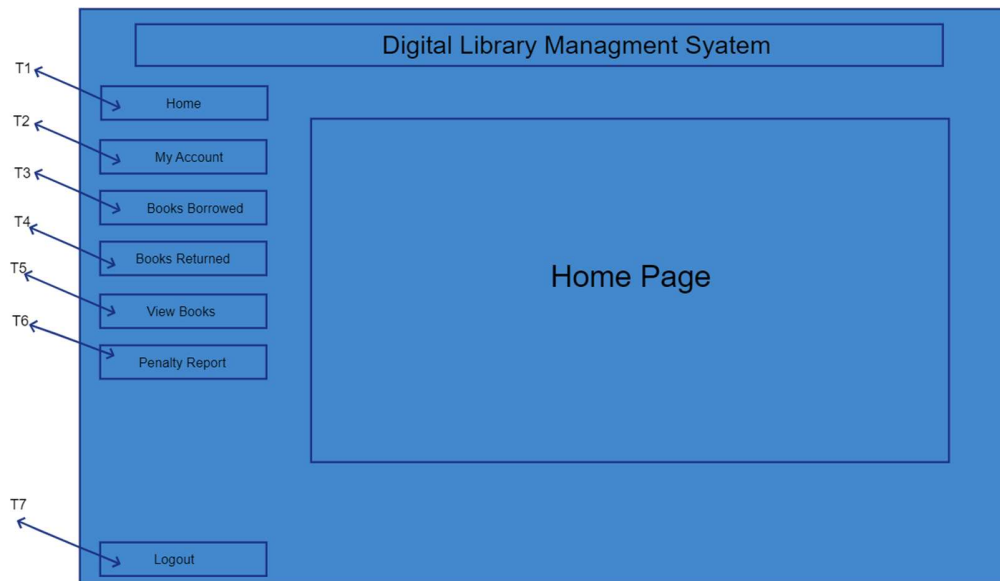
Requirements	Librarian
T1	As a Librarian, I shall have a drop down to select a student.
T2	As a Librarian, I shall have a drop down to select a Book.
T3	As a Librarian, I shall click on reset filter button to reset selected page.
T4	As a Librarian, I shall click view button to see selected student.
T5	As a Librarian, I shall click view button to see selected book.

View Librarian Report:

The mockup shows a web interface titled "View librarian Report". It features a table with columns "ID", "Name", "Username", "Password", "Mobile", "Address", and "Gender". Below the table are two buttons: "Remove Librarian" and "Update Librarian". Test points T1 and T2 are indicated with arrows pointing to the "Remove Librarian" and "Update Librarian" buttons respectively.

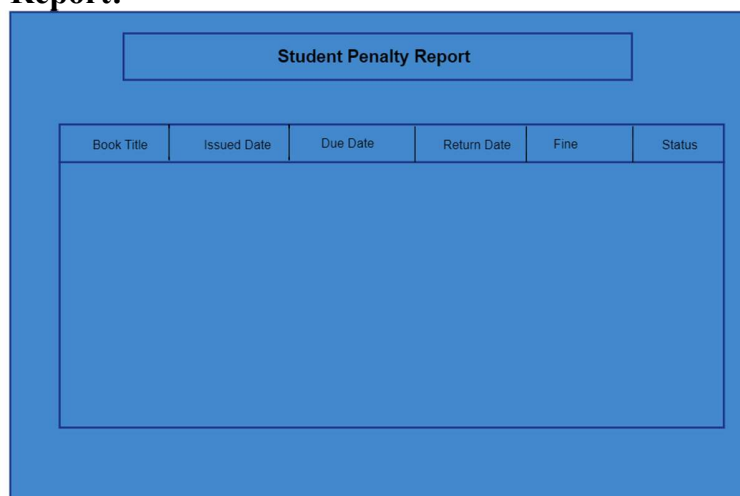
Requirements	Admin
T1	As an admin I shall click on Remove librarian button to remove a librarian.
T2	As an admin I shall click on update librarian button to update a librarian info.

Student Home Page:



Requirements	Student
T1	As a student, I shall view the Home page after logging in.
T2	As a student, I shall be able to access my account details through the 'My Account' option.
T3	As a student, I shall view a list of books I have borrowed.
T4	As a student, I shall view a list of books I have returned.
T5	As a student, I shall view the available books.
T6	As a student, I shall be able to see my penalty report.
	As a student, I shall be able to securely logout from the system.

Student Penalty Report:



Librarian Penalty Report:

The screenshot shows a web interface titled "Penalty Report". It features a "Student Roll No" dropdown menu, a "View" button, a "Reset Filter" button, and a table with columns: Roll No, Book Title, Issued Date, Due Date, Return Date, and Fine. Below the table are two buttons: "Waive Fine" and "Paid Fine". Test cases are indicated by arrows: T1 points to the "Penalty Report" title, T2 points to the "Student Roll No" dropdown, T3 points to the "Reset Filter" button, T4 points to the "Waive Fine" button, and T5 points to the "Paid Fine" button.

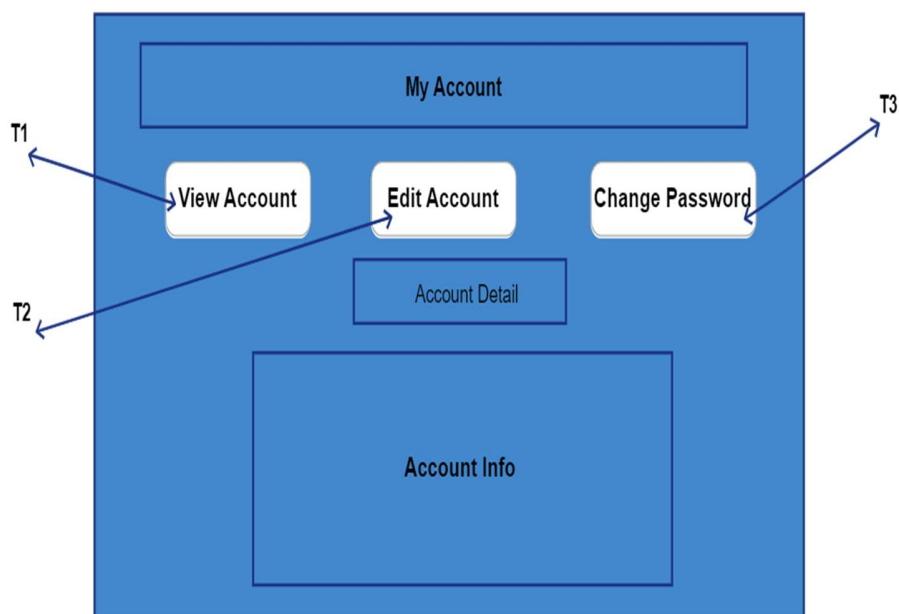
Requirements	User
T1	As a Librarian, I shall have a drop down to select a student.
T2	As a Librarian, I shall view selected student.
T3	As a Librarian, I click on reset filter button.
T4	As a Librarian, I shall click on Waive fine button
T5	As a Librarian, I shall click on Paid fine button.

Update Librarian:

The screenshot shows a web interface titled "Update Librarian". It contains a form with fields for Librarian ID, Name, Gender (with radio buttons for Male and Female), Mobile Number, Address, New Password, and Confirm Password. There is an "Update Librarian" button at the bottom. Test cases are indicated by arrows: T1 points to the "Update Librarian" title, T2 points to the "Librarian ID" field, T3 points to the "Name" field, T4 points to the "Gender" label, T5 points to the "Male" radio button, T6 points to the "Mobile Number" field, T7 points to the "Address" field, T8 points to the "New Password" field, and T9 points to the "Update Librarian" button.

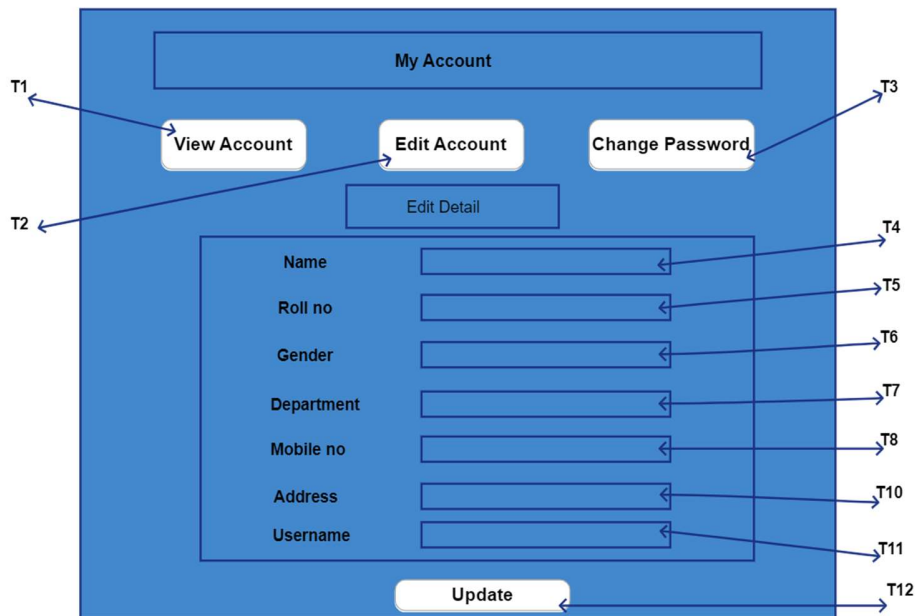
Requirements	Admin
T1	As an Admin, I shall be able to enter librarian id.
T2	As an Admin, I shall be able to provide my name.
T3/T4	As an Admin, I shall be able to Select my gender.
T5	As an Admin, I shall be able to enter my mobile no.
T6	As an Admin, I shall be able to enter my address.
T7	As an Admin, I shall be able to enter the username.
T8	As an Admin, I shall be able to enter password.
T9	As an Admin, I shall add Librarian to the system by clicking the 'Update Librarian'.

Student Account:



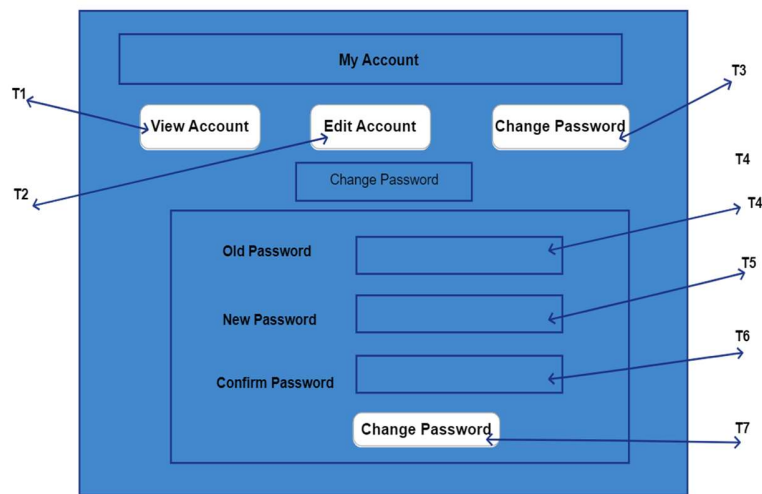
Requirements	Student
T1	As a Student, I shall view my account detail.
T2	As a Student, I shall view my account detail.
T3	As a Student, I shall change my password.

Edit Account Detail:



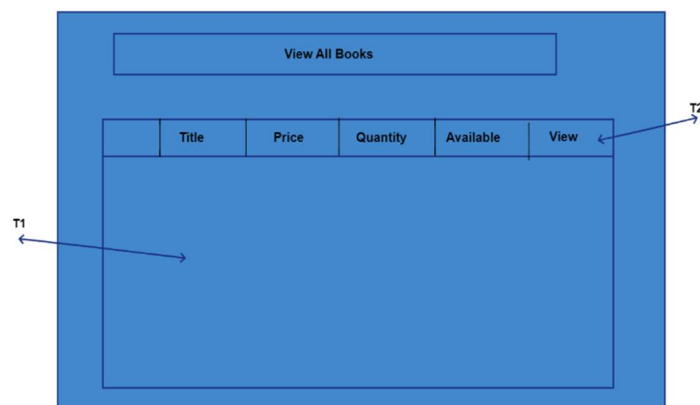
Requirements	Student
T1	As a Student, I shall view my account detail.
T2	As a Student, I shall view my account detail.
T3	As a Student, I shall change my password.
T4	As a Student, I shall be able to edit my name.
T5	As a Student, I shall be able to edit my roll number.
T6	As a Student, I shall be able to select or change my gender.
T7	As a Student, I shall be able to edit my department.
T8	As a Student, I shall be able to update my mobile number.
T10	As a Student, I shall be able to edit my address.
T11	As a Student, I shall be able to change my username
T12	As a Student, I shall click on update button to submit my changes.

Student Change Password:



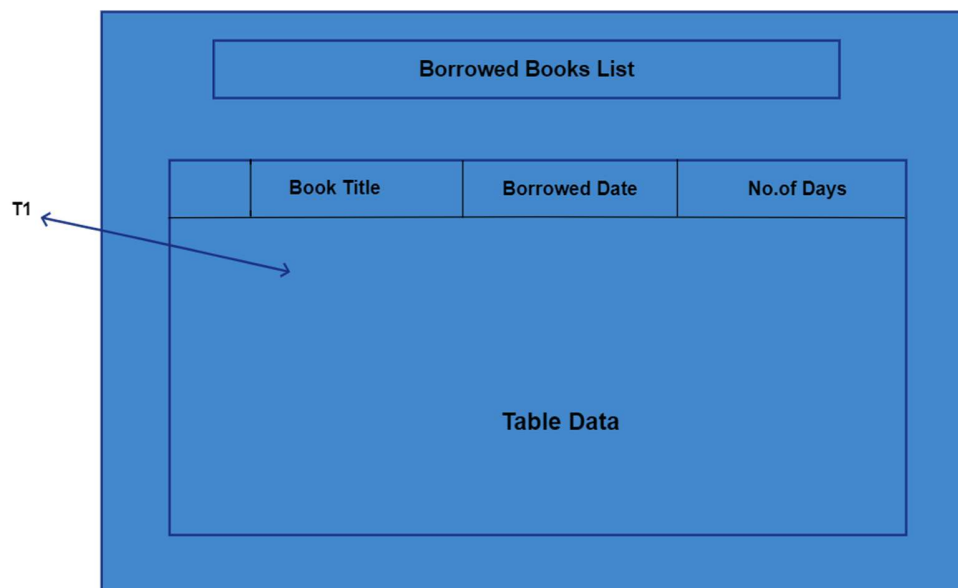
Requirements	Student
T1	As a Student, I shall view my account detail.
T2	As a Student, I shall view my account detail.
T3	As a Student, I shall change my password.
T4	As a Student I shall be required to enter my old password to change my password.
T5	As a Student I shall be able to enter new password when changing password.
T6	As a Student I shall confirm new password by entering it again.
T7	As a Student, I shall click change password button to submit the changes.

View All Books:



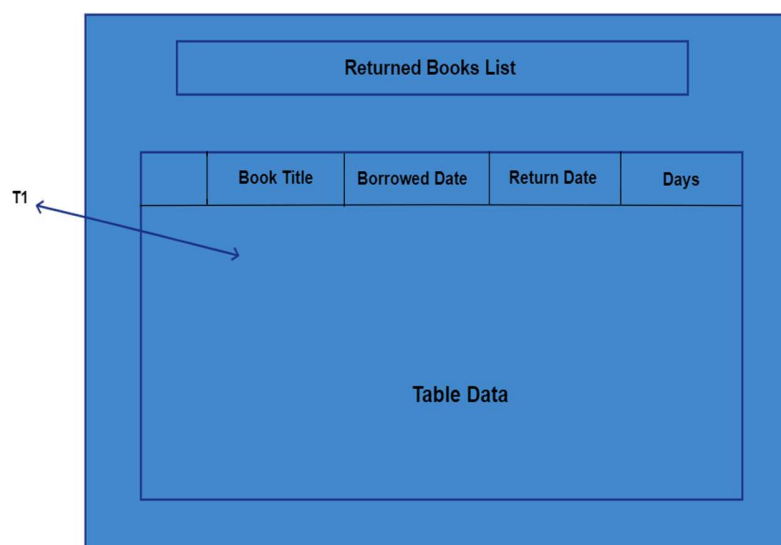
Requirements	Student
T1	As a Student, I shall be able to view a table displaying the Book Title, Price, Quantity, Available status, and a View option for each book
T2	As a Student, I shall be able to view individual Data.

Borrowed Books List:



Requirements	Student
T1	As a Student, I shall be able to view a table displaying the Book Title, Number of Days, and Borrowed Date for the books I have borrowed.

Returned Books List:



Requirements	Student
T1	As a Student, I shall be able to view a table displaying the Book Title, Borrowed Date, Return Date, and Number of Days for each borrowed book.

User Stories:

Administrator:

- As an Admin, I want to log in to the system so that I can access and manage librarian accounts and oversee library operations.
- As an Admin, I want to add new librarians to the system so that the library staff can be effectively managed.
- As an Admin, I want to update librarian details so that I can keep the librarian records accurate and up-to-date.
- As an Admin, I want to remove librarians from the system so that I can manage staff efficiently.
- As an Admin, I want to view librarian reports so that I can monitor their activity and ensure smooth library operations.
- As an Admin, I want to update my account details so that I can keep my profile information up-to-date.
- As an Admin, I want to log out from the system so that I can securely end my session.

Librarian:

- As a Librarian, I want to log in to the system so that I can access and manage the library resources.
- As a Librarian, I want to add new books to the system so that students and staff can easily find and borrow them.
- As a Librarian, I want to generate a book report so that I can keep track of the library inventory and usage statistics.
- As a librarian, I want to view the details of a specific book so that I can access important information about the book
- As a Librarian, I want to view all departments in the system so that I can quickly access and manage departmental information.
- As a Librarian, I want to add departments to the system so that I can add students according to their respective department.
- As a Librarian, I want to edit departments to the system so that I can keep my department information latest.
- As a Librarian, I want to delete departments to the system so that I can manage my department record.

- As a Librarian, I want to add students to the system so that they can borrow books and access library resources.
- As a Librarian, I want to generate a student report so that I can track student activity and manage library usage effectively.
- As a Librarian, I want to view detail of a student so that I can manage their account effectively.
- As a Librarian, I want to issue books to students so that they can borrow the resources they need for their studies.
- As a Librarian, I want to generate an issued book report so that I can keep track of the library inventory.
- As a Librarian, I want to process the return of books so that the library's inventory remains accurate and up-to-date.
- As a Librarian, I want to generate and view a returned books report so that I can track the books that have been returned and ensure inventory accuracy.
- As a Librarian, I want to generate a penalty report so that I can manage overdue fines for students.
- As a Librarian, I want to mark penalties as paid or waive them so that I can handle fines in accordance with library policies.
- As a Librarian, I want to change my information so that I can maintain the information of my account and keeps my record accurate and up-to-date.
- As a Librarian, I want to log out from the system so that I can securely end my session.

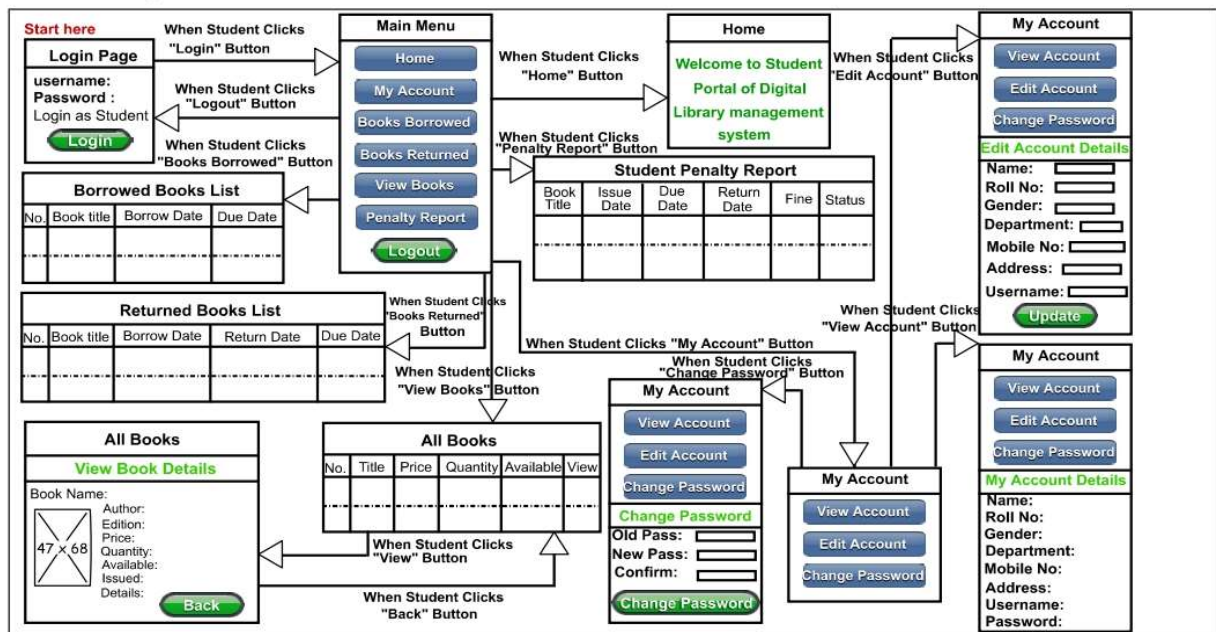
Student:

- As a student, I want to log in to the system so that I can access my library account and use available resources.
- As a student, I want to view my account details so that I can see my personal information and borrowing history.
- As a student, I want to edit my account details so that I can keep my information up-to-date.
- As a student, I want to change my password so that I can maintain the security of my account.
- As a student, I want to view all the books in the library so that I can find and borrow the books I need.

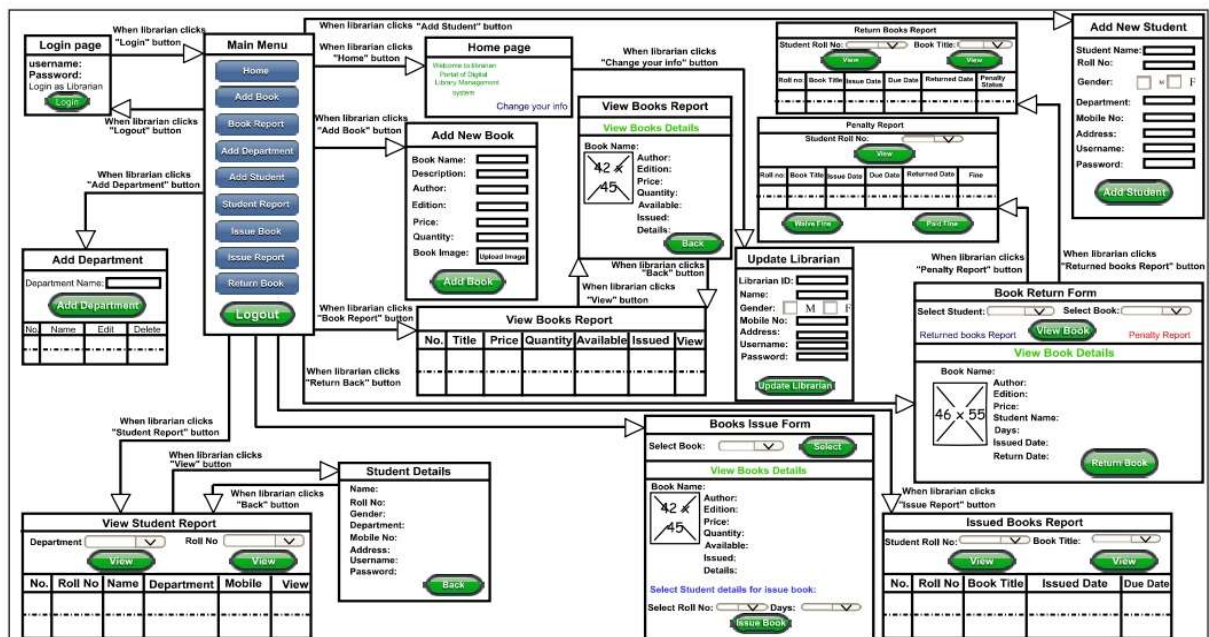
- As a student, I want to view detail of any book so that I can see if it is useful for my needs.
- As a student, I want to see the list of books I have borrowed so that I can keep track of the due dates and return them on time.
- As a student, I want to see the list of books I have returned so that I can confirm that my borrowing record is accurate.
- As a Student, I want to view my penalty report so that I can stay informed about any fines I need to pay.
- As a student, I want to log out from the system so that I can securely end my session.

Story Board:

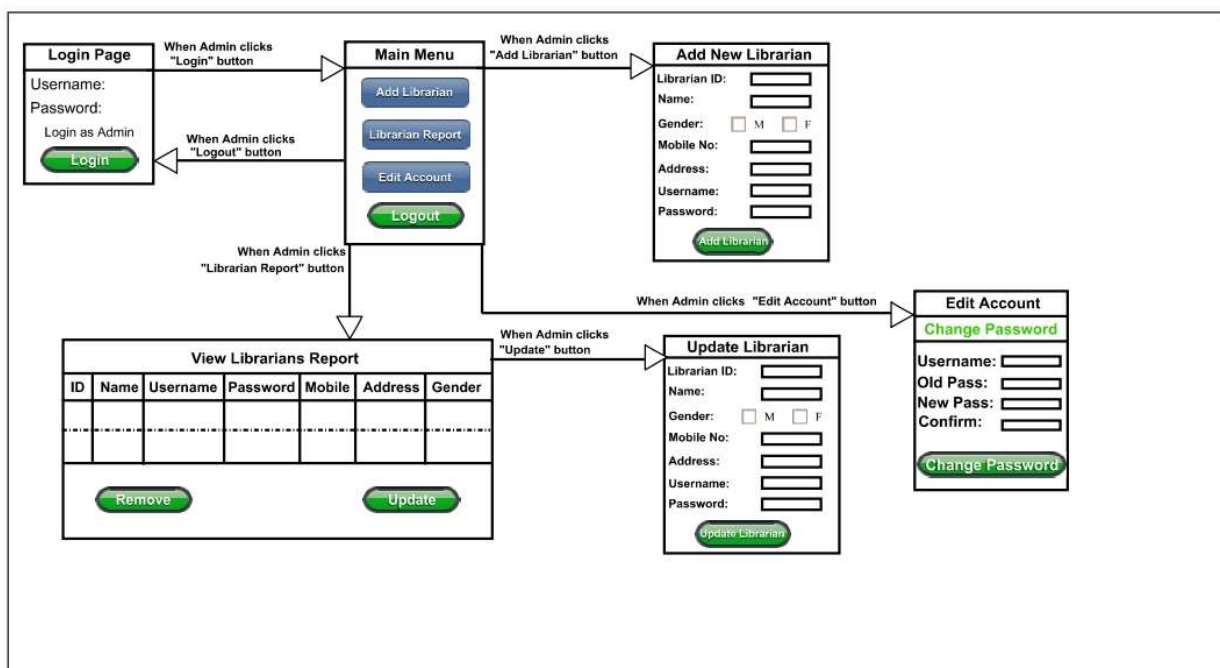
Student:



Librarian:



Admin:



Uses Cases:

Login

Name	System Login		
Participating Actor	Admin, Librarian, Student		
Goals	I want to Login to System so that I can access library’s functionalities.		
Triggers	When user open the Library Management System.		
Pre-conditions	1. Librarian or Student Must logged in 2. Librarian or Student must have internet availability.		
Post-conditions	Librarian or Student successfully login his account.		
Basic flow		Librarian, Student, Admin	System
		1. Admin, Librarian or Student Enter username and password. 2. Click on login button. 3. Librarian or Student logged in successfully.	1. The system store and display the credentials on screen. 2. The system analyzes and validate the credentials. 3. The system successfully login after verification.
Alternative flow	Not Available		
Exceptions	1. Admin, Librarian or Student do not have Account. Solution: Admin, Librarian or Student must have account. 2. Admin, Librarian or Student enter the invalid username. Solution: Admin, Librarian or Student must enter the correct username. 3. Admin, Librarian or Student enter the wrong password. Solution: Admin, Librarian or Student must enter the correct password. 4. Admin, Librarian or Student select the wrong role. Solution: Admin, Librarian or Student must select the correct role.		
Qualities	• Security: The system ensures secure access with encrypted data transmission and multi-level authentication for user credential protection. • Response: The system guarantees login actions complete within 2-3 seconds for a smooth user experience. • Usability: The interface is intuitive and requires minimal training, enabling efficient login for both Librarians and Students.		

Add Book

Name	Add Book	
Participating Actor	Librarian	
Goals	I want to add Book in my Library so that Students can access it.	
Triggers	When Librarian click on Add book button.	
Pre-conditions	1. Librarian Must logged in 2. Librarian must have internet availability.	
Post-conditions	The Librarian successfully added the book.	
Basic flow	Librarian	System
	1. Librarian enters all detail of Book. 2. Librarian upload the image of Book. 3. Librarian click on add Book button	1. The system store and display the details on screen. 2. The system stores the Image. 3. The system successfully adds the book.
Alternative flow	Not Available.	
Exceptions	1. Librarian does not fill All information Solution: Librarian have to fill all information 2. Librarian does not upload the image. Solution: Librarian have to upload the image. 3. Librarian does not click on Add Book Button. Solution: Librarian have to click on button to add the book successfully.	
Qualities	• Performance: The system should store and display the book details within 2 seconds of submission. • Usability: The interface should be user-friendly and allow the librarian to easily add books with minimal training. • Security: Only authenticated and authorized users (librarians) should be able to access the "Add Book" functionality.	

Books Report

Name	View Book Report	
Participating Actor	Librarian, Student	
Goals	I want to view report of All Books.	
Triggers	When Librarian or Student click on View Book Report Button.	
Pre-conditions	1. Librarian or Student Must logged in 2. Librarian or Student must have internet availability.	
Post-conditions	The reports of the book are displayed successfully	
Basic flow	Librarian, Student	System
	1. Librarian or Student click the button of Report Book	1. System shows the table of All Books.
Alternative flow	Not Available	
Exceptions	1. Librarian or Student does not press the view Issued book Report button. Solution: Librarian or Student Must press the view book report button	
Qualities	• Performance: The system must generate and display the book report within 2-3 seconds to ensure a smooth user experience. • Scalability: The system must be able to handle growing datasets, ensuring the book report can accommodate large volumes of books as the library expands. • Usability: The report interface must be intuitive and easy to navigate, allowing the Librarian to access and interpret the report with minimal effort.	

Book Details

Name	View book details	
Participating Actor	Librarian, Student	
Goals	I want to view Book details.	
Triggers	When Librarian or Student click on view button of specific Book.	
Pre-conditions	1. Librarian or student must log in the system. 2. Librarian or student must have internet availability.	
Post-conditions	Book details will be displayed on the screen	
Basic flow	Librarian, Student	System
	1. Librarian or student click on Book Report 2. Librarian or student click on view button of specific book to view its detail	1. The system will show the table of All Books. 2. The System will show the detail of that book.
Alternative flow	Not Available	
Exceptions	1. No Book Found Solution: The Librarian must add the books 2. The Book details are not showing. Solution: The Librarian must select the book and press view button.	
Qualities	• Performance: The system should display book details within 2 seconds after the librarian selects a book. • Usability: The interface should be easy to navigate, allowing the librarian to quickly locate and view book details. • Reliability: The system should ensure continuous access to book details, even during peak usage times, without failure.	

Add department

Name	Add Department	
Participating Actor	Librarian	
Goals	I want to add department so that I can add students in it.	
Triggers	When Librarian click on add department button.	
Pre-conditions	1. Librarian Must logged in 2. Librarian must have internet availability.	
Post-conditions	Department will be added to the Digital Library	
Basic flow	Librarian	System
	1. Librarian enter the name of Department. 2. Librarian Click on Add Department button 3. Librarian successfully add the button	4. The system display the department name on screen. 5. The system stores the name of department 6. The system successfully adds the department.
Alternative flow	Not Available.	
Exceptions	1. Librarian does not enter name of department. Solution: Librarian must enter the name. 2. Librarian does not click on add department button. Solution: Librarian click on add department button.	
Qualities	• Security: The system ensures secure access with role-based access control, allowing only authenticated Librarians to add departments. • Response: The system guarantees that the add department action completes within 2-3 seconds, ensuring a fast and efficient user experience. • Reliability: The system must consistently perform the department addition function without failure, accurately storing and displaying the department name.	

Edit department

Name	Edit department	
Participating Actor	Librarian	
Goals	I want to edit the name of department.	
Triggers	When Librarian click on Edit option.	
Pre-conditions	1. Librarian Must logged in the system. 2. Librarian must have internet availability.	
Post-conditions	Librarian successfully edit the name of department	
Basic flow	Librarian	System
	1. Librarian click on Add Department Button. 2. Librarian click on edit button of specific department. 3. Librarian enter the new name of Department	1. The System show the table of all departments. 2. The system show option to edit department name. 3. The system successfully updates the name of department.
Alternative flow	Not Available	
Exceptions	1. No department Found Solution: The Librarian must add the department 2. The department name is not change. Solution: The Librarian must click on the edit option.	
Qualities	• Performance: The system should update the department name within 2 seconds after the librarian submits the changes. • Usability: The interface should allow the librarian to easily find and edit department details with minimal effort. • Security: Only authorized users (librarians) should have permission to edit department details, ensuring data integrity.	

View department

Name	View All department		
Participating Actor	Librarian		
Goals	I want to View All department.		
Triggers	When Librarian click on Add Department Button.		
Pre-conditions	1. Librarian Must logged in the system. 2. Librarian must have internet availability.		
Post-conditions	Librarian delete the department successfully.		
Basic flow	Librarian	System	
	1. Librarian click on Add Department Button.	2. The System show the table of all departments.	
Alternative flow	Not Available		
Exceptions	1. No department Found Solution: The Librarian must add the department 2. The department does not click Add Department button. Solution: The Librarian must click on the Add department button.		
Qualities	• Performance: The system must generate and display the Departments report within 2-3 seconds to ensure a smooth user experience. • Scalability: The system must be able to handle growing datasets, ensuring the Department report can accommodate large volumes of department as the library expands. • Usability: The report interface must be intuitive and easy to navigate, allowing the Librarian to access and interpret the report with minimal effort.		

Delete department

Name	Delete department		
Participating Actor	Librarian		
Goals	I want to Delete department.		
Triggers	When Librarian click on delete option.		
Pre-conditions	3. Librarian Must logged in the system. 4. Librarian must have internet availability.		
Post-conditions	Librarian delete the department successfully.		
Basic flow	Librarian	System	
	1. Librarian click on Add Department Button. 2. Librarian click on delete button of specific department.	4. The System show the table of all departments. 5. The system deletes the department successfully.	
Alternative flow	Not Available		
Exceptions	3. No department Found Solution: The Librarian must add the department 4. The department does not delete. Solution: The Librarian must click on the delete option.		
Qualities	• Performance: The system should remove the department within 2 seconds after the librarian confirms the deletion. • Usability: The system should provide a confirmation prompt to prevent accidental deletions and ensure clarity in the process. • Security: Only authorized users (librarians) should be allowed to delete departments, and an audit trail should be maintained for record-keeping.		

Add Student

Name	Add Student	
Participating Actor	Librarian	
Goals	I want to Add Student so that student can borrow and return books	
Triggers	When Librarian click on add Student Button.	
Pre-conditions	1. Librarian Must logged in 2. Librarian must have internet availability.	
Post-conditions	Student will be added successfully in the Library Management System	
Basic flow	Librarian	System
	1. Librarian enter all the detail of student. 2. Librarian click on Add Student button. 3. Librarian successfully add the student.	1. The system displays all details on screen. 2. The system stores all details of student. 3. The system successfully adds the student.
Alternative flow	Not Available	
Exceptions	1. Librarian does not enter all information of student. Solution: Librarian must enter all information of student. 2. Librarian add the existing Student. Solution: Librarian must not enter the existing Student.	
Qualities	• Security: The system ensures secure access with role-based access control, allowing only authenticated Librarians to add students. • Response: The system guarantees that the add student action completes within 2-3 seconds, ensuring a fast and efficient user experience. • Reliability: The system must consistently perform the student addition function without failure, accurately storing and displaying student details.	

Add Librarian

Name	Add Librarian	
Participating Actor	Admin	
Goals	I want to Add Librarian so that Librarian can manage the library	
Triggers	When Admin click on add Librarian Button.	
Pre-conditions	3. Admin Must logged in 4. Admin must have internet availability.	
Post-conditions	Librarian will be added successfully in the Library Management System	
Basic flow	Admin	System
	4. Admin enter all the detail of Librarian. 5. Admin click on Add Librarian button. 6. Admin successfully add the Librarian.	4. The system displays all details on screen. 5. The system stores all details of Librarian. 6. The system successfully adds the Librarian.
Alternative flow	Not Available	
Exceptions	3. Admin does not enter all information of student. Solution: Admin must enter all information of student. 4. Admin add the existing Librarian. Solution: Admin must not enter the existing Librarian.	
Qualities	• Security: The system ensures secure access with role-based access control, allowing only authenticated Admin to add Librarian. • Response: The system guarantees that the add Librarian action completes within 2-3 seconds, ensuring a fast and efficient user experience. • Reliability: The system must consistently perform the Librarian addition function without failure, accurately storing and displaying student details.	

Remove Librarian

Name	Remove Librarian	
Participating Actor	Admin	
Goals	I want to remove a Librarian.	
Triggers	When Admin click on Librarian Report Button.	
Pre-conditions	1. Admin Must logged in 2. Admin must have internet availability.	
Post-conditions	Admin successfully remove the librarian.	
Basic flow	Admin	System
	1. Admin click on Librarian Report Button. 2. Admin Select the Librarian 3. Admin Click on Remove Librarian button.	1. The system will show the table of All Librarian. 2. The system will select the librarian 3. The system successfully remove the librarian
Alternative flow		
Exceptions	1. No Librarian Found. Solution: The admin must add the librarian 2. No Librarian Selected. Solution: The admin must select the Librarian.	
Qualities	• Usability: The interface for removing a librarian must be user-friendly, enabling the admin to quickly identify and remove the desired librarian. • Performance: The system must process the removal of a librarian within 2-3 seconds to ensure a smooth and efficient experience. • Reliability: The system must accurately update the database to reflect the removed librarian without errors or inconsistencies	

Edit Librarian Info

Name	Edit Librarian	
Participating Actor	Admin	
Goals	I want to Edit a Librarian Info.	
Triggers	When Admin click on Librarian Report Button.	
Pre-conditions	3. Admin Must logged in 4. Admin must have internet availability.	
Post-conditions	Admin successfully remove the librarian.	
Basic flow	Admin	System
	4. Admin click on Librarian Report Button. 5. Admin Select the Librarian 6. Admin Click on Update Librarian button.	4. The system will show the table of All Librarian. 5. The system will select the librarian 6. The system show form and successfully update the librarian information.
Alternative flow		
Exceptions	3. No Librarian Found. Solution: The admin must add the librarian 4. No Librarian Selected. Solution: The admin must select the Librarian.	
Qualities	• Usability: The interface for updating information of a librarian must be user-friendly, enabling the admin to quickly identify and remove the desired librarian. • Performance: The system must process the update of a librarian info within 2-3 seconds to ensure a smooth and efficient experience. • Reliability: The system must accurately update the database to reflect the updated librarian info without errors or inconsistencies.	

Students Report

Name	View Student Report	
Participating Actor	Librarian	
Goals	I want to view reports of all students.	
Triggers	When Librarian click on Student Report Button.	
Pre-conditions	5. Librarian Must logged in 6. Librarian must have internet availability.	
Post-conditions	Student Reports will be displayed on the screen	
Basic flow	Librarian	System
	1. Librarian click on Student Report Button.	7. The system will show the table of All Students.
Alternative flow	Librarian	System
	1. Librarian click on Student Report Button. 2. Librarian select the student Roll No or Department. 3. Librarian press the View button.	1. The system shows table of All issued Books. 2. The system filters the table based on selected student Roll No or Department. 3. The system displays the filtered rows.
Exceptions	1. No Department Found. Solution: The Librarian just select the added Departments 2. The Librarian enter wrong Roll No of Student. Solution: The Librarian must select the valid and authentic Student Roll No.	
Qualities	• Usability: The interface for viewing student reports must be intuitive, allowing the Librarian to easily search by Roll No. or Department. • Performance: The system must display the student reports within 2-3 seconds to provide a seamless user experience. • Scalability: The system must be able to handle a growing number of student records as the library expands.	

Librarian Report

Name	View Librarian Report	
Participating Actor	Admin	
Goals	I want to view reports of all Librarian.	
Triggers	When Admin click on Librarian Report Button.	
Pre-conditions	7. Admin Must logged in 8. Admin must have internet availability.	
Post-conditions	Librarian Reports will be displayed on the screen	
Basic flow	Admin	System
	7. Admin click on Librarian Report Button.	8. The system will show the table of All Librarian.
Alternative flow		
Exceptions	5. No Librarian Found. Solution: The admin must add the librarian	
Qualities	• Usability: The interface for viewing Librarian reports must be intuitive, allowing the admin to easily view the librarian • Performance: The system must display the Librarian reports within 2-3 seconds to provide a seamless user experience. • Scalability: The system must be able to handle a growing number of Librarian records as the library expands.	

Student Details

Name	View student details	
Participating Actor	Librarian	
Goals	I want to view student details.	
Triggers	When Librarian click on view Button from the tables of all Students	
Pre-conditions	1. Librarian Must logged in the system. 2. Librarian must have internet availability.	
Post-conditions	Student details will be displayed on the screen	
Basic flow	Librarian	System
	1. Librarian click on student report Button. 2. Librarian click on view button of specific student from the student report table	1. The system will show the table of all students. 2. The system will show the details of specific student.
Alternative flow	Not Available	
Exceptions	1. No Student Found Solution: The Librarian must add the students 2. The student details are not showing. Solution: The Librarian must click the button of view to get detail of specific student.	
Qualities	• Performance: The system should display student details within 2 seconds after the librarian selects a student. • Usability: The interface should be easy to navigate, enabling the librarian to quickly locate and view student details. • Reliability: The system should ensure that student details are accessible without interruption, even during times of high demand.	

Book Issue

Name	Book issue	
Participating Actor	Librarian	
Goals	I want to issue Book to the Student	
Triggers	When Librarian click on Issue Book Button.	
Pre-conditions	1. Librarian Must logged in 2. Librarian must have internet availability.	
Post-conditions	The Book issues to the student successfully.	
Basic flow	Librarian	System
	1. Librarian select the Book. 2. Librarian select the student. 3. Librarian enter no of Days. 4. Librarian successfully issues Book to the student.	1. System shows information of selected book. 2. System Show Roll No of Selected Student. 3. System store no of days. 4. System successfully issues the book.
Alternative flow	Not Available	
Exceptions	1. Librarian does not select the Book. Solution: Librarian must select the book to issue. 2. Librarian issue the book having 0 available books. Solution: Librarian must display message to not issue book having zero availability. 3. Librarian does not select the student. Solution: Librarian must select the student. 4. Librarian does not enter no of days of Book issued. Solution: Librarian must enter no of days.	
Qualities	• Security: The system ensures secure access with role-based access control, allowing only authenticated Librarians to issue books. • Response: The system guarantees that the book issue action completes within 2-3 seconds, providing a fast and efficient user experience. • Reliability: The system must consistently perform the book issue function without failure, accurately recording the transaction and displaying relevant information.	

Issued books report

Name	Issued Books Report	
Participating Actor	Librarian	
Goals	I want to generate the reports of Books Issued.	
Triggers	When librarian click on the issue book report button	
Pre-conditions	1. Librarian Must logged in 2. Librarian must have internet availability.	
Post-conditions	The Book report will be issued successfully	
Basic flow	Librarian	System
	1. Librarian click on Issue Report Button.	1. The system will show the table of Books issued to student.
Alternative flow	Librarian	System
	1. Librarian click on Issue Report Button 2. Librarian select the student or Book. 3. Librarian press the View button.	1. The system shows table of All issued Books. 2. The system filters the table based on selected student or Books. 3. The system displays the filtered rows.
Exceptions	1. No Books Issued Solution: The Librarian must issue the book to view its report 2. The student does not Issued any report. Solution: The Librarian must issue book to student to view its issued reports.	
Qualities	• Performance: The system must generate and display the issued book report within 2-3 seconds to ensure a smooth experience for the Librarian. • Scalability: The system must handle large datasets, ensuring the issued book report can accommodate a growing number of issued books as the library expands. • Usability: The interface for generating reports must be simple and intuitive, allowing the Librarian to filter and view reports with ease.	

Return Book

Name	Return Book	
Participating Actor	Librarian	
Goals	I want to return the book borrowed by student.	
Triggers	When Librarian click on the Return book button	
Pre-conditions	1. Librarian Must logged in 2. Librarian must have internet availability.	
Post-conditions	The Book will be return to Digital library successfully	
Basic flow	Librarian	System
	1. Librarian select the student. 2. Librarian select the book. 3. Librarian press the Select button. 4. Librarian presses the return book button.	1. System gets the information 2. System gets the information 3. System shows the information of issued book to respective student. 4. System successfully returns the book.
Alternative flow	Not Available	
Exceptions	1. Librarian does not select the student Solution: Librarian must select the student 2. Librarian does not select the book Solution: Librarian must select the book 3. Librarian does not return the book Solution: Librarian must select press the return book button to return book	
Qualities	• Reliability: The system must consistently process the book return function without failure, ensuring that the book is properly marked as returned. • Security: The system ensures secure access with role-based permissions, allowing only authorized Librarians to process book returns. • Performance: The system must complete the return process within 2-3 seconds, ensuring a smooth and fast transaction for the Librarian.	

Return books report

Name	Return Books Report	
Participating Actor	Librarian	
Goals	I want to generate the reports of Books Returned.	
Triggers	When librarian click on the Return book report button	
Pre-conditions	3. Librarian Must logged in 4. Librarian must have internet availability.	
Post-conditions	The Book report will be issued successfully	
Basic flow	Librarian	System
	2. Librarian click on Return Book Button. 3. Librarian Click on return Book report button	2. The system will show the return book form. 3. The system will show the table of Books returned by student.
Alternative flow	Librarian	System
	4. Librarian click on Return book report Button 5. Librarian select the student or Book. 6. Librarian press the View button.	4. The system shows table of All Return Books. 5. The system filters the table based on selected student or Books. 6. The system displays the filtered rows.
Exceptions	3. No Books Returned Solution: The Librarian must returned the book to view its report 4. The student does not Issued any report. Solution: The Librarian must issue book to student to return book and view report	
Qualities	• Performance: The system must generate and display the return book report within 2-3 seconds to ensure a smooth experience for the Librarian. • Scalability: The system must handle large datasets, ensuring the issued book report can accommodate a growing number of issued books as the library expands. • Usability: The interface for generating reports must be simple and intuitive, allowing the Librarian to filter and view reports with ease.	

Paid Penalty Fine

Name	Paid Penalty Fine	
Participating Actor	Librarian	
Goals	I want to Paid the fine of Penalty.	
Triggers	When librarian click on the Paid Fine button	
Pre-conditions	1. Librarian Must logged in 2. Librarian must have internet availability.	
Post-conditions	The Librarian successfully paid the fine.	
Basic flow	Librarian	System
	4. Librarian click on Return Book Button. 5. Librarian Click on Penalty Report Button 6. Librarian select the penalty 7. Librarian click on paid the fine button.	4. The system will show the return book form. 5. The system will show the table of Books returned by student. 6. The system selects that penalty. 7. The system successfully paid the fine
Alternative flow		
Exceptions	5. No Penalty Found Solution: There must be the penalty in record. 6. The Librarian did not select the penalty. 7. Solution: The Librarian must select the penalty to paid the fine.	
Qualities	• Usability: The interface for paying penalties must be user-friendly, allowing the librarian to quickly identify penalties and complete the payment process with minimal effort. • Performance: The system must process and confirm the payment of penalties within 2-3 seconds to ensure an efficient workflow. • Reliability: The system must ensure accurate and error-free updates to the records after a penalty is paid.	

Waive Penalty Fine

Name	Waive Penalty Fine	
Participating Actor	Librarian	
Goals	I want to waive the fine of Penalty.	
Triggers	When librarian click on the waive Fine button	
Pre-conditions	3. Librarian Must logged in 4. Librarian must have internet availability.	
Post-conditions	The Librarian successfully waive the fine.	
Basic flow	Librarian	System
	8. Librarian click on Return Book Button. 9. Librarian Click on Penalty Report Button 10. Librarian select the penalty 11. Librarian click on waive the fine button.	8. The system will show the return book form. 9. The system will show the table of Books returned by student. 10. The system selects that penalty. 11. The system successfully waive the fine
Alternative flow		
Exceptions	8. No Penalty Found Solution: There must be the penalty in record. 9. The Librarian did not select the penalty. 10. Solution: The Librarian must select the penalty to waive the fine.	
Qualities	• Usability: The interface for waive penalties must be user-friendly, allowing the librarian to quickly identify penalties and waive them with minimal effort. • Performance: The system must process the waive of penalties within 2-3 seconds to ensure an efficient workflow. • Reliability: The system must ensure accurate and error-free updates to the records after a penalty is waive.	

Change Password

Name	Change Password	
Participating Actor	Student	
Goals	I want to change my account password	
Triggers	When the student or librarian clicks on the "Change Password" button.	
Pre-conditions	1. The student must be logged into the system. 2. The student must have internet availability.	
Post-conditions	The student successfully changes his password.	
Basic flow	Librarian	System
	1. The student clicks the Change Password button. 2. The student enters the old password 3. The student enters the new password 4. The student enters the confirm password 5. The student clicks on change password button	1. The system shows the option of change password. 2. The system store and display the old password. 3. The system store and display the new password. 4. The system store and display the confirm password. 5. The system check validity and change the password successfully.
Alternative flow	Not Available	
Exceptions	1. The old password is incorrect. Solution: The system notifies the student that the old password is incorrect. 2. The new password is not match with confirm password. Solution: The student or librarian must enter the same password in new and confirm password.	
Qualities	• Usability: The password change process should be straightforward, with fields for the old and new passwords clearly labelled. • Security: The system should ensure secure transmission and storage of the new password using encryption. • Performance: The system should confirm the password change within 2-3 seconds.	

Logout

Name	Logout from system		
Participating Actor	Admin, Librarian, Student		
Goals	I want to logout from the system.		
Triggers	When Admin, Librarian or Student click on logout button.		
Pre-conditions	1. Admin, Librarian or Student Must logged in the system. 2. Admin, Librarian or Student must have internet availability.		
Post-conditions	The Admin, Librarian or Student successfully logged out of the system and taken to the login page.		
Basic flow	Admin, Librarian, Student	System	
	1. Admin, Librarian, Student click on Logout Button.	1. The System successfully logout the system and move to login page.	
Alternative flow	Not Available		
Exceptions	1. Admin, Librarian, Student is still in system Solution: The Admin, Librarian must click the logout button again.		
Qualities	• Performance: The system should log the user out and redirect them to the login page within 1 second of clicking the logout button. • Usability: The logout option should be easily accessible and clearly visible, ensuring a smooth logout process for the user. • Security: The system should terminate the user session immediately to prevent unauthorized access and ensure that sensitive data is cleared.		

View Account

Name	View Account	
Participating Actor	Student	
Goals	I want to view my account details.	
Triggers	When the student clicks on the "View Account" button	
Pre-conditions	1. The student must be logged into the system. 2. Student must have internet availability.	
Post-conditions	The student's account details are displayed on the screen.	
Basic flow	Librarian	System
	1. The student clicks the View Account button.	1. The system displays the account detail of the student.
Alternative flow	Not Available	
Exceptions	1. No account information is found. Solution: The student must click the view account button.	
Qualities	• Usability: The interface should make viewing account details intuitive by offering clear buttons for navigation. • Security: The system must ensure that only the logged-in student can view their account details. • Performance: The system must display the student's details within 2-3 seconds after the button is clicked.	

Edit Account

Name	Edit Account	
Participating Actor	Student, Librarian, Admin	
Goals	I want to edit my account details	
Triggers	When the student, admin and Librarian clicks on the Edit Account button.	
Pre-conditions	1. The student, admin and Librarian must be logged into the system. 2. The student, admin and Librarian must have internet availability.	
Post-conditions	The student, admin and Librarian successfully updates his account detail.	
Basic flow	Librarian	System
	1. The student, admin and Librarian clicks the "Edit Account" button. 2. The student, admin and Librarian enters the detail to edit it.	1. The system shows the edit detail options. 2. The system updates the details successfully.
Alternative flow	Not Available	
Exceptions	1. The student, admin and Librarian tries to save incomplete details. Solution: The system should ask the student to fill all the necessary fields before saving.	
Qualities	• Usability: Editing account details should be simple, with clear labels and editable fields. • Reliability: The system must reliably save updated information without data loss. • Performance: The system should allow edits and save changes within 2-3 seconds.	

Borrowed book list

Name	View borrowed book list	
Participating Actor	Student	
Goals	I want to view the list of books that I have borrowed.	
Triggers	When the student clicks on the "Books borrowed" button.	
Pre-conditions	1. The student must be logged into the system. 2. The student must have internet availability.	
Post-conditions	The system displays a list of borrowed books, including the book title, borrowed date, and number of days borrowed.	
Basic flow	Librarian	System
	1. The student clicks the Books borrowed button.	1. The system will show the list of books that student has been borrowed.
Alternative flow	Not Available	
Exceptions	1. There is no book in list of borrowed books. Solution: The student must have to borrow book.	
Qualities	• Performance: The system should display the list of borrowed books within 2 seconds after the student clicks the "Books borrowed" button. • Usability: The list of borrowed books should be presented in a clear and organized format, with easy-to-read details. • Reliability: The system should provide accurate and up-to-date information about the borrowed books, ensuring data consistency at all times.	

Return book list

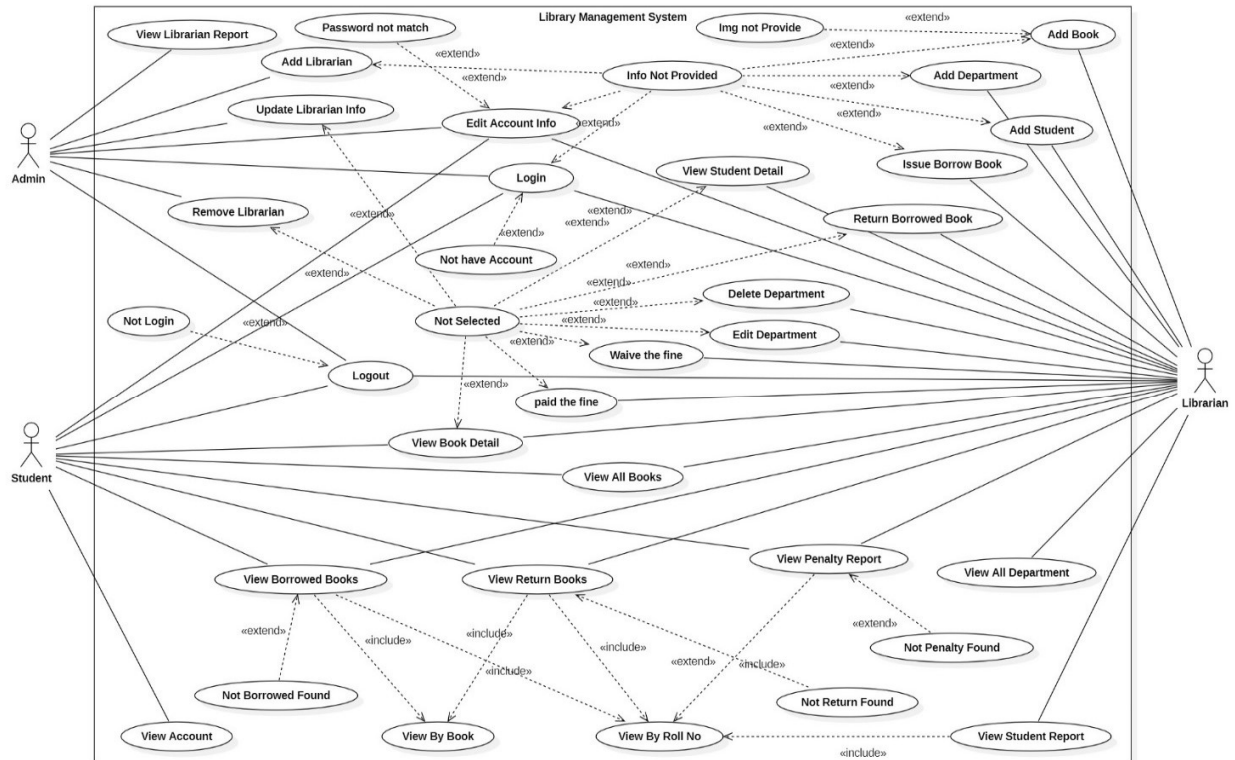
Name	View Returned book list	
Participating Actor	Student	
Goals	I want to view the list of books they have Returned.	
Triggers	When the student clicks on the "Books Returned" button.	
Pre-conditions	1. The student must be logged into the system. 2. The student must have internet availability.	
Post-conditions	The system displays a list of Returned books, including the book title, borrowed date, returned date and number of days borrowed.	
Basic flow	Student	System
	1. The student clicks the Books returned button.	1. The system will show the list of books that have been returned.
Alternative flow	Not Available	
Exceptions	1. There is no book in list of returned books. Solution: The student must have to return book	
Qualities	• Performance: The system should display the list of returned books within 2 seconds after the student clicks the Books Returned button. • Usability: The list should be well-organized and easy to understand, and in an attractive way. • Reliability: The system should ensure that all returned book data is accurate and consistently updated in real-time.	

Penalty Report list

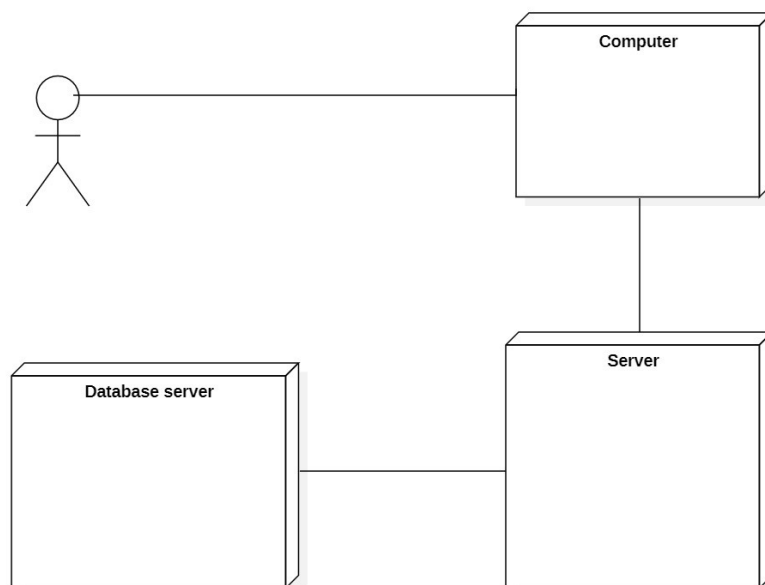
Name	View Penalty Report	
Participating Actor	Student, Librarian	
Goals	I want to view the Penalty Reports.	
Triggers	When the student clicks on the "View Penalty Report" button.	
Pre-conditions	3. The student, Librarian must be logged into the system. 4. The student, Librarian must have internet availability.	
Post-conditions	The system displays a list of Penalty, including the book title, borrowed date, returned date and due date and fine status.	
Basic flow	Student, Librarian	System
	2. The student clicks the Penalty Report button.	2. The system will show the list of Penalty that have been recorded.
Alternative flow	Not Available	
Exceptions	2. There is no Penalty Record. Solution: There must Penalty Record of student.	
Qualities	• Performance: The system should display the list of Penalty within 2 seconds after the student clicks the Penalty Report button. • Usability: The list should be well-organized and easy to understand, and in an attractive way. • Reliability: The system should ensure that all Penalty data is accurate and consistently updated in real-time.	

UML Diagrams

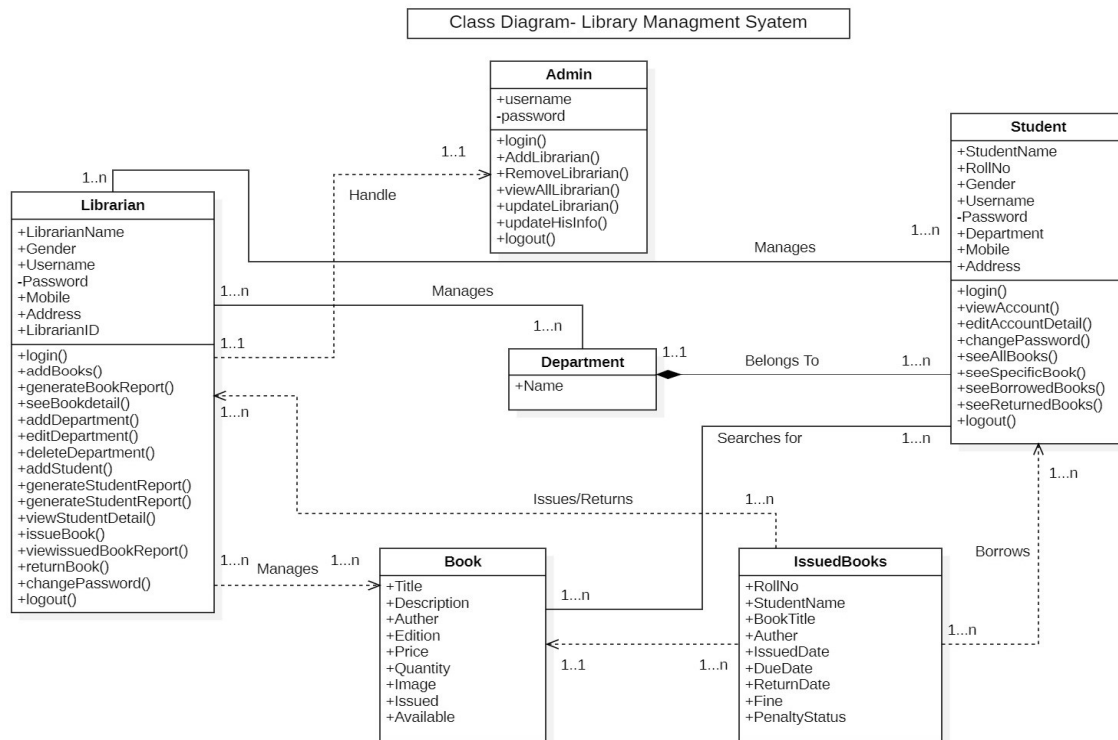
Use Case Diagram:



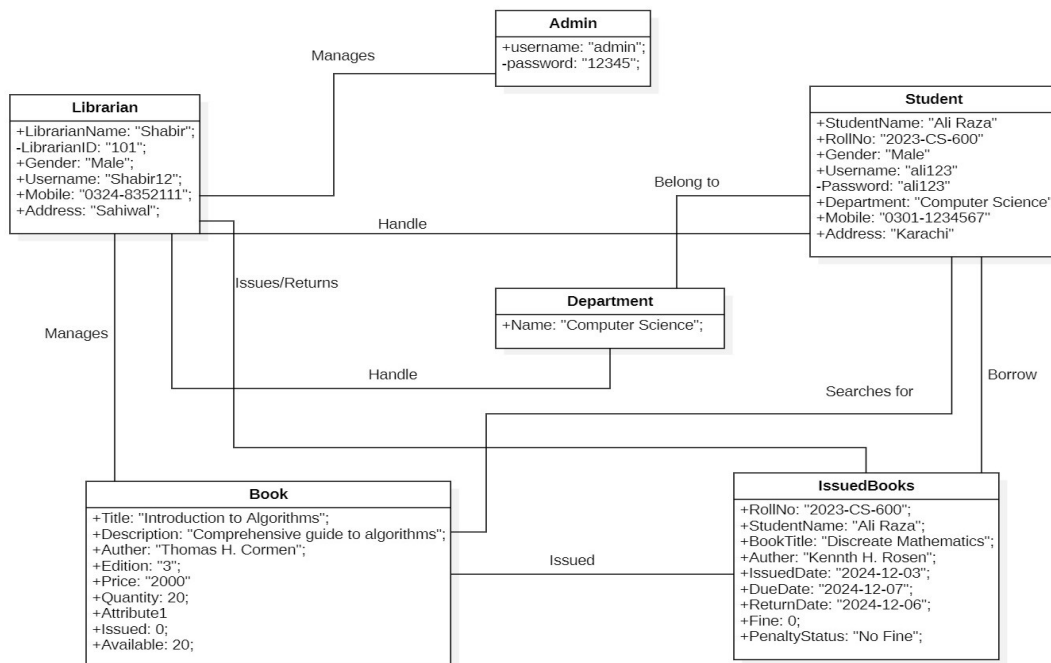
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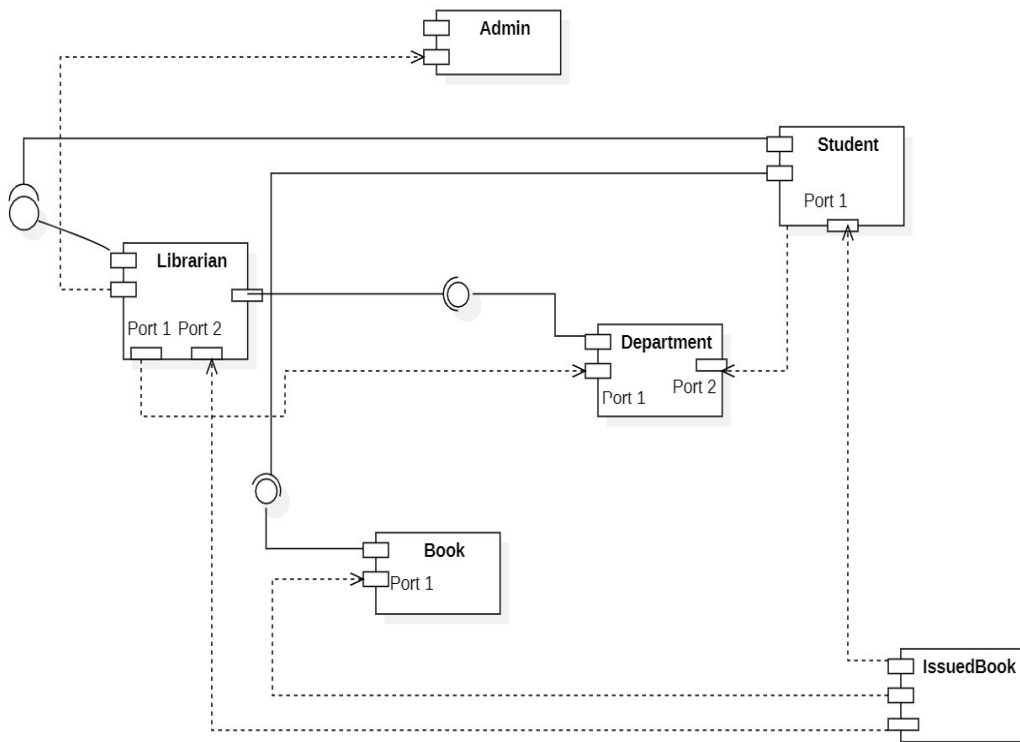
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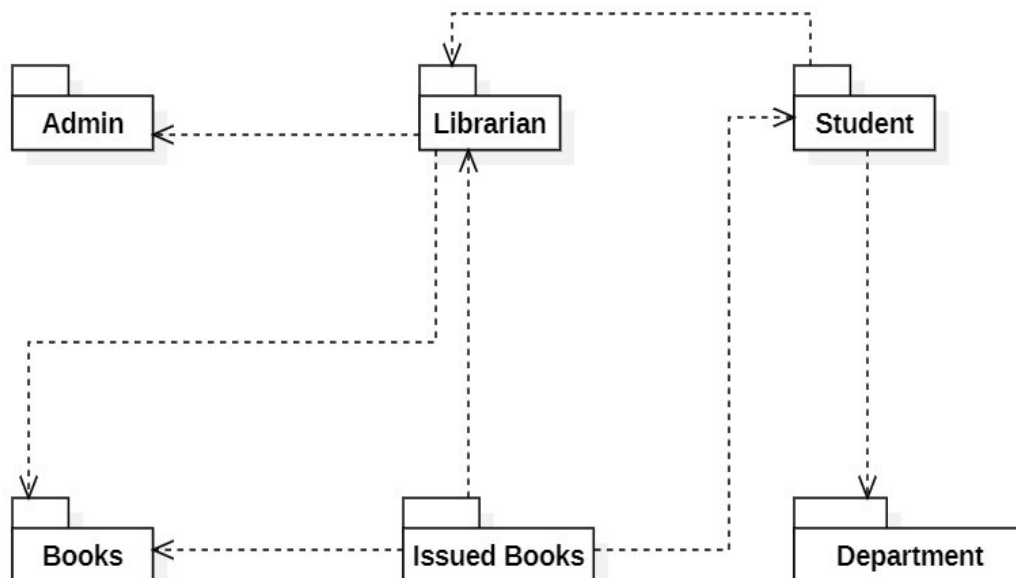
Object Diagram:



Component Diagram:

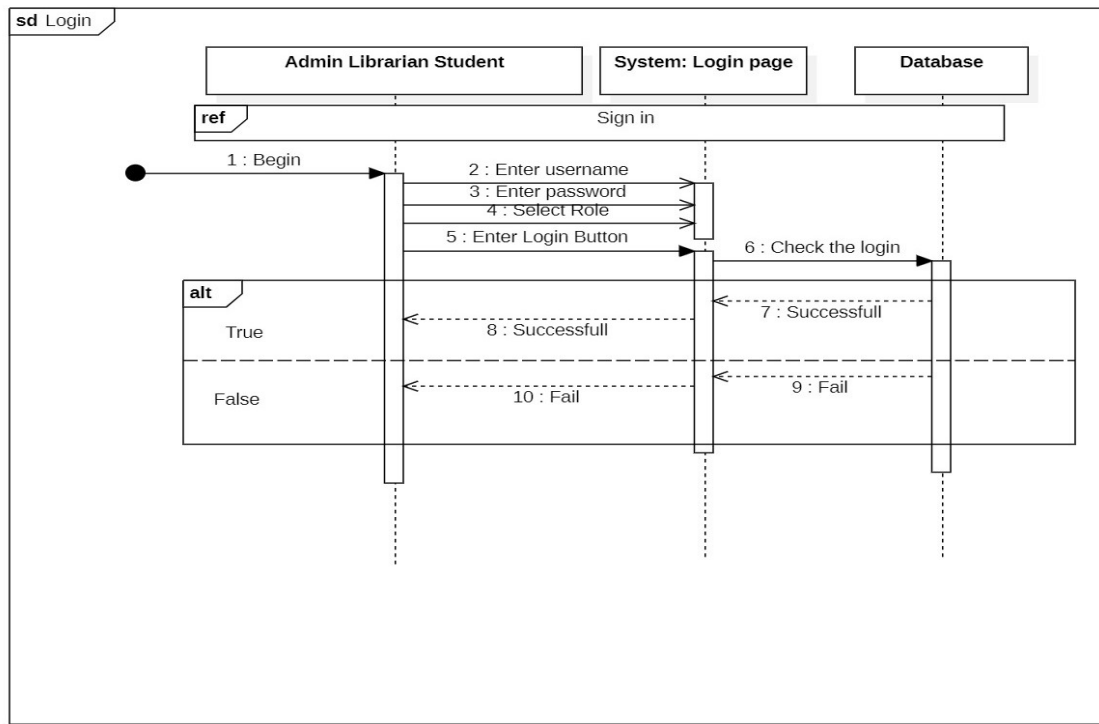


Package Diagram:

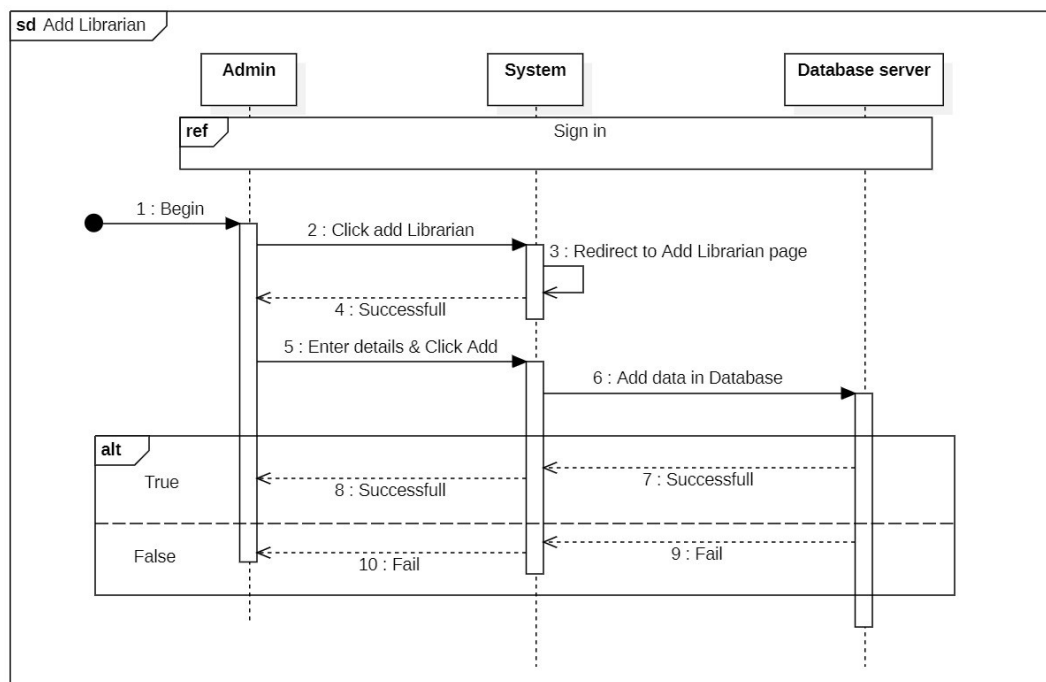


Sequence Diagram:

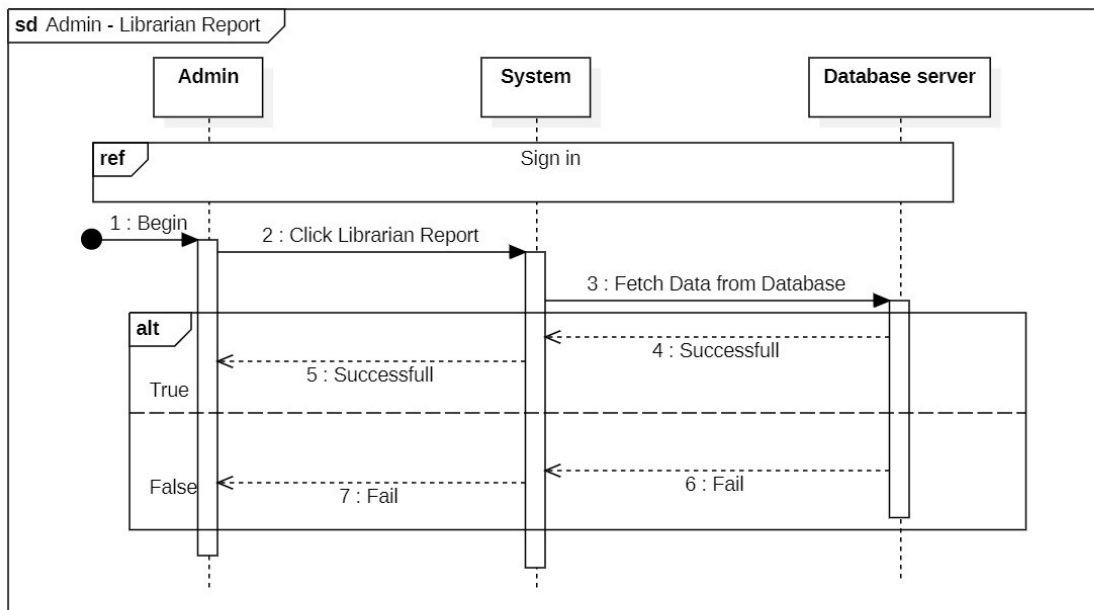
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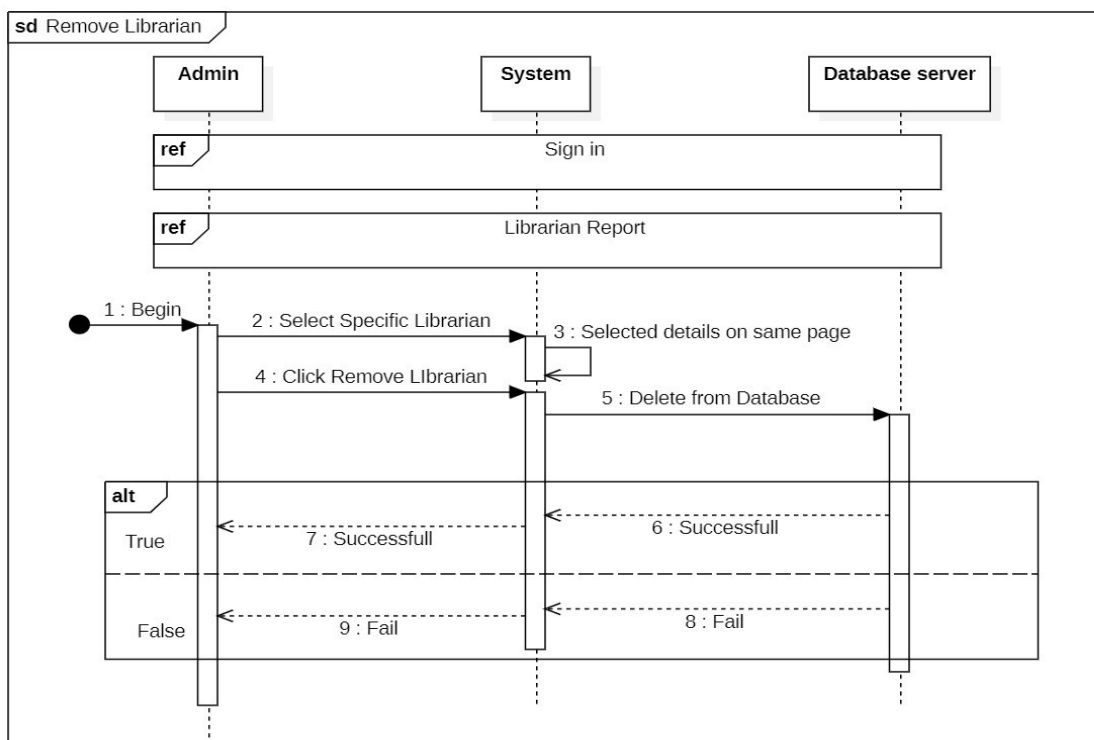
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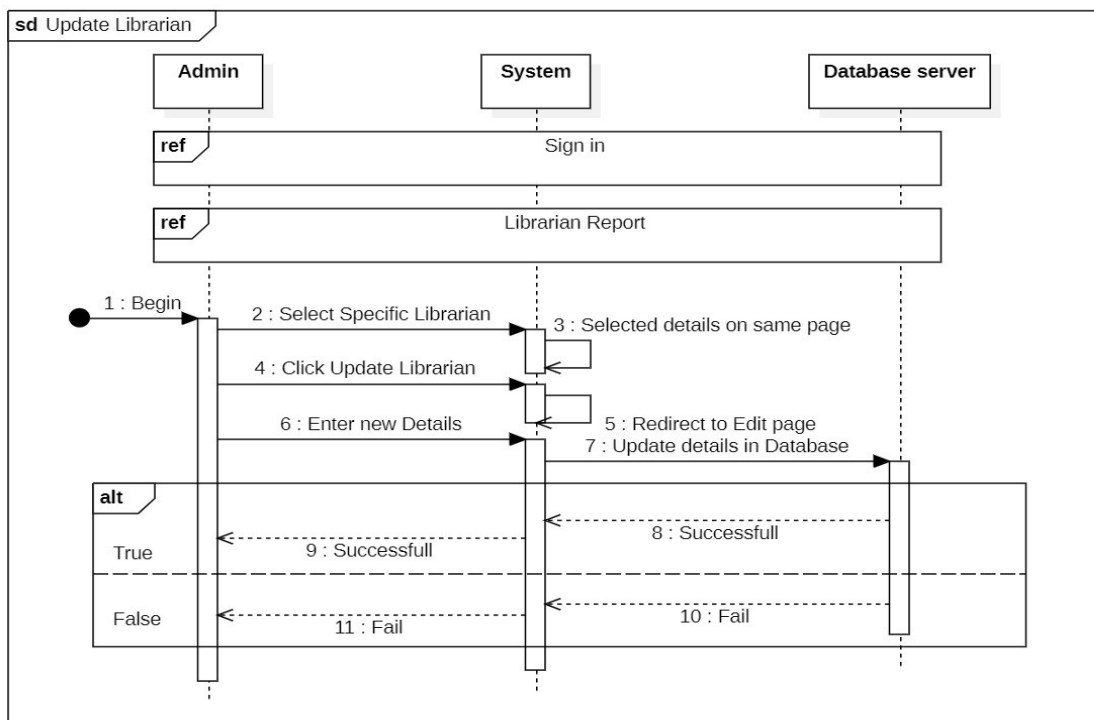
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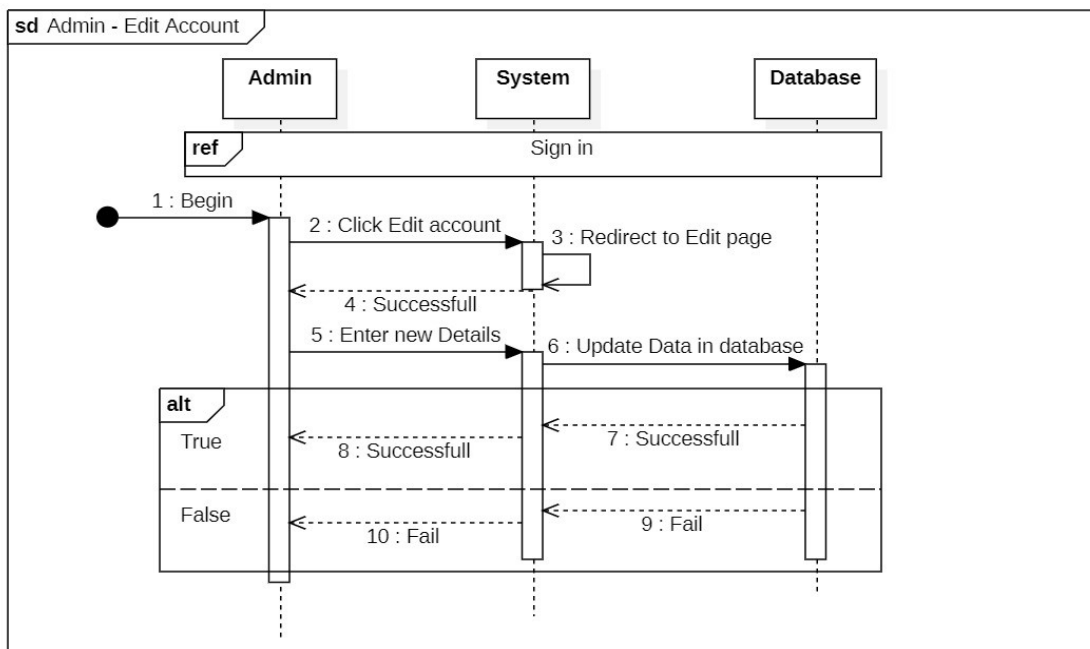
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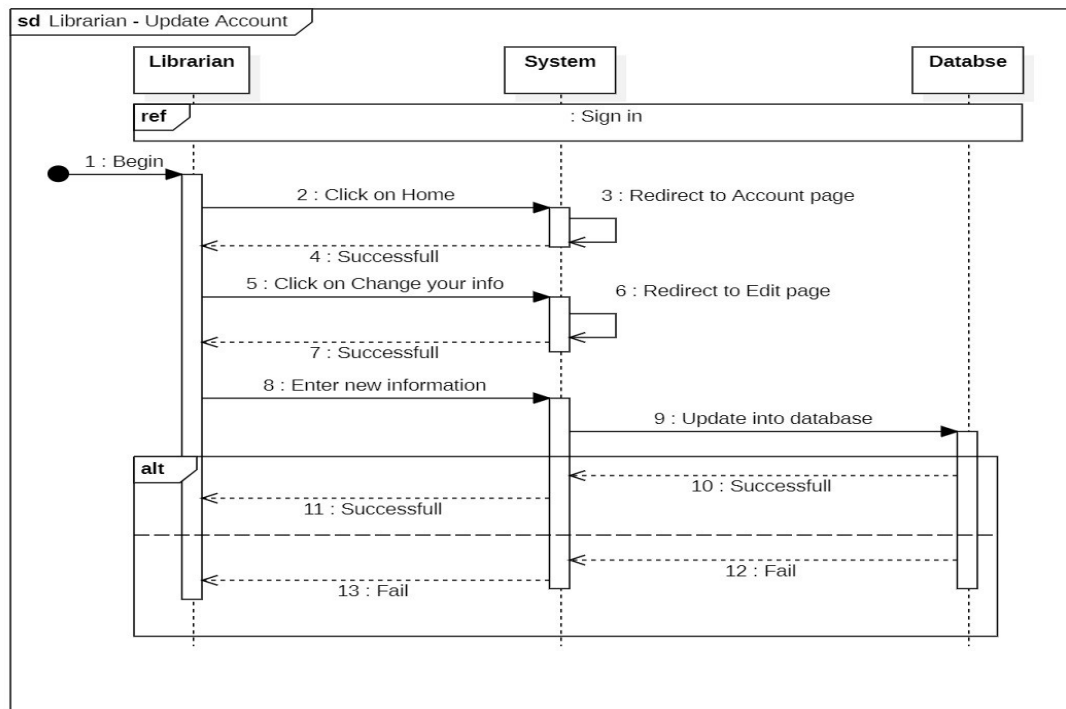
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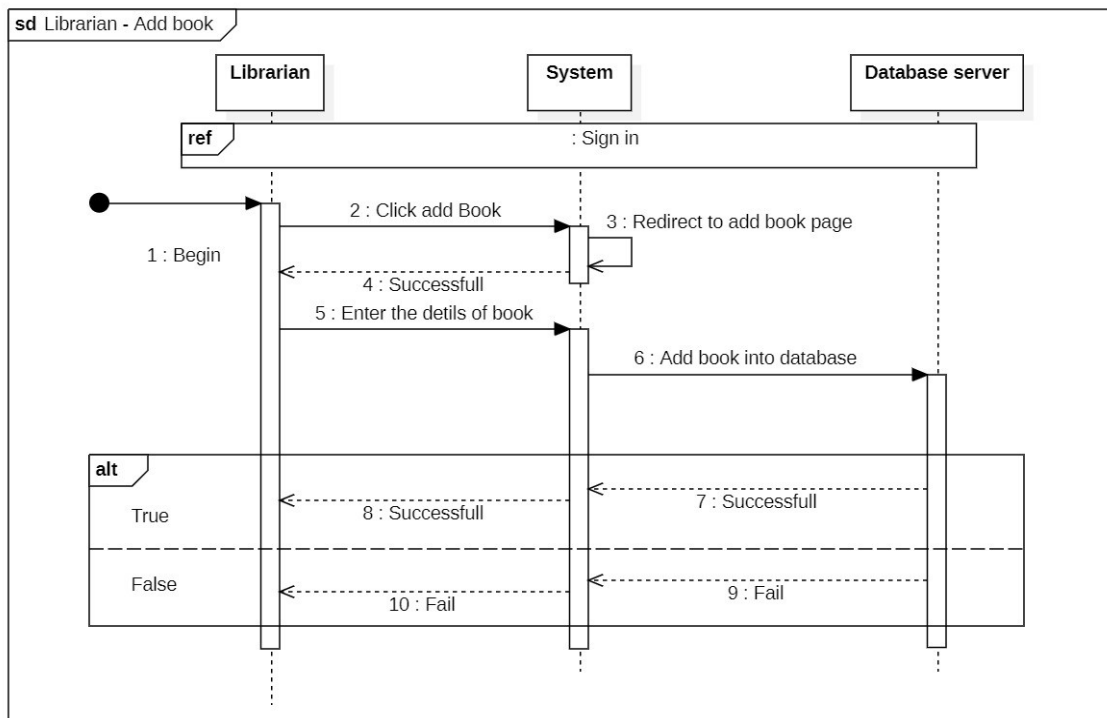
Admin - Update his Information:



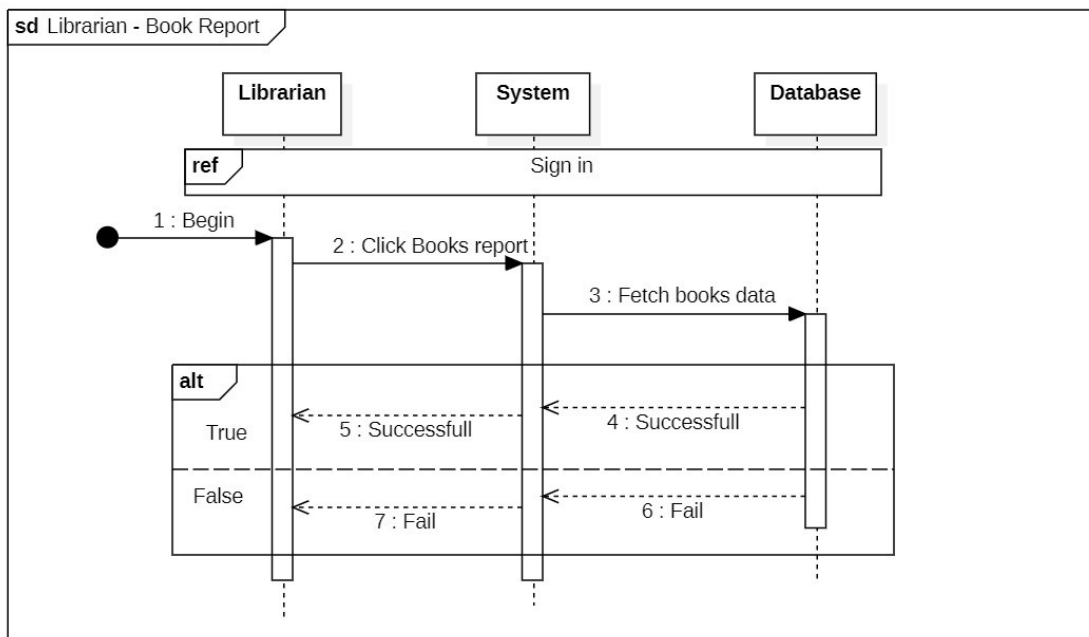
Librarian - update his Information:



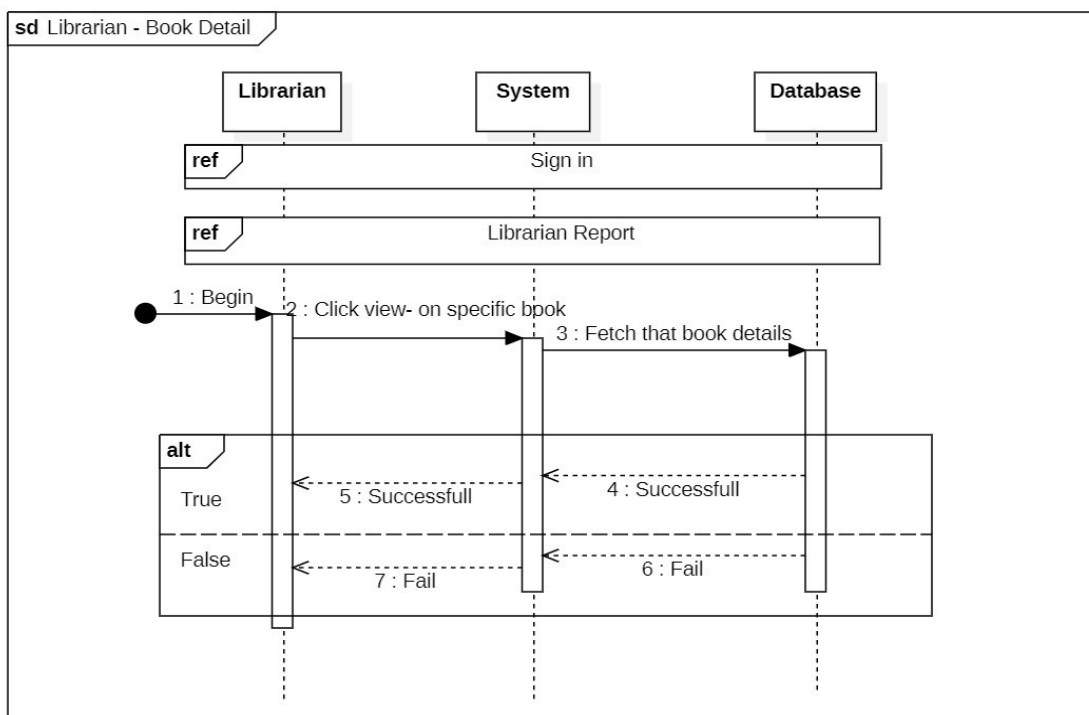
Add Books:



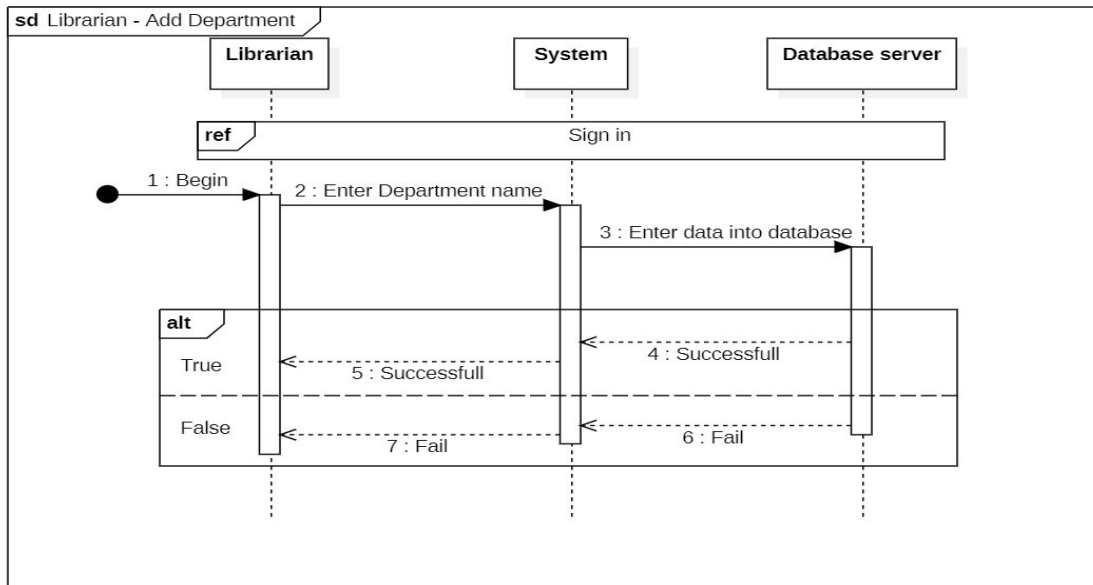
Librarian - View all Books Report:



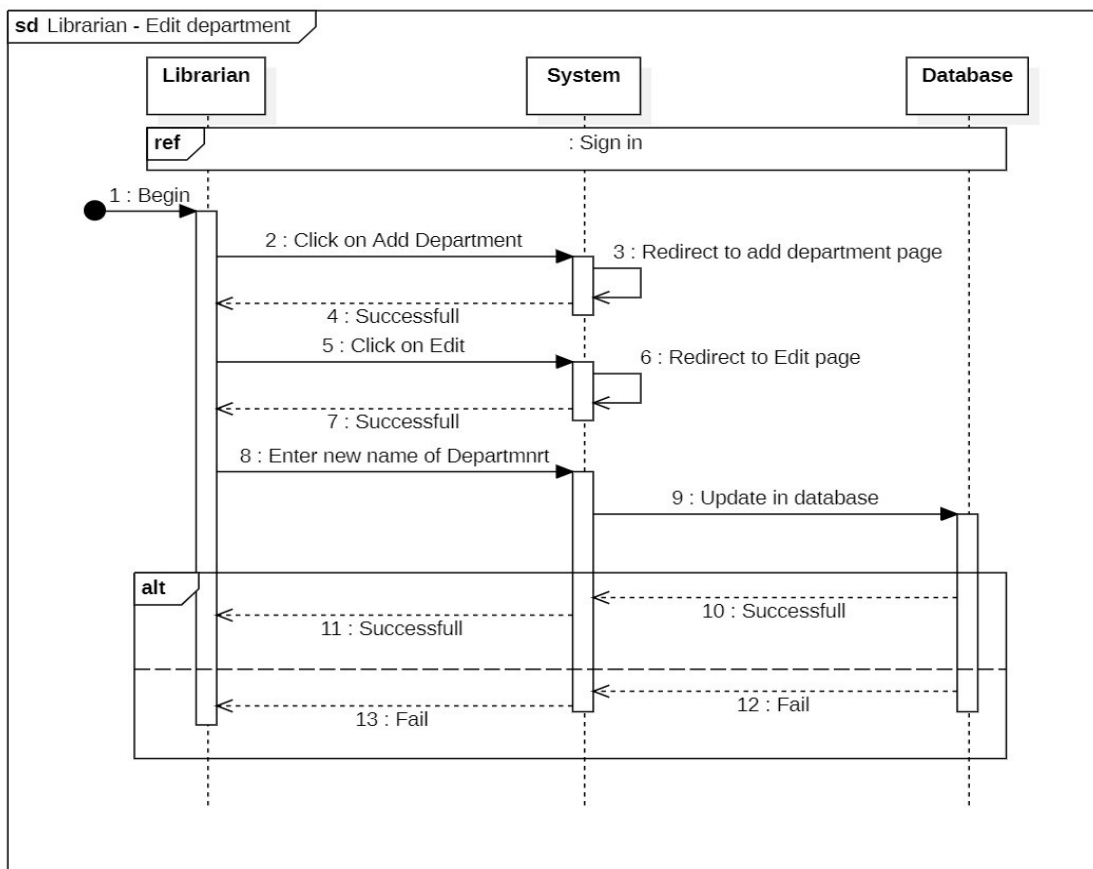
Librarian - Book Details:



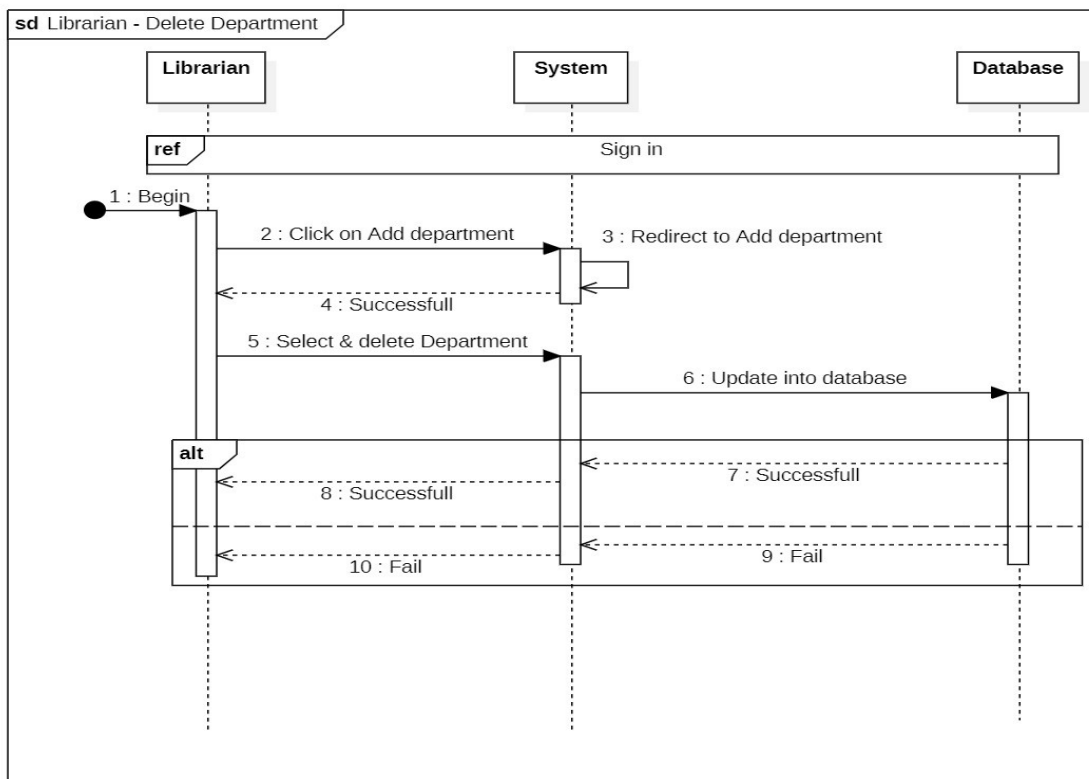
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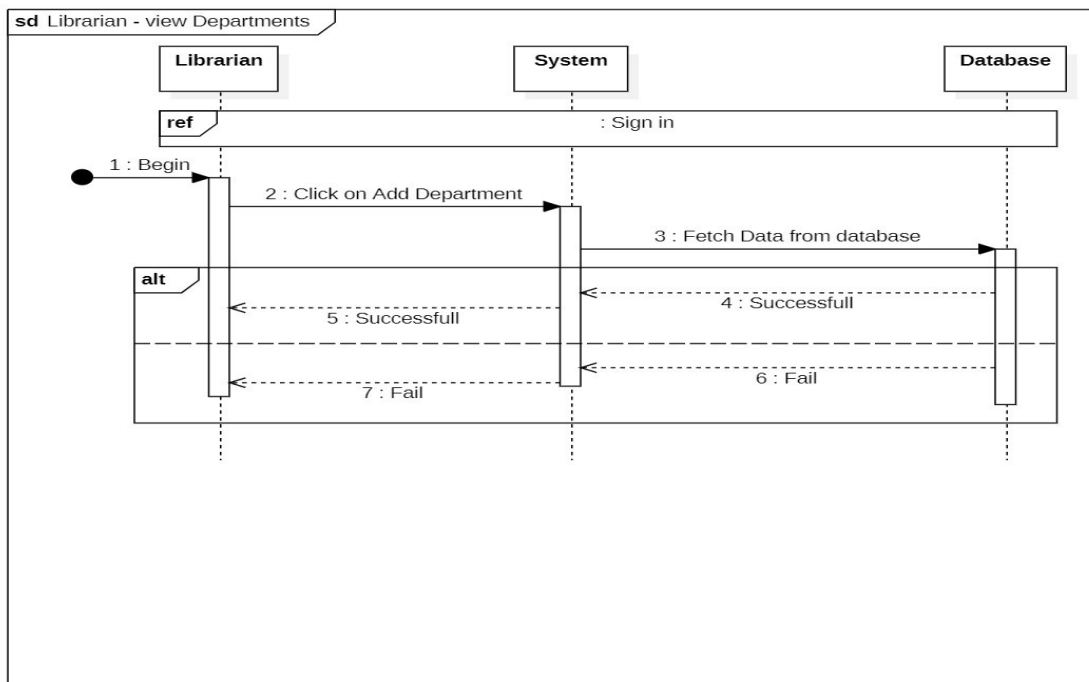
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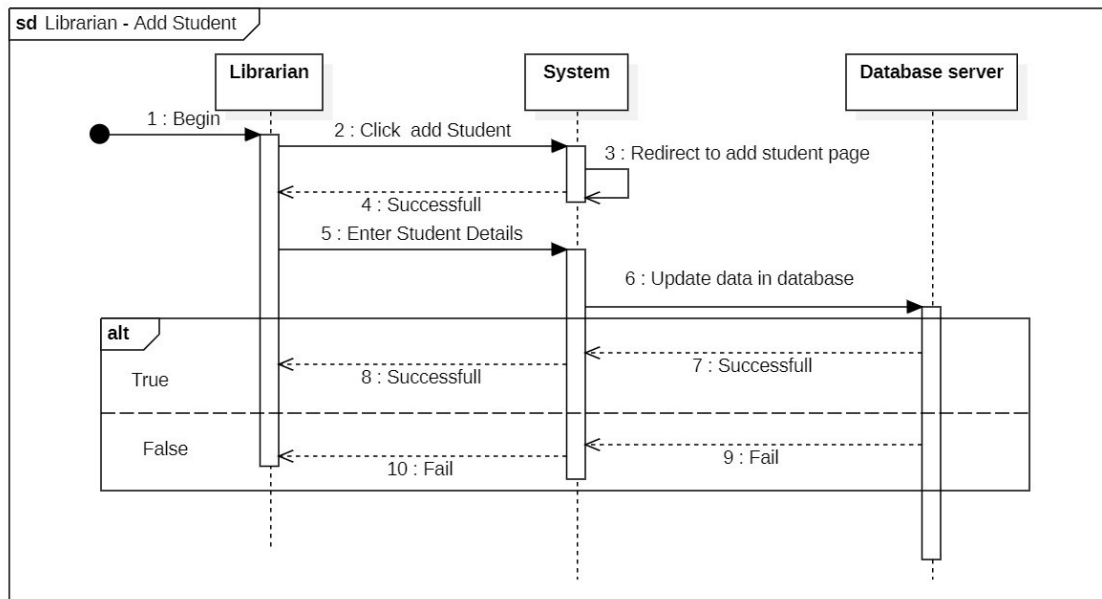
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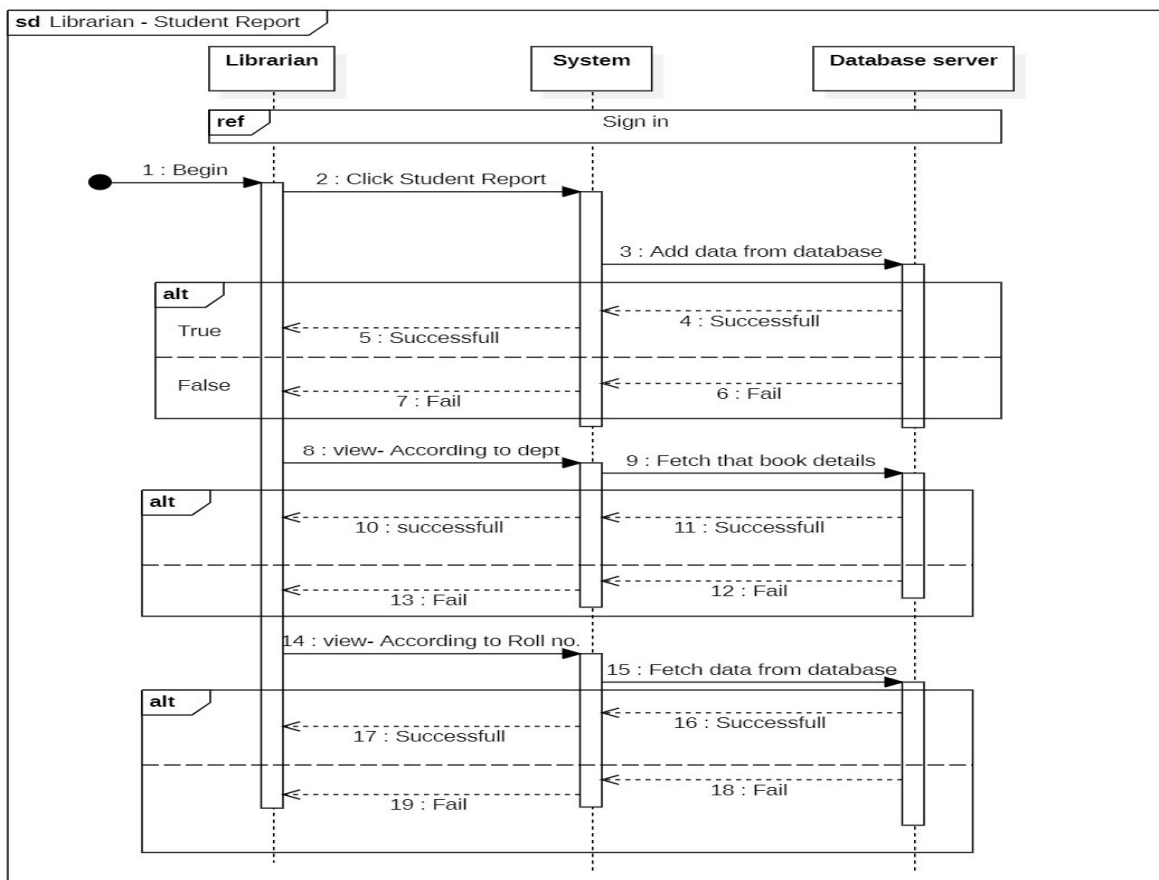
All departments:



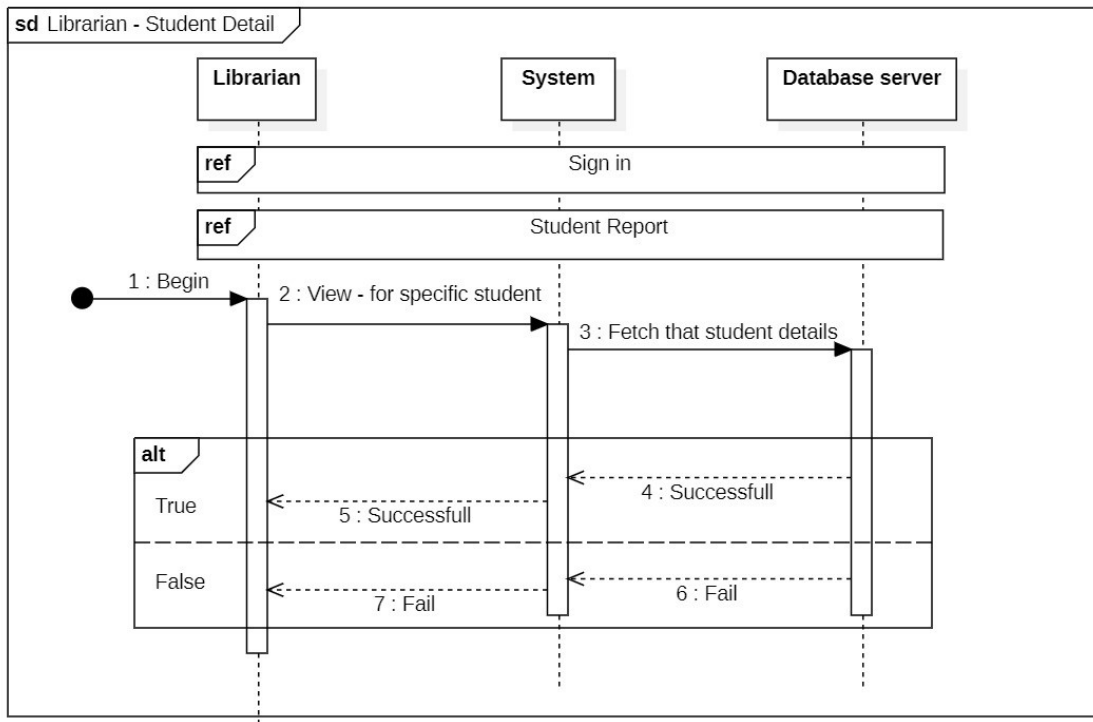
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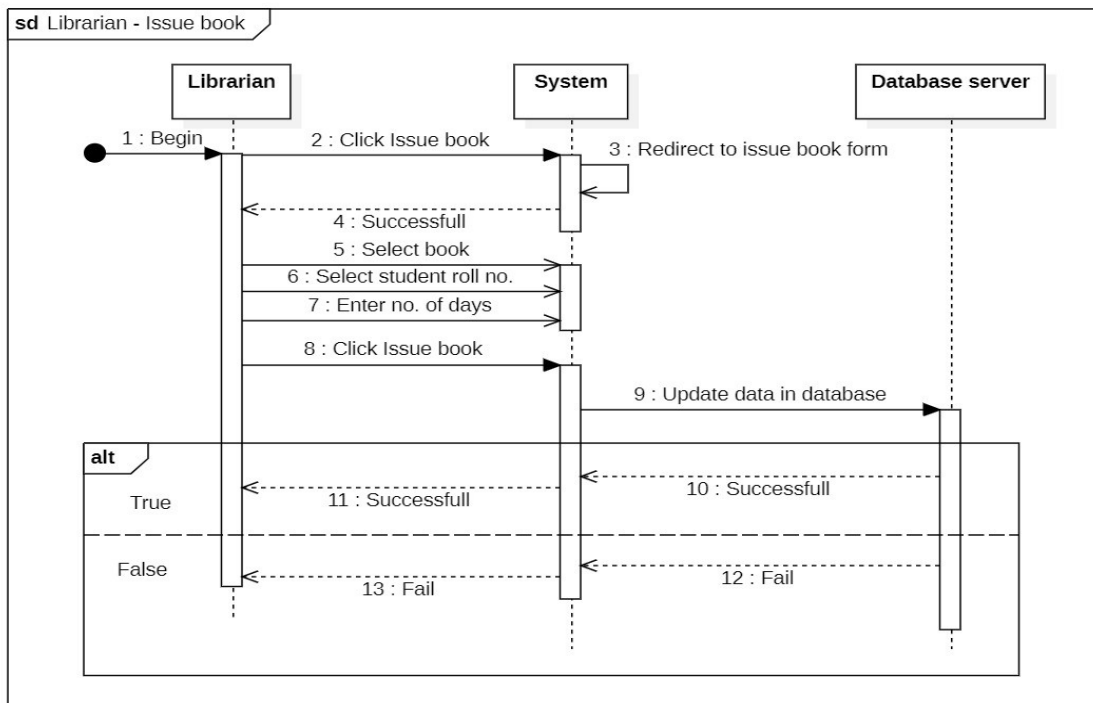
View students report:



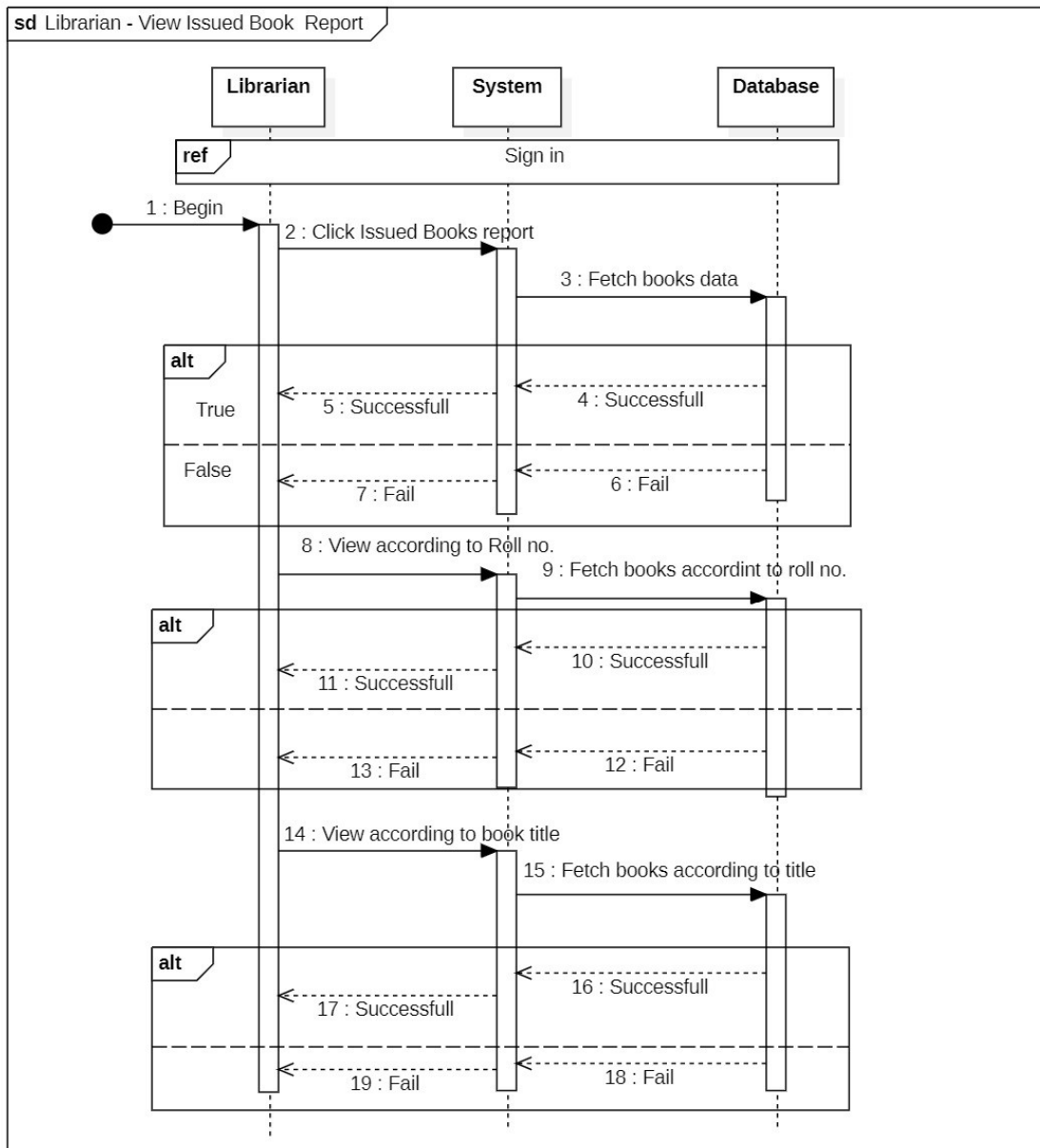
Student Detail:



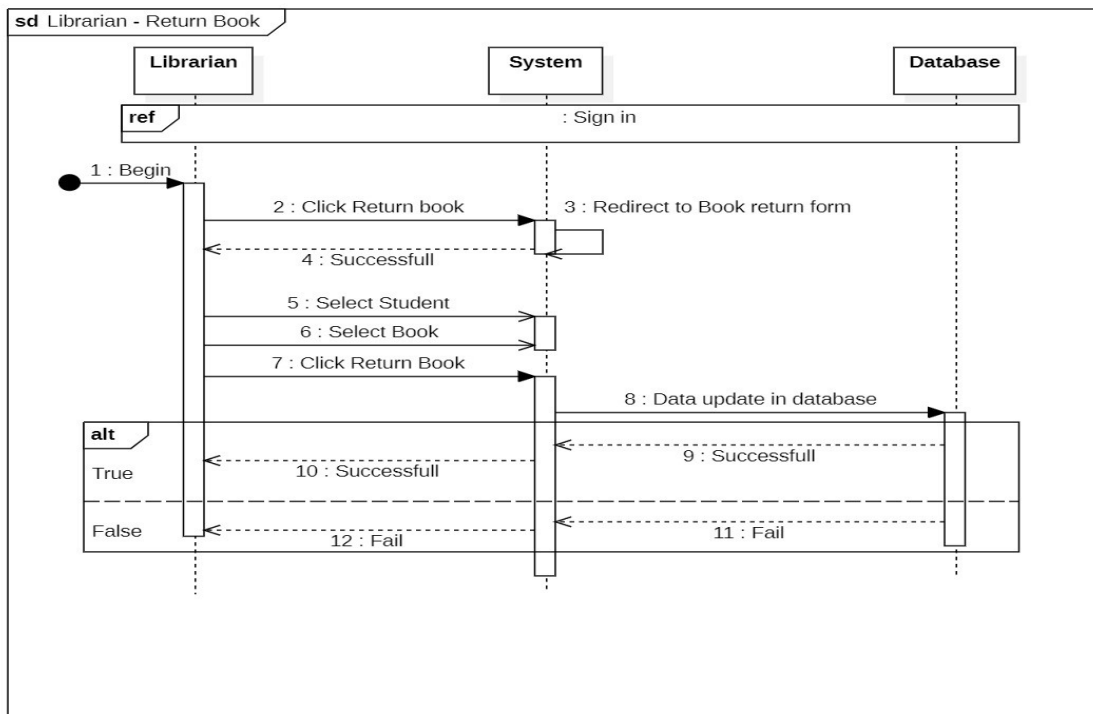
Issue Book:



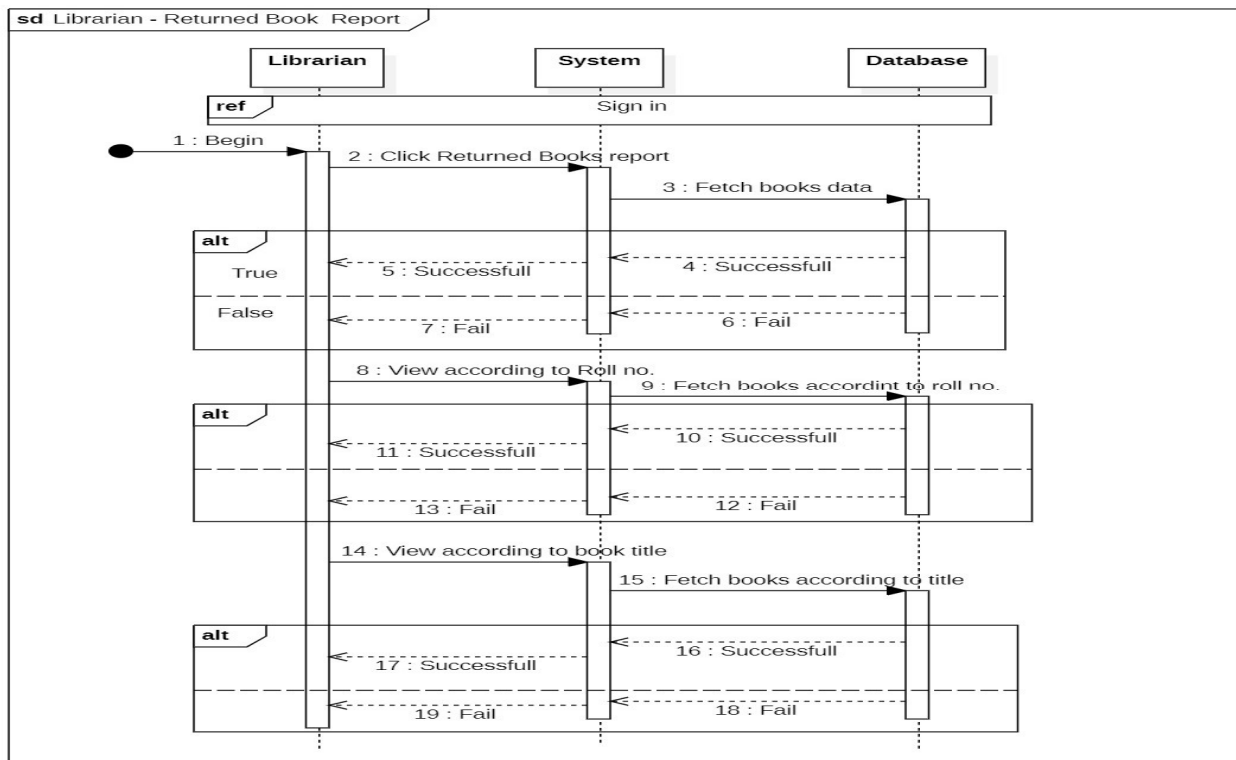
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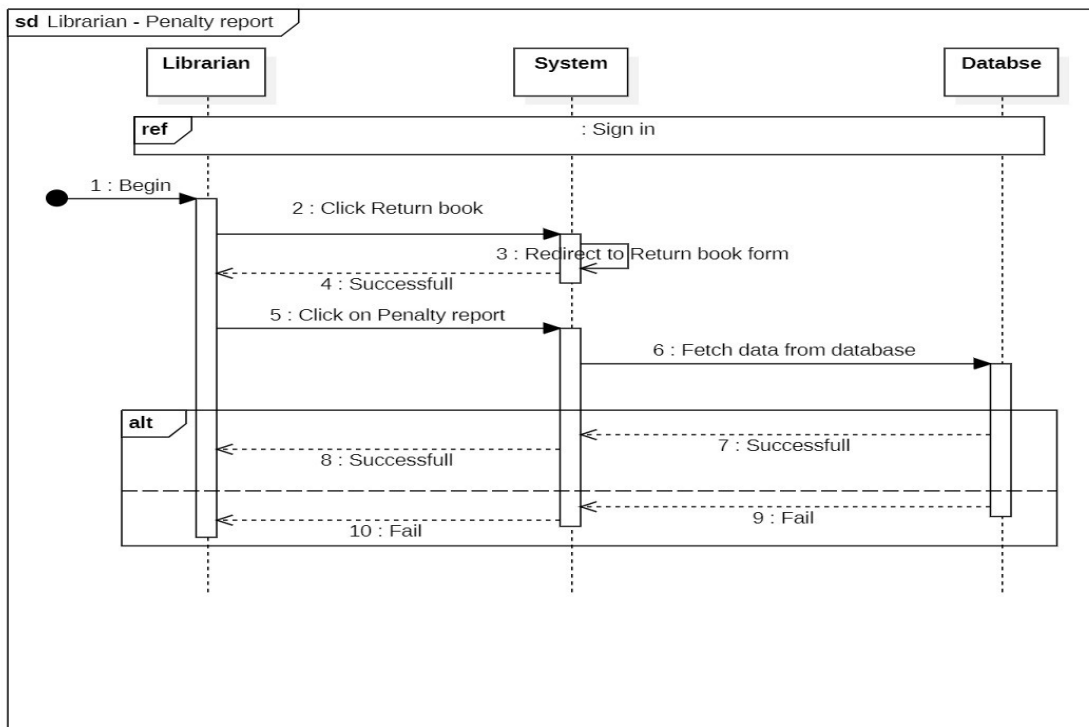
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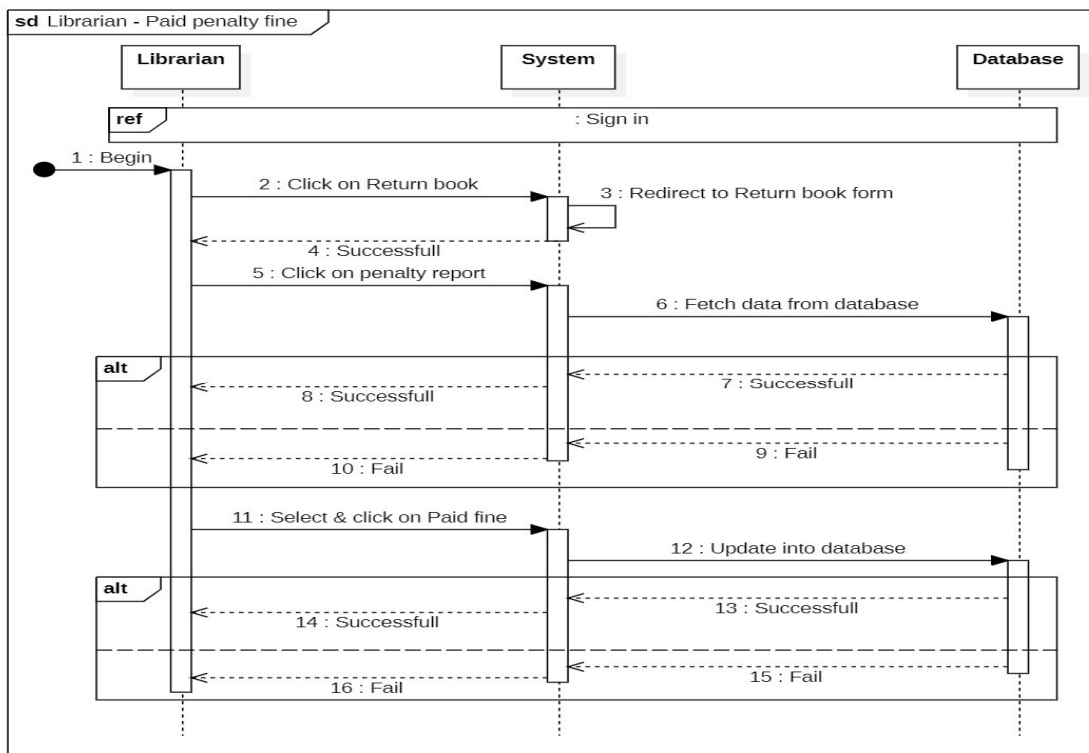
Librarian - Returned Books Report



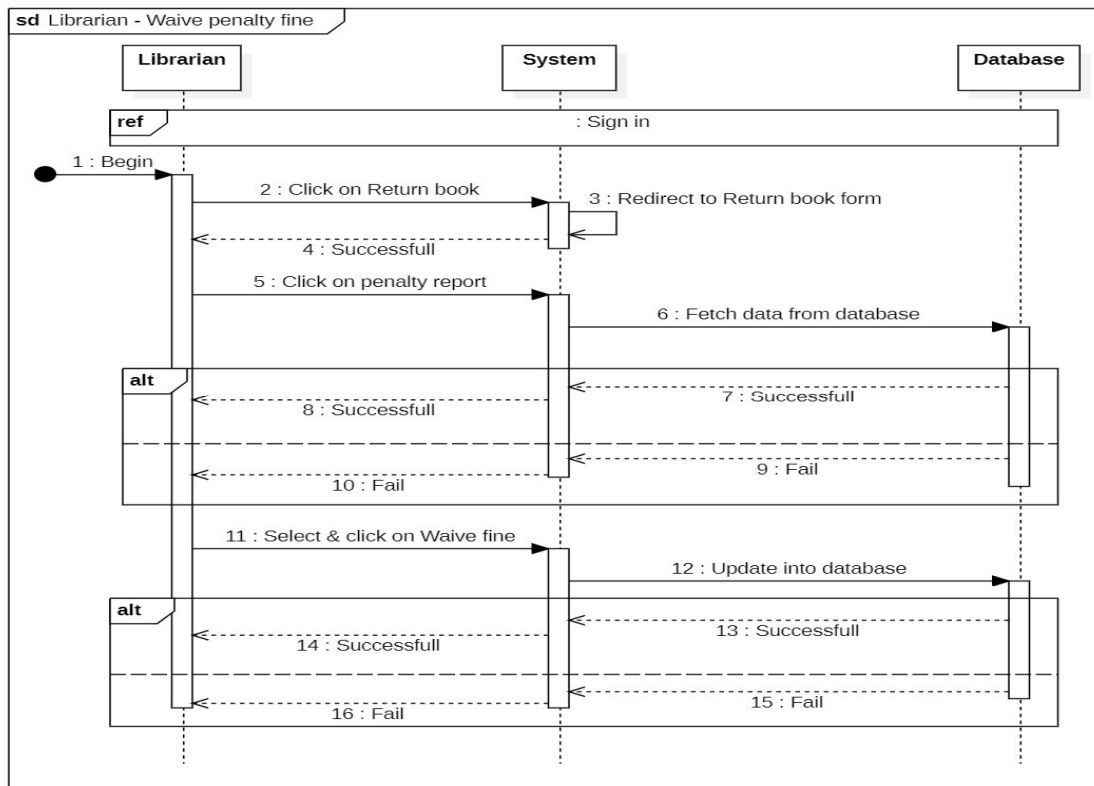
Librarian view Penalty Report:



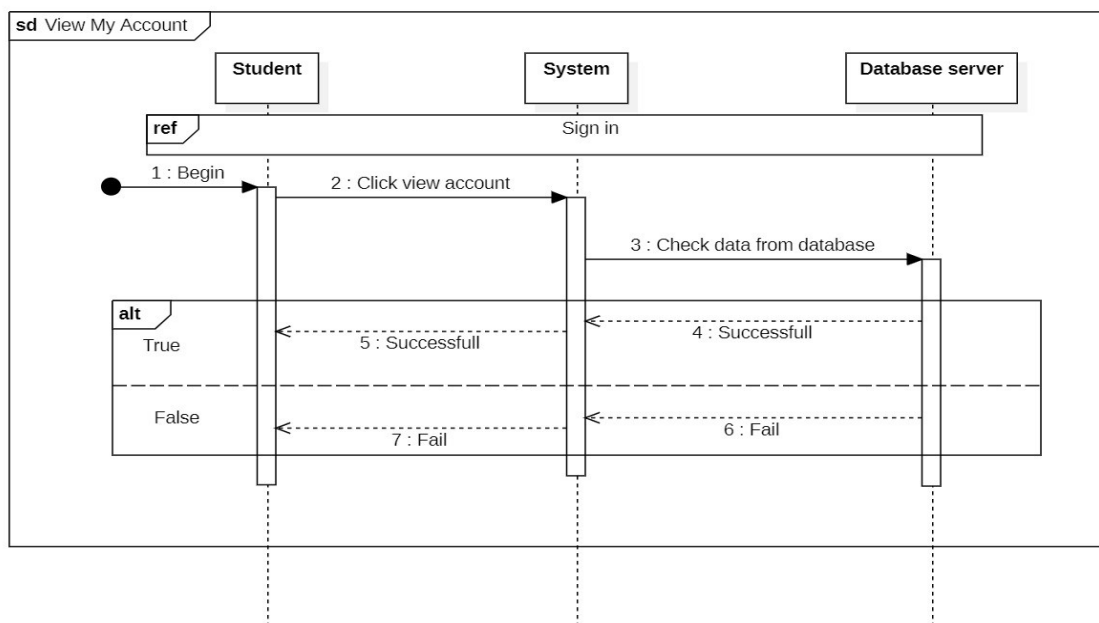
Paid the Penalty Fine:



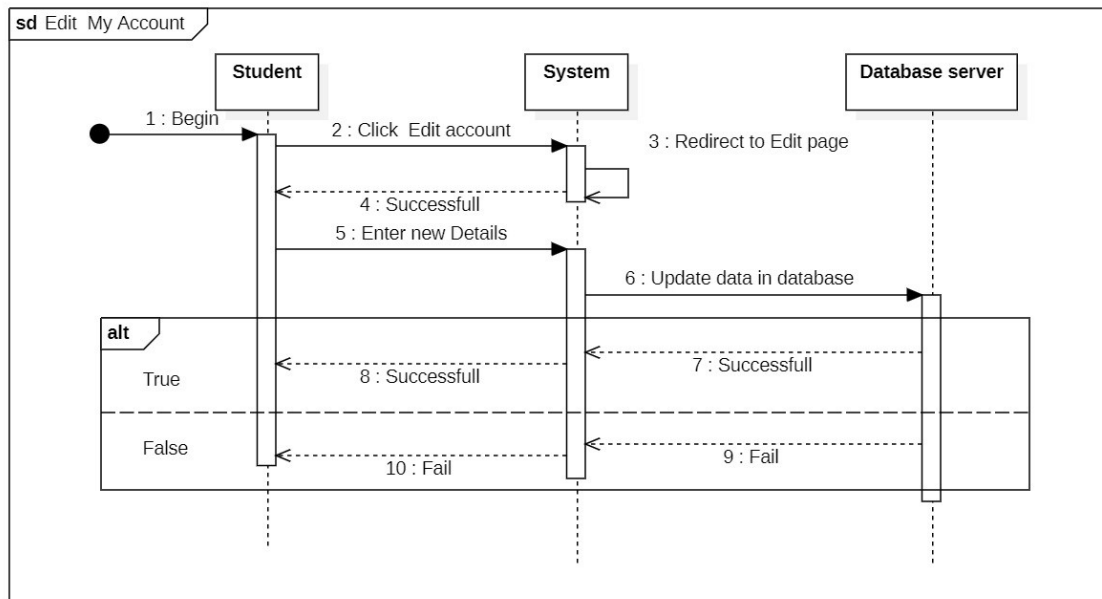
Waive the Penalty Fine:



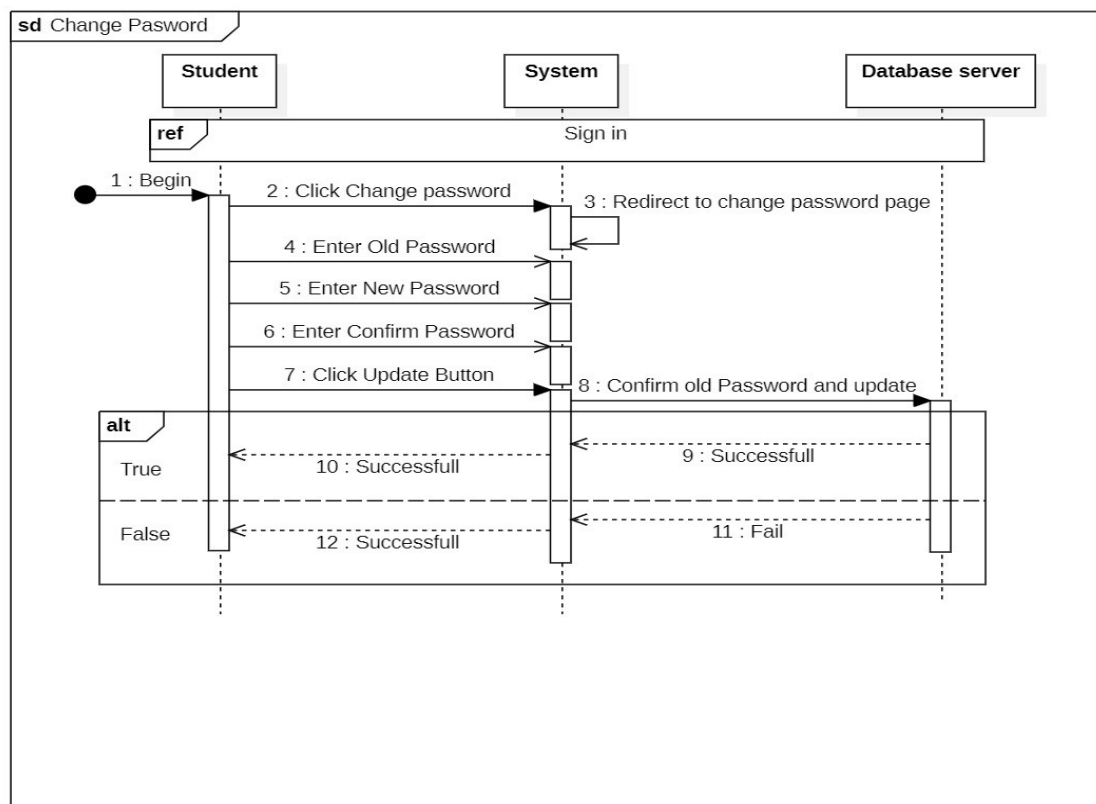
Student - view his Account:



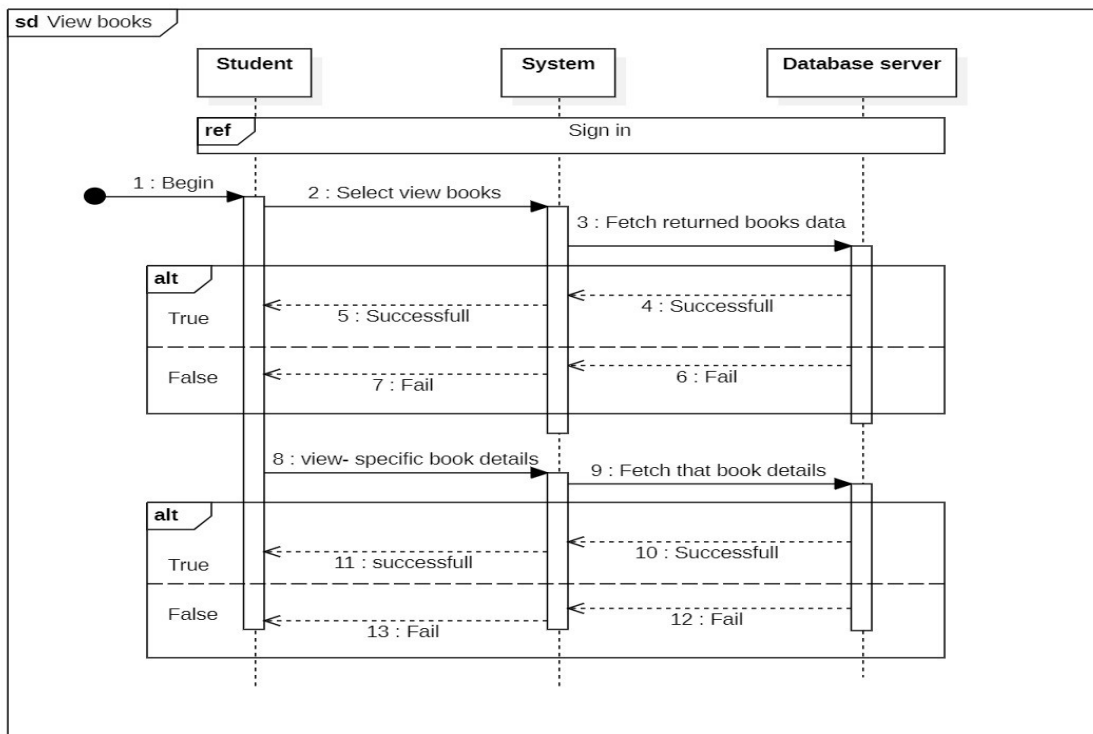
Student - Edit his Account Detail:



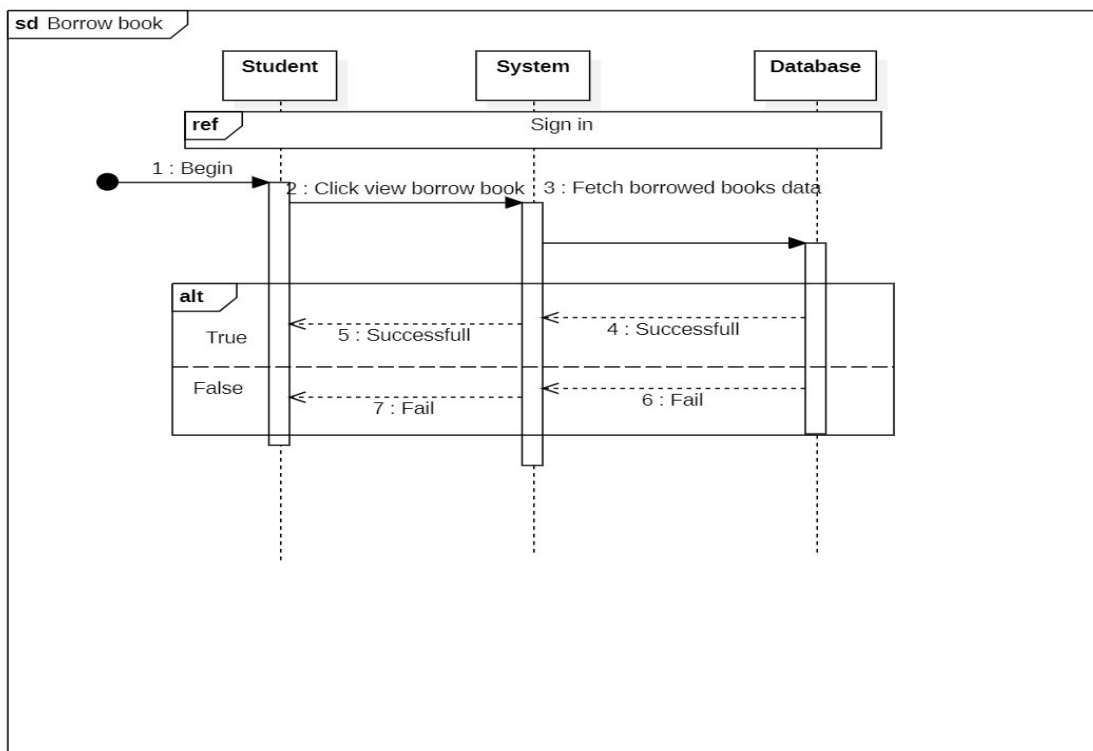
Student - Change his Password:



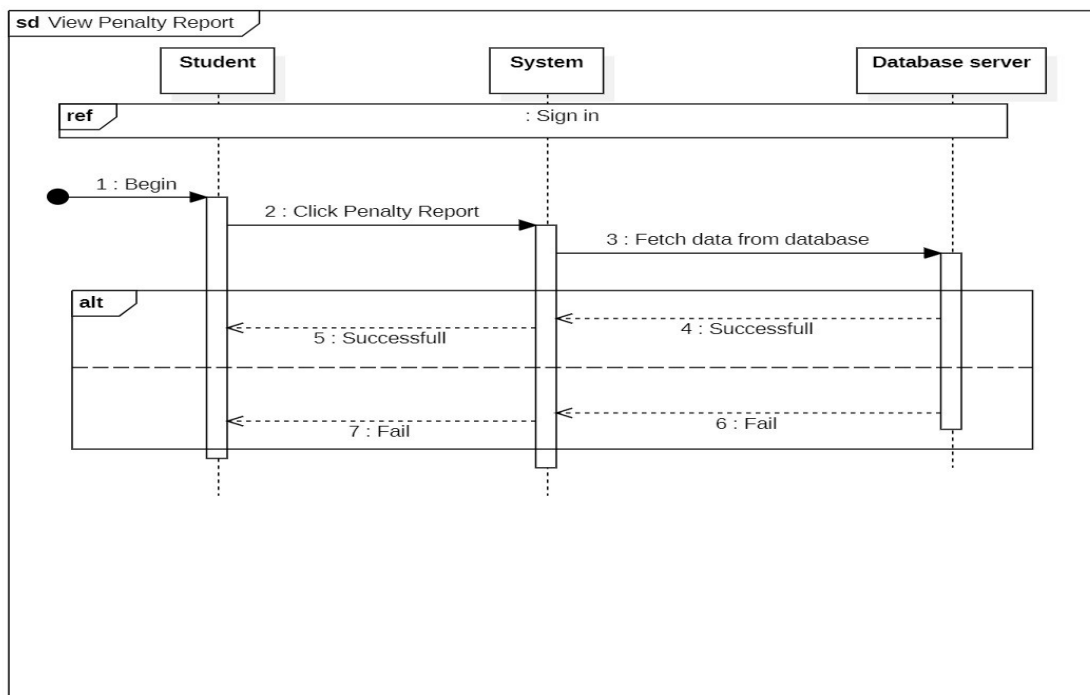
Student - View All Books:



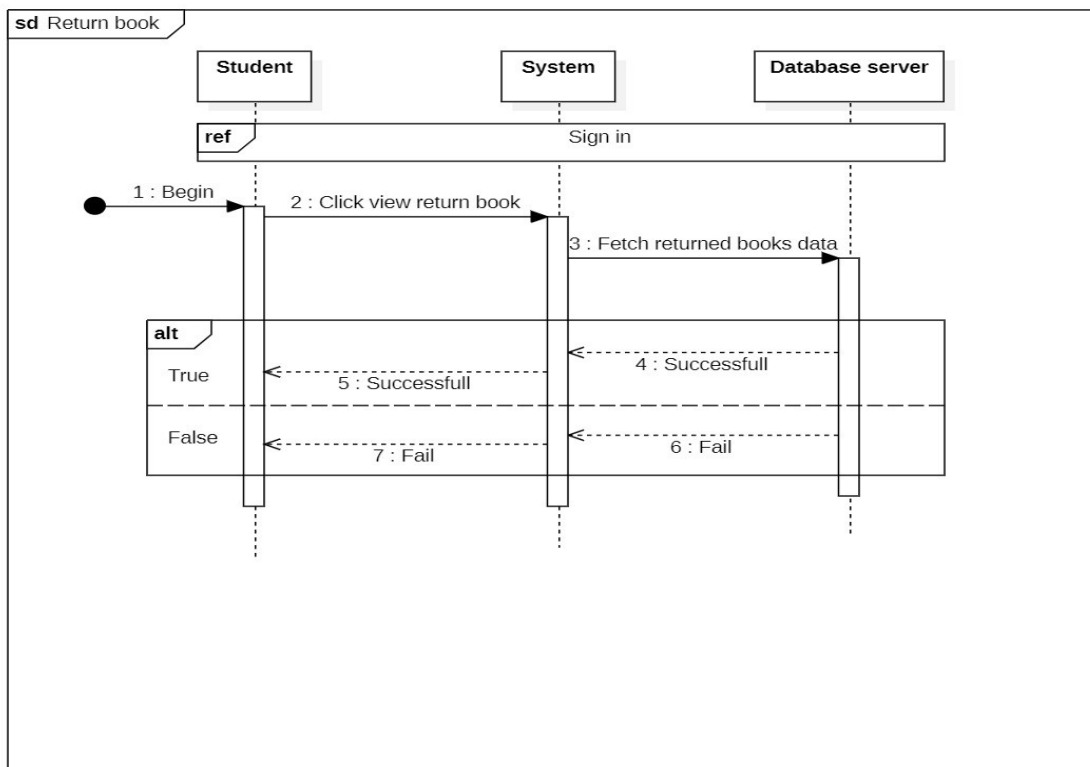
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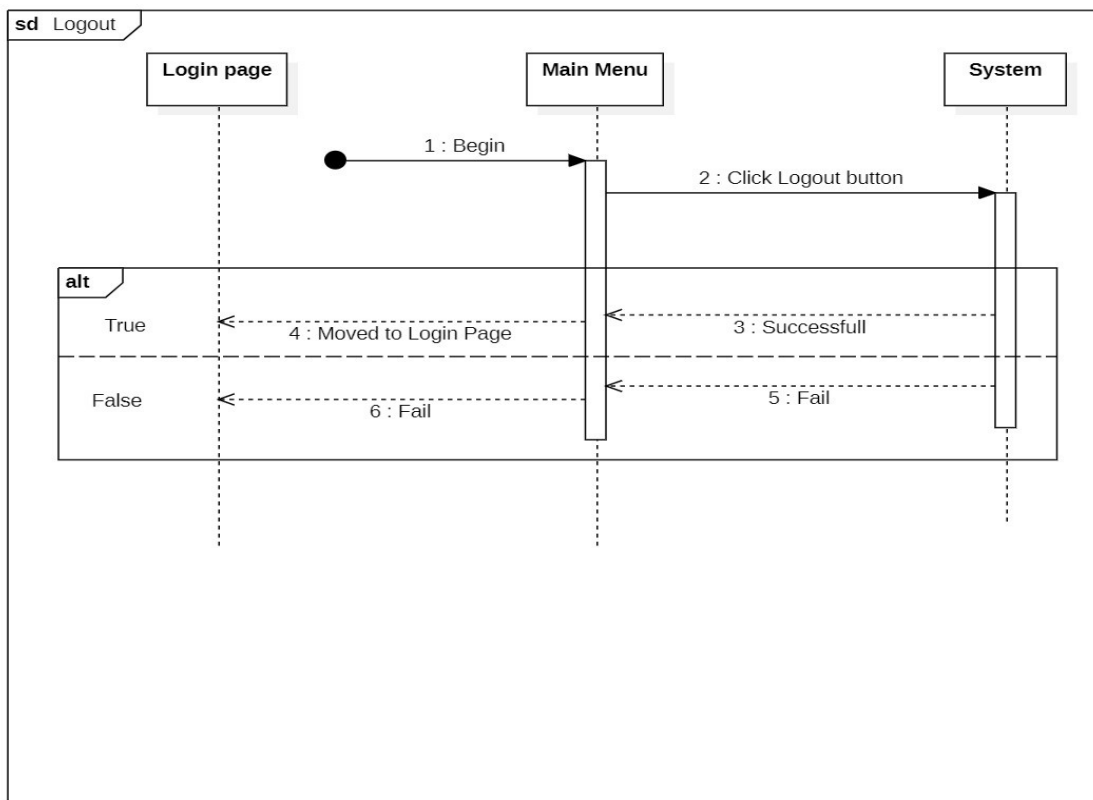
Student - View Penalty Reports



Student - view Return Books:

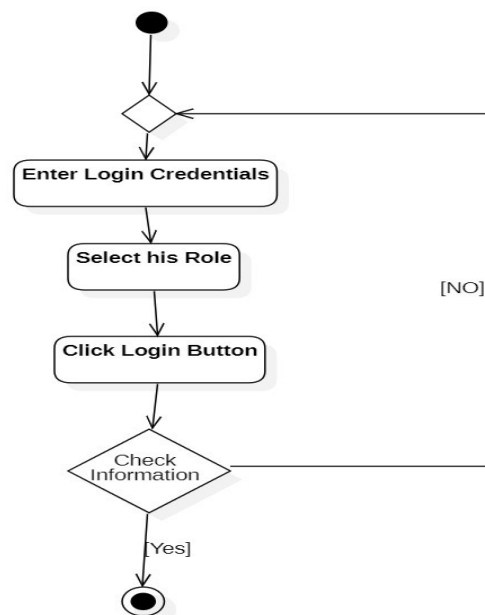


Logout:

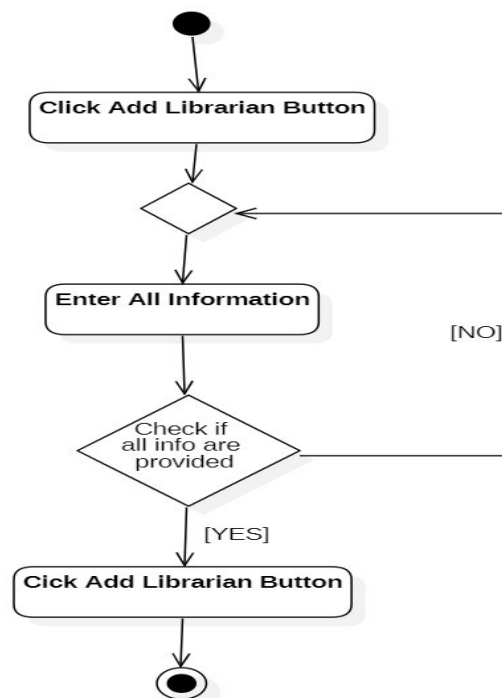


Activity Diagram:

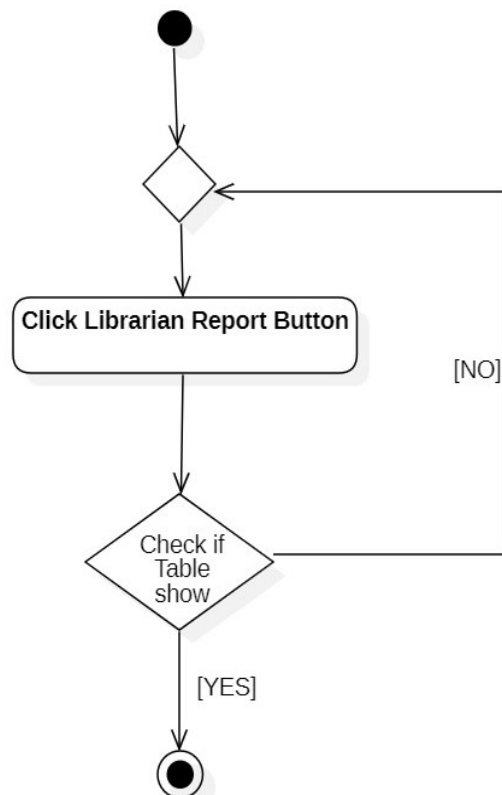
Login:



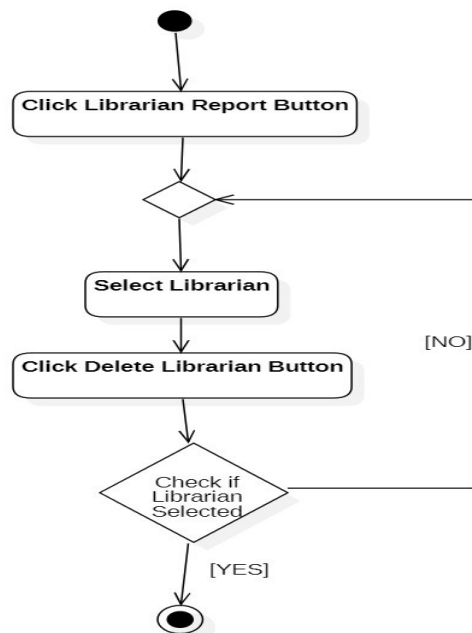
Add Librarian:



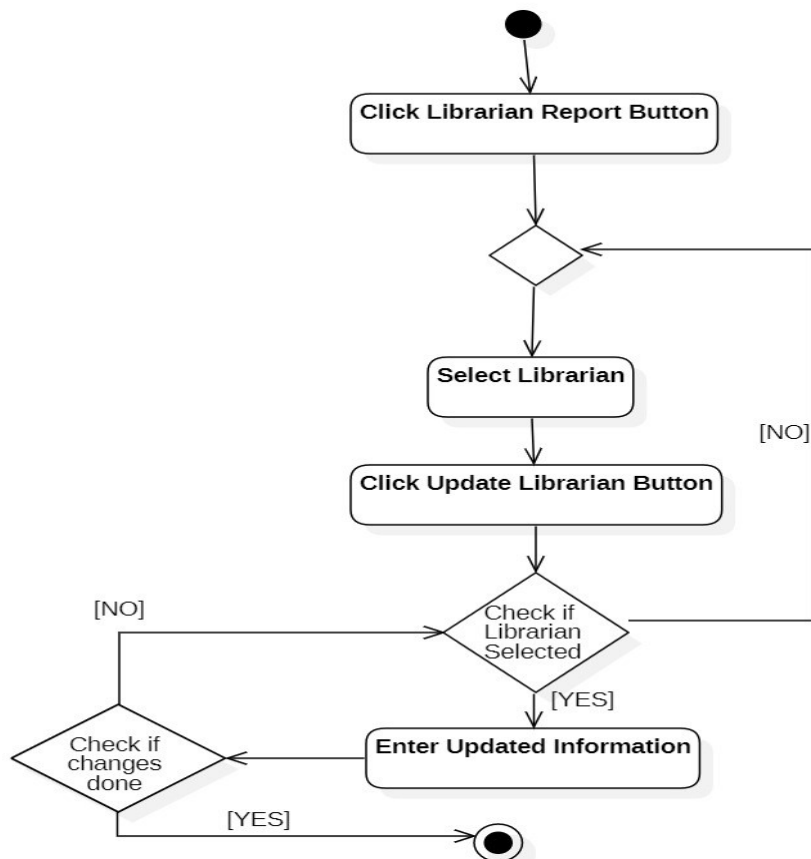
View Librarian Report:



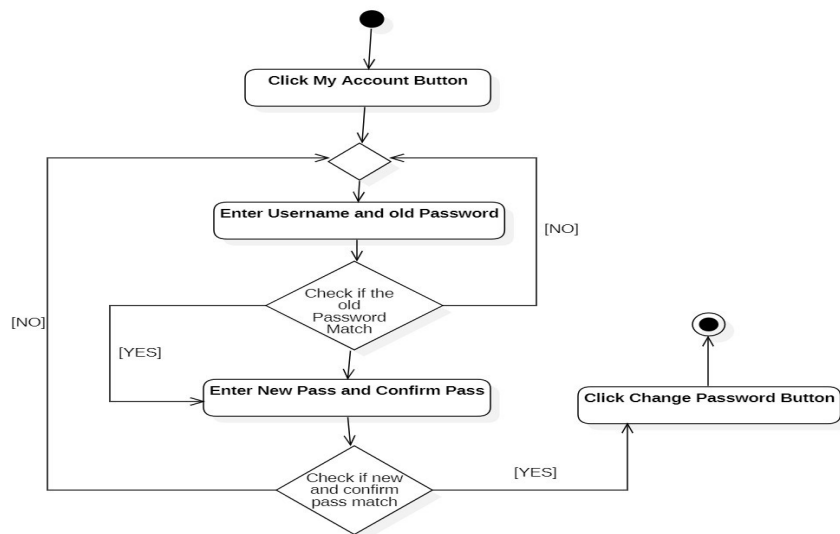
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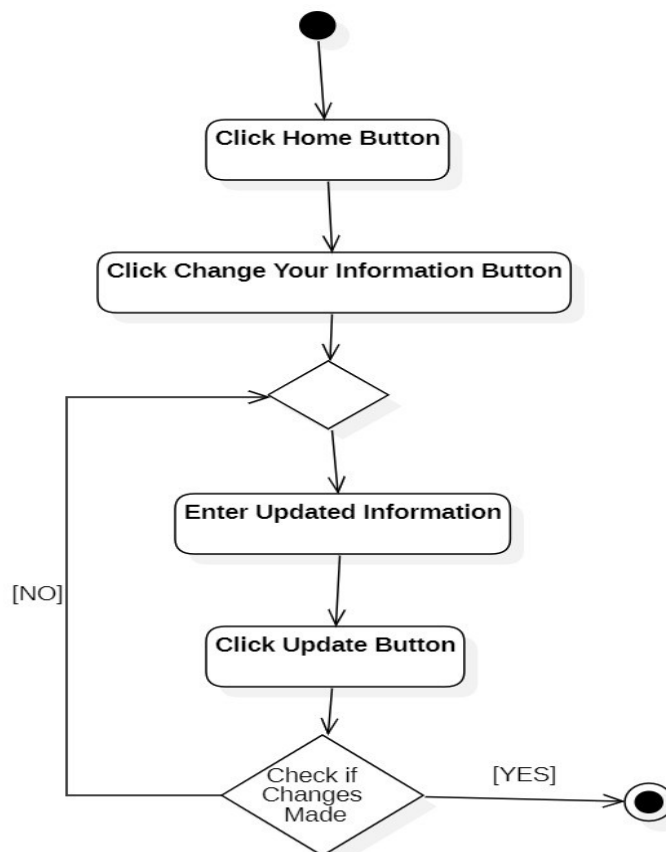
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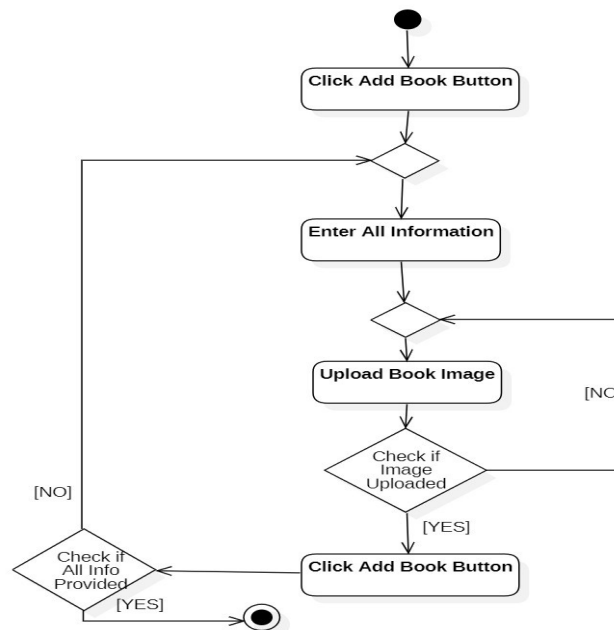
Admin - Update his Information:



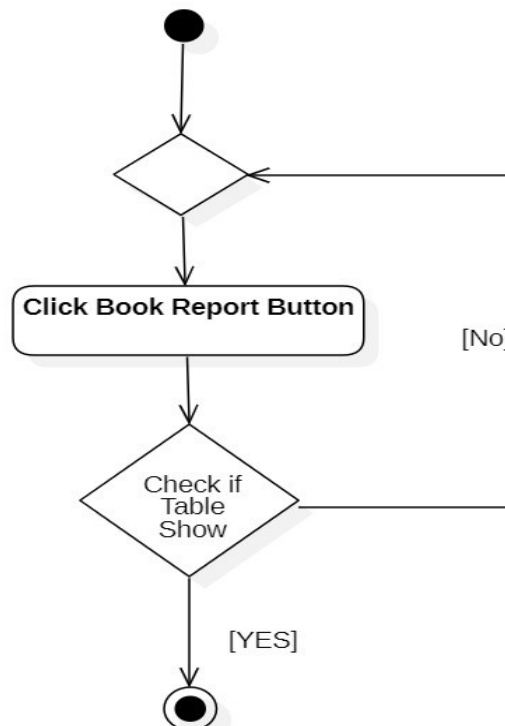
Librarian - update his information:



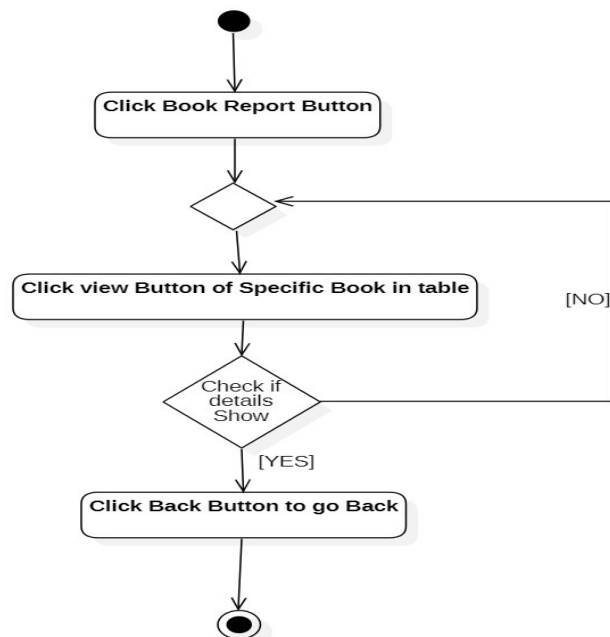
Add Books:



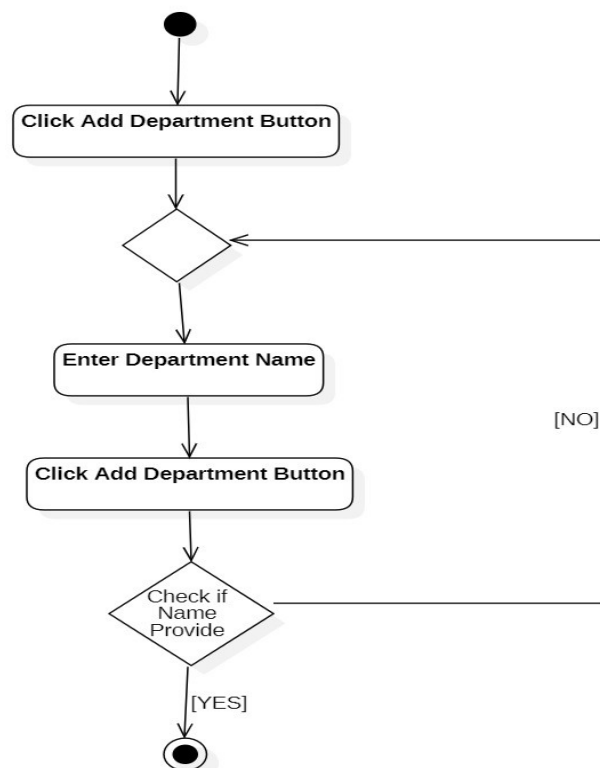
Librarian - view all books report:



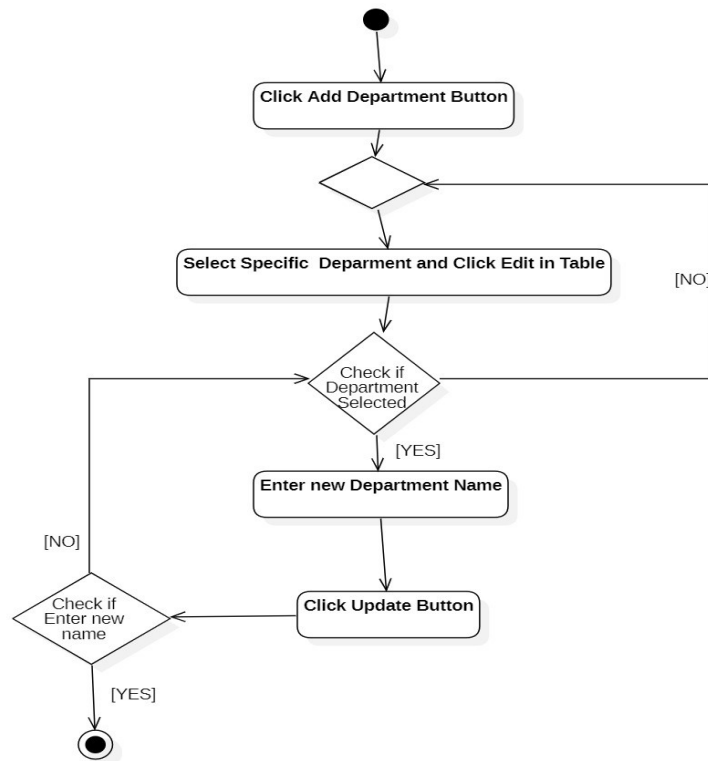
Librarian - Book Details:



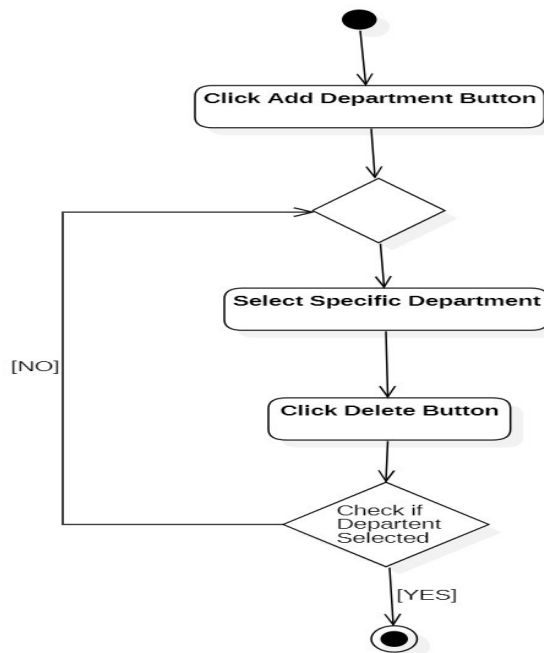
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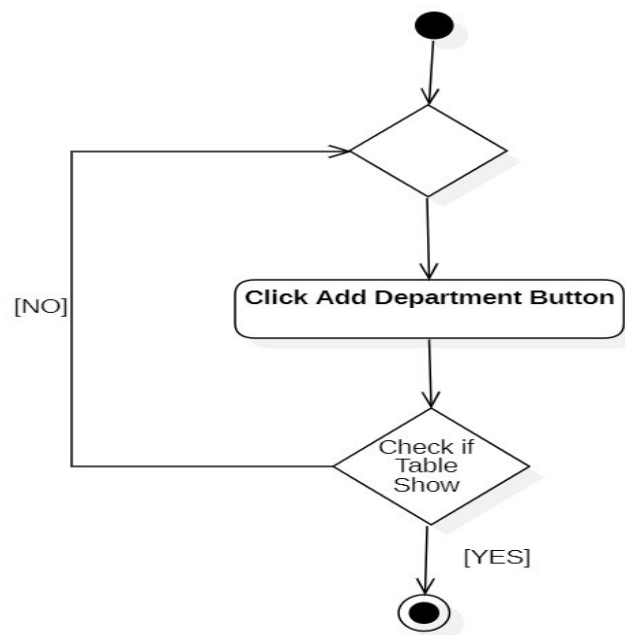
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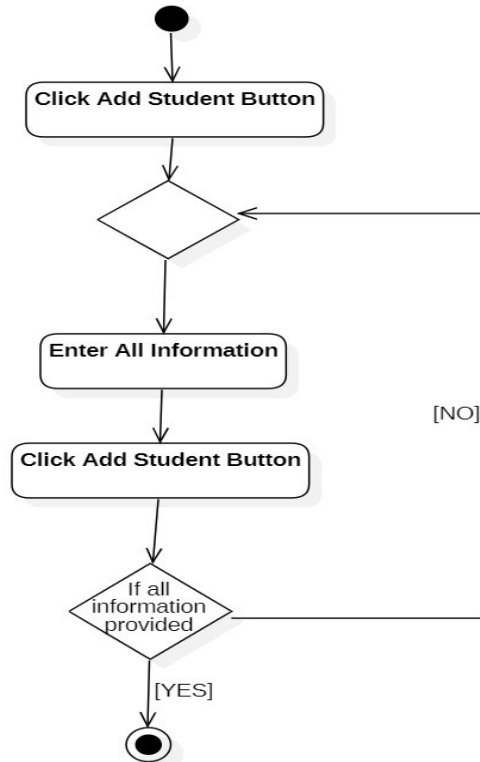
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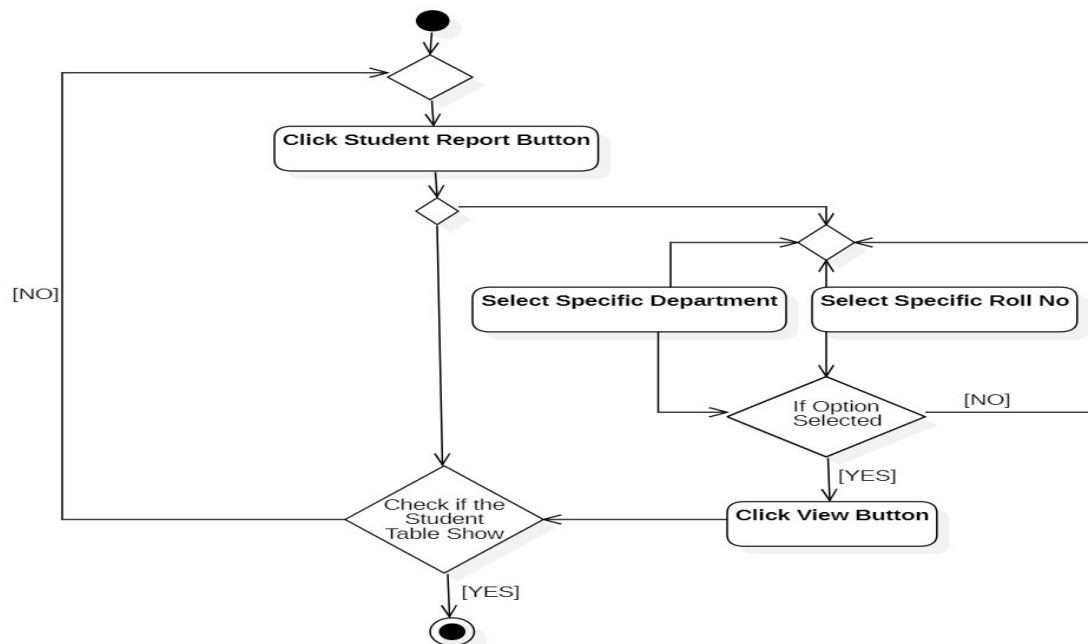
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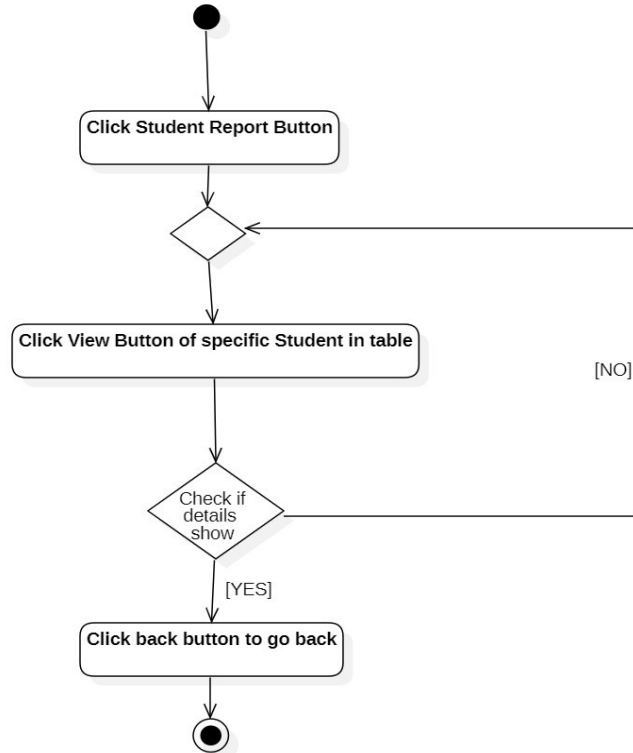
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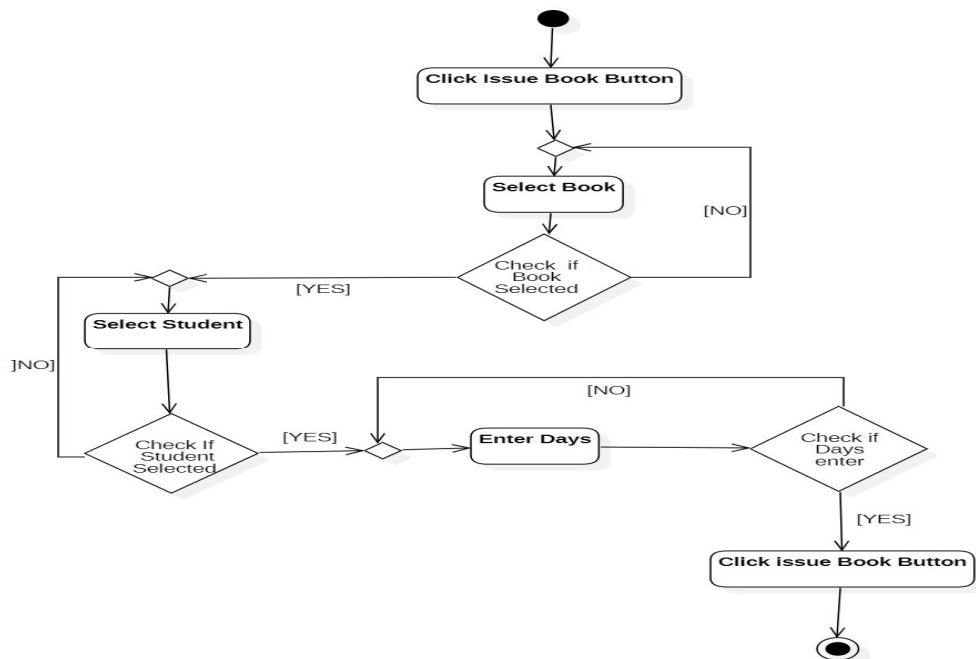
View students report:



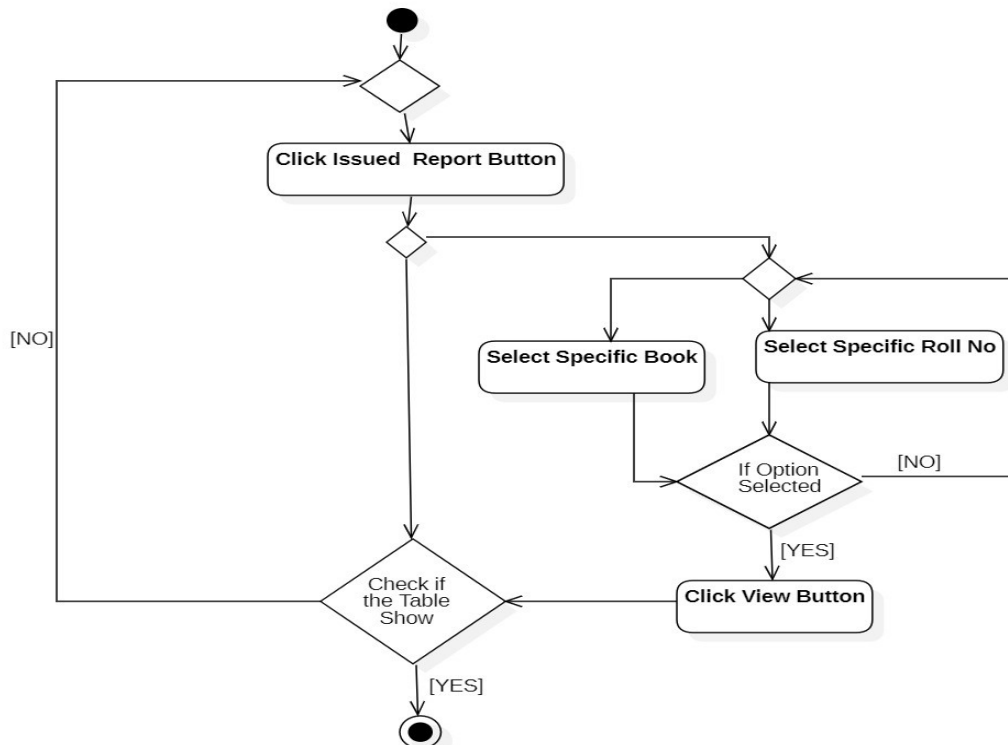
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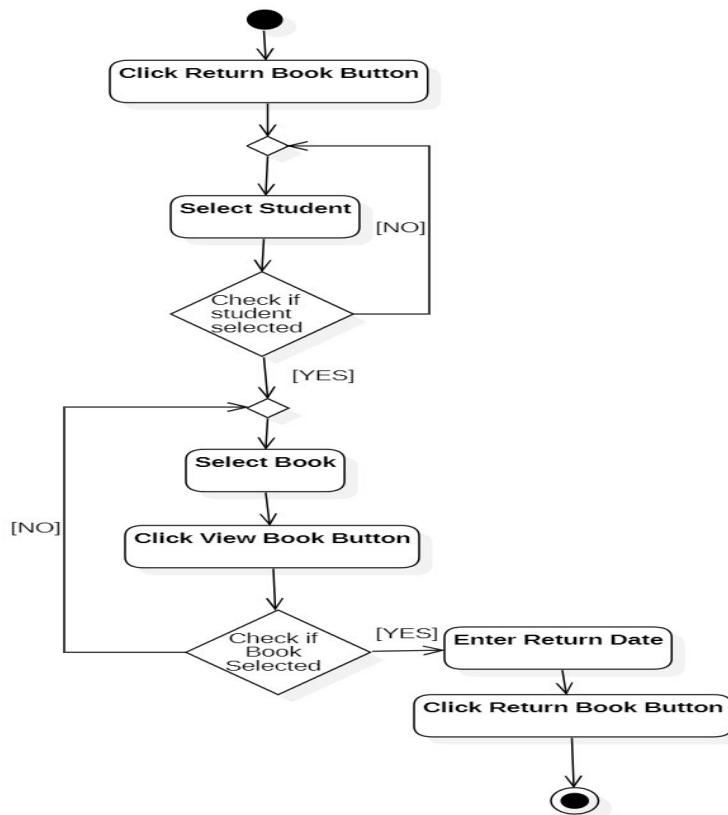
Issue Book:



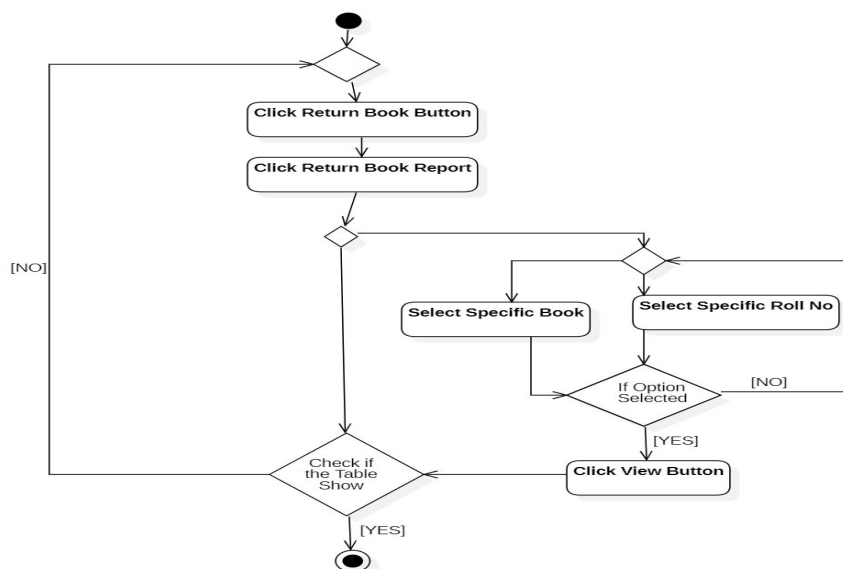
Librarian - Issued Books Report:



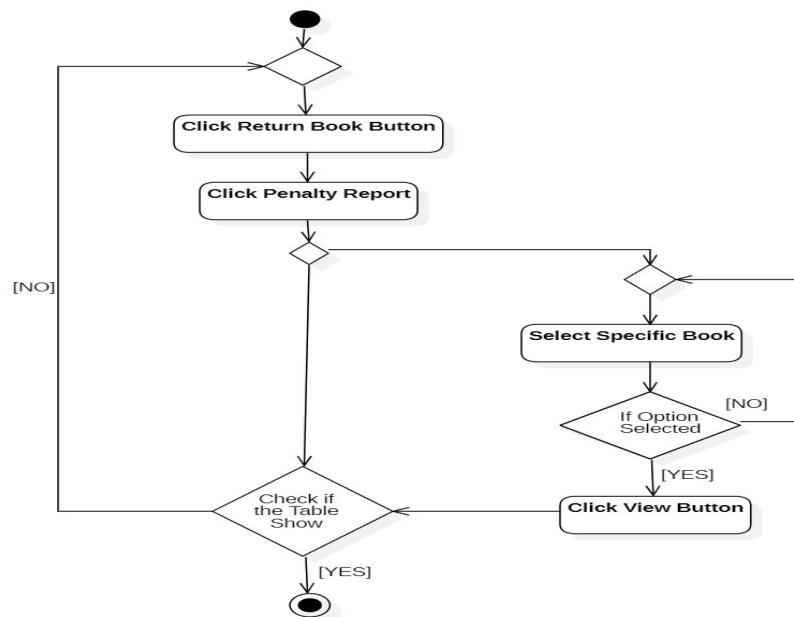
Return Borrow Book:



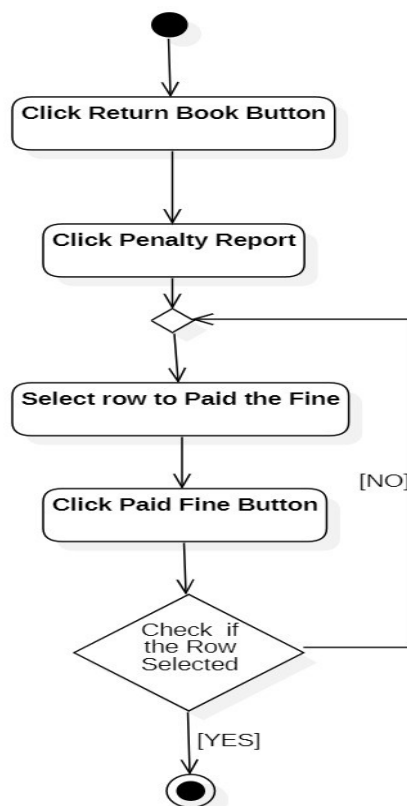
Librarian - Returned Books Report:



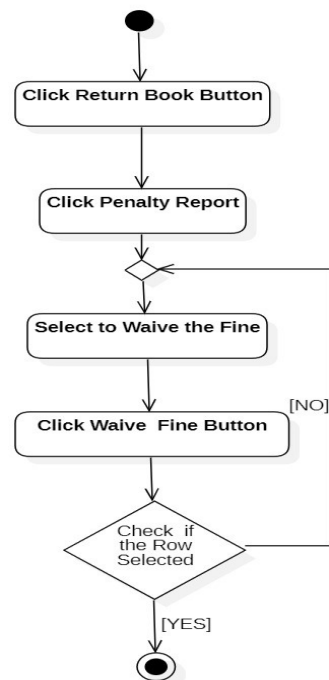
Librarian view Penalty Report:



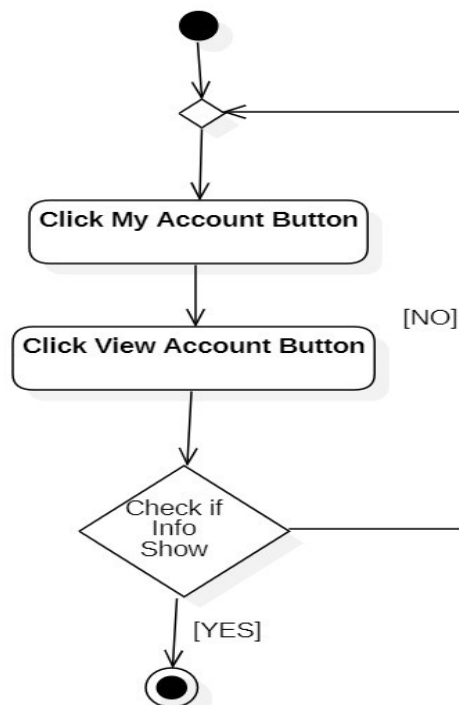
Paid the Penalty Fine:



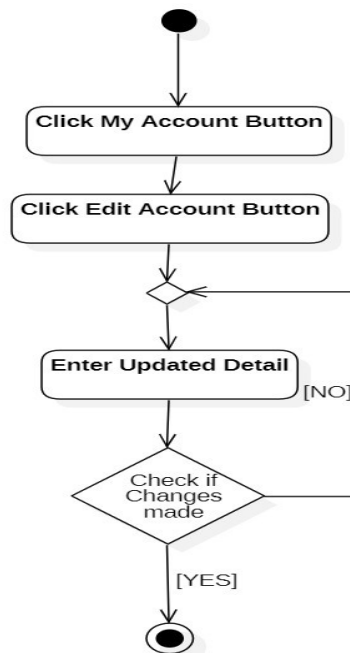
Waive the Penalty Fine:



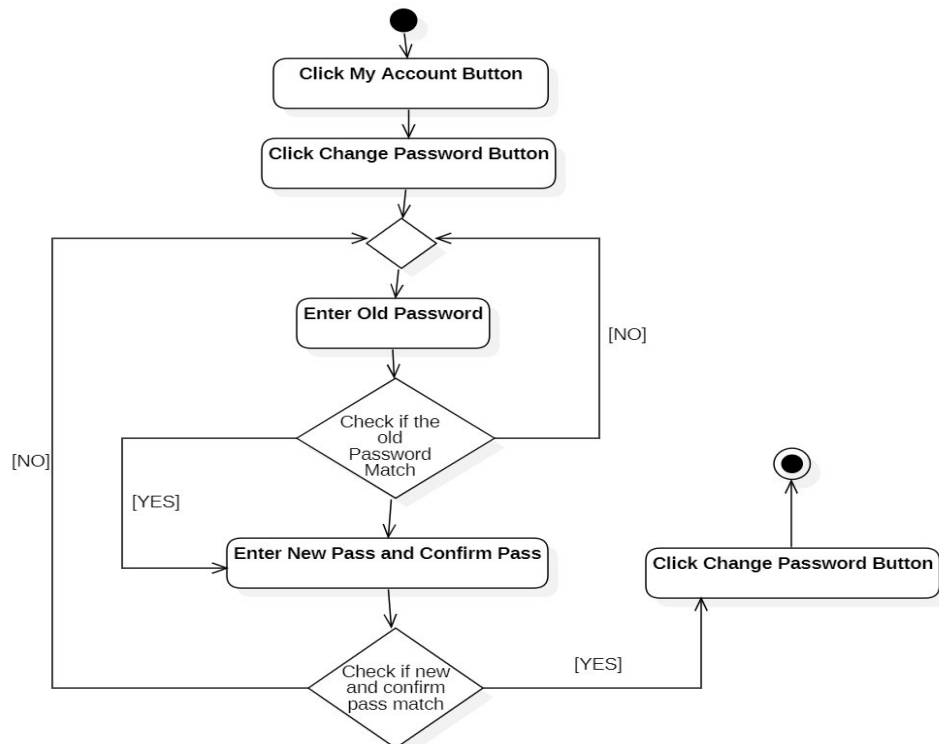
Student - view his Account:



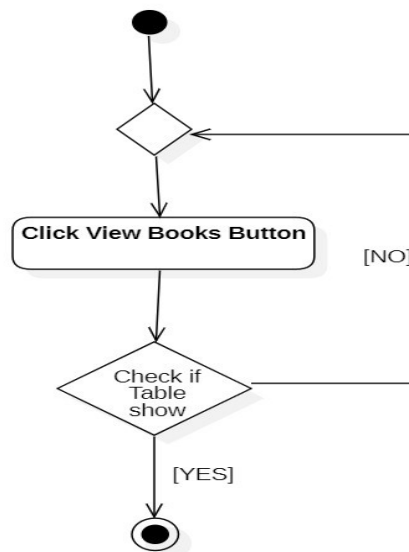
Student - Edit his Account Detail:



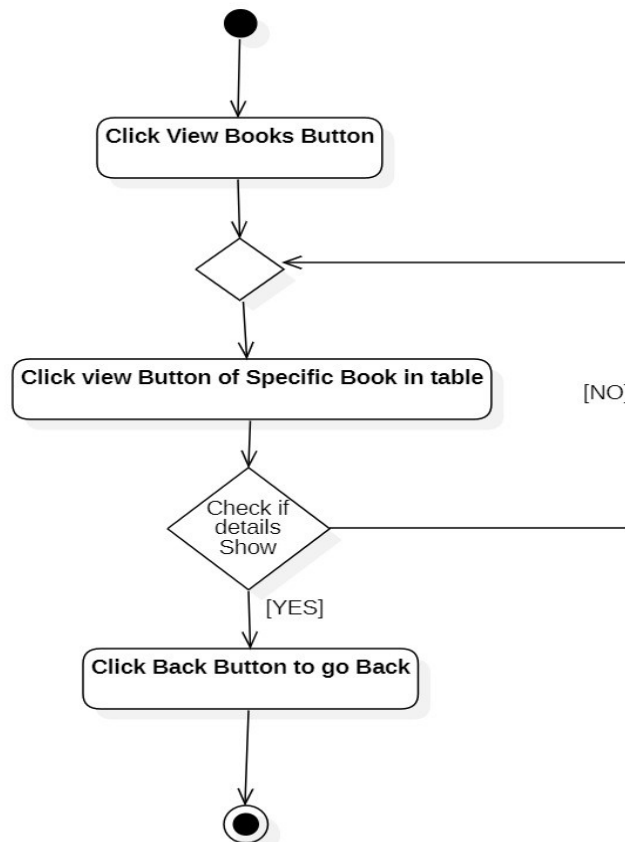
Student - Change his Password:



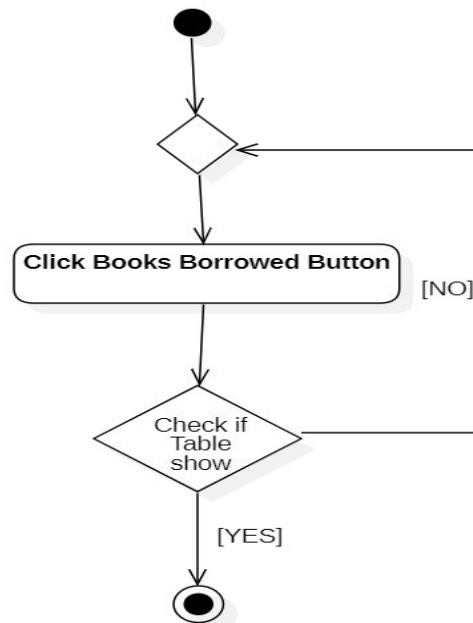
Student - View All Books:



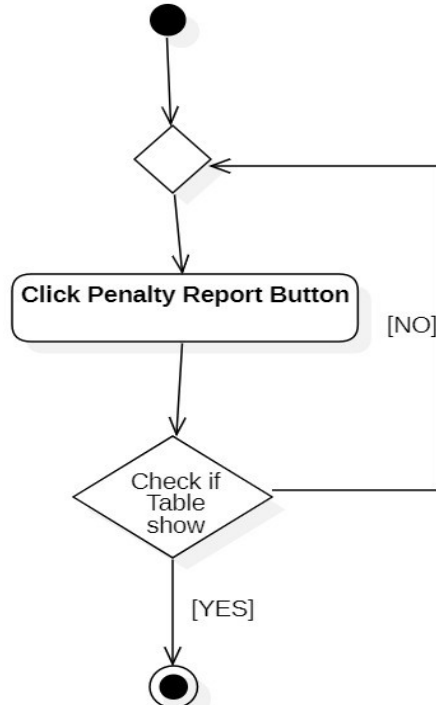
Student - view detail of specific Book:



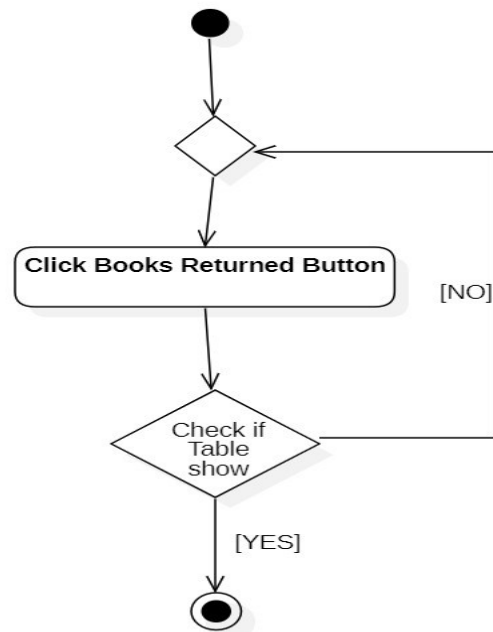
Student - view Borrow Books:



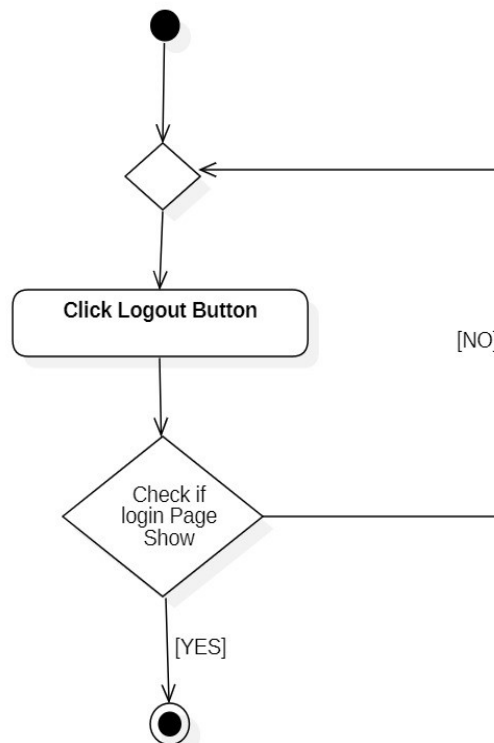
Student - View Penalty Reports:



Student - view Return Books:

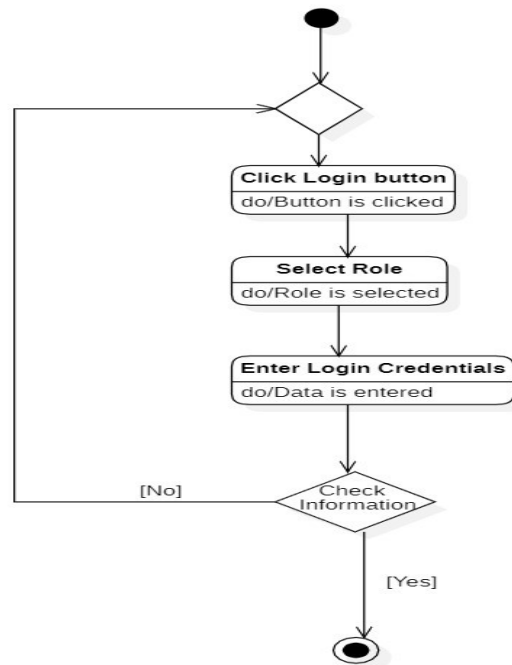


Logout:

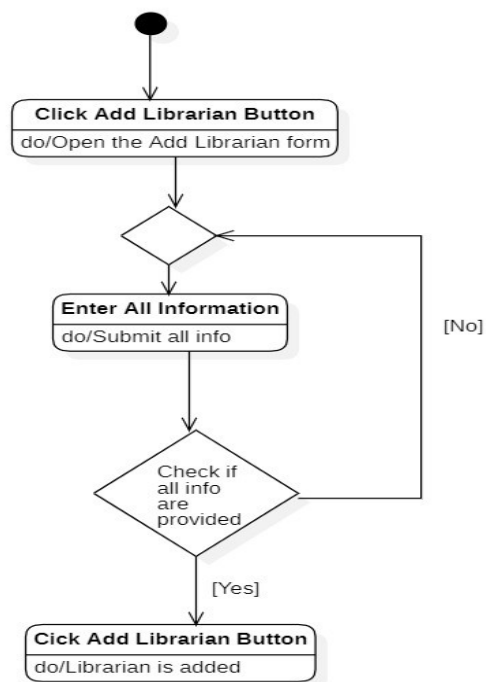


State Machine Diagram:

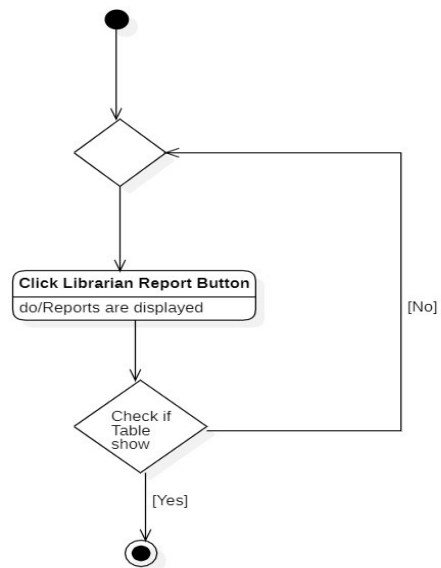
Login:



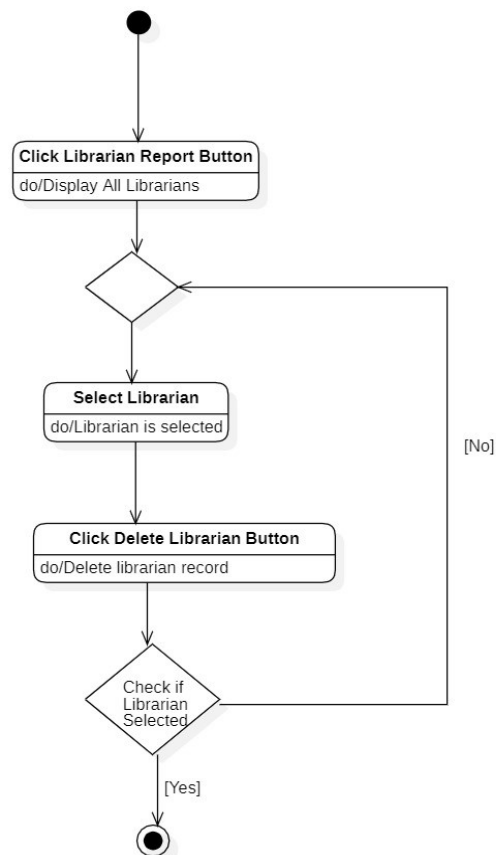
Add Librarian:



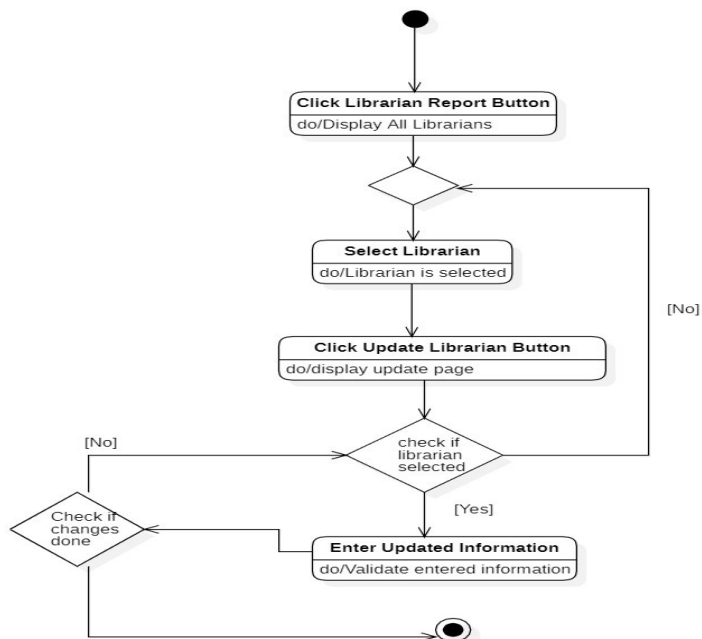
View Librarian Report:



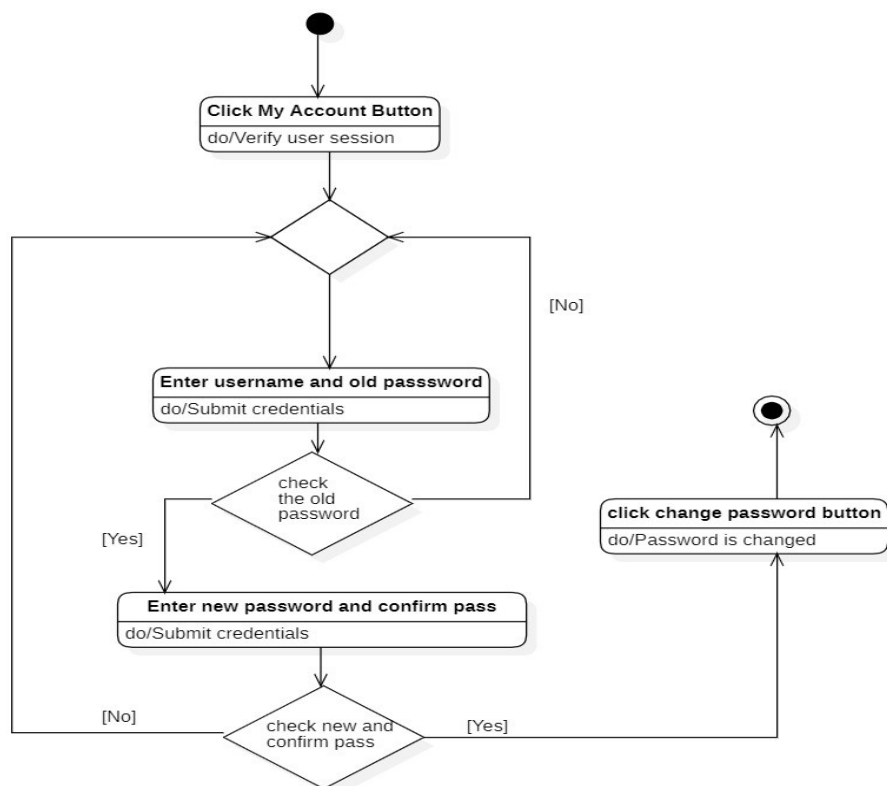
Remove Librarian:



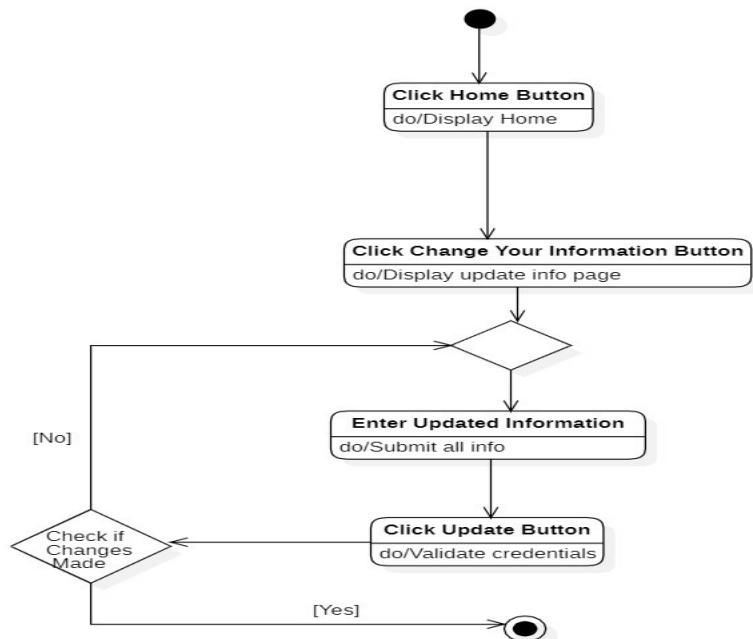
Update Librarian Information:



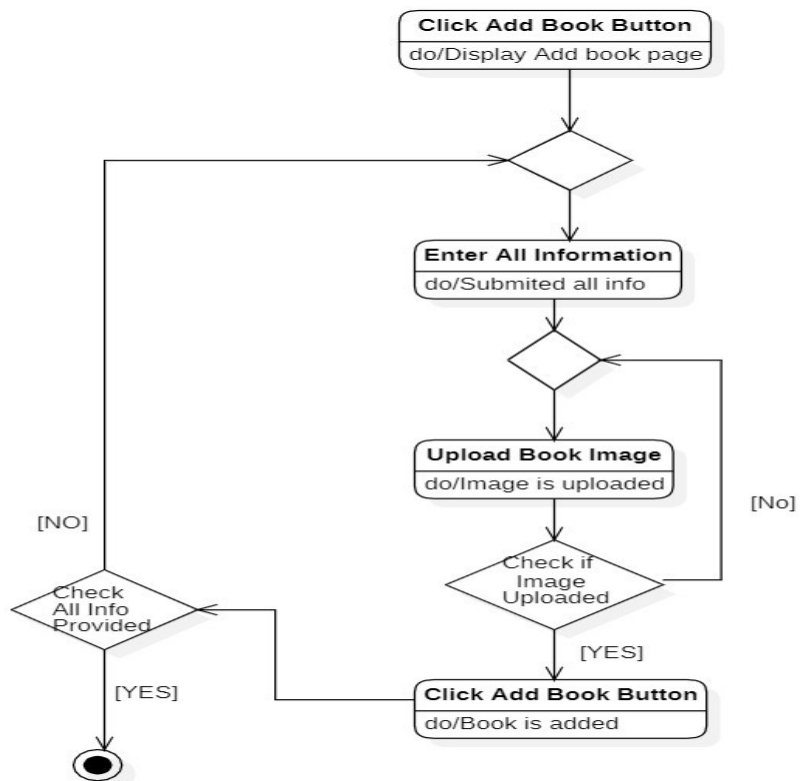
Admin - Update his Information:



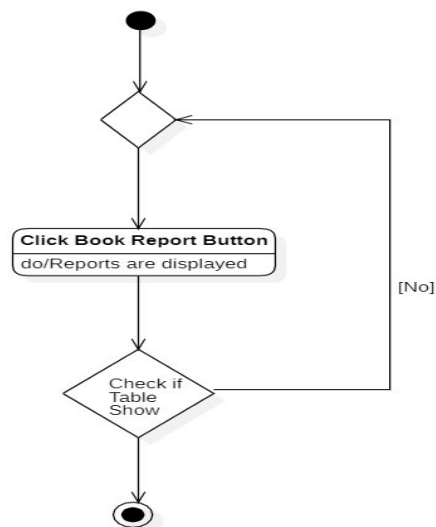
Librarian - update his information:



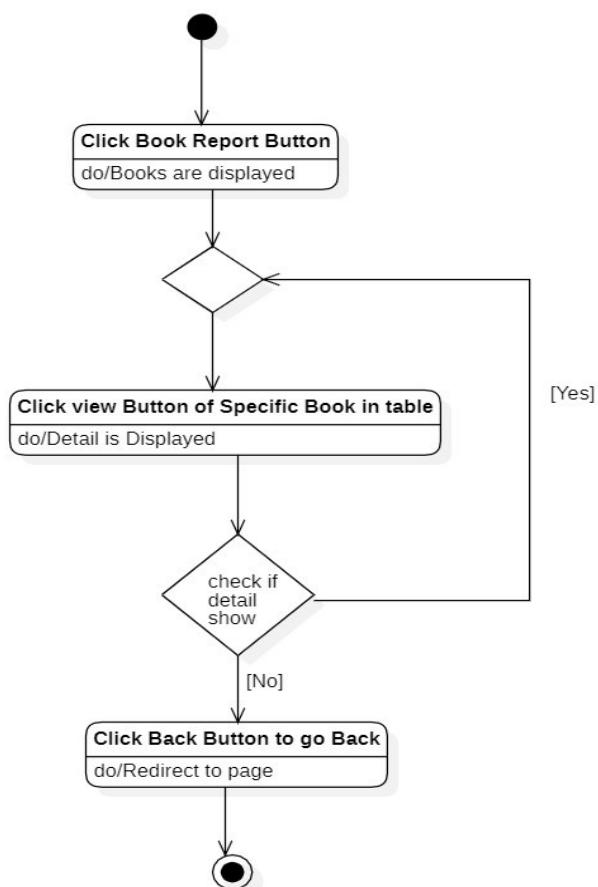
Add Books:



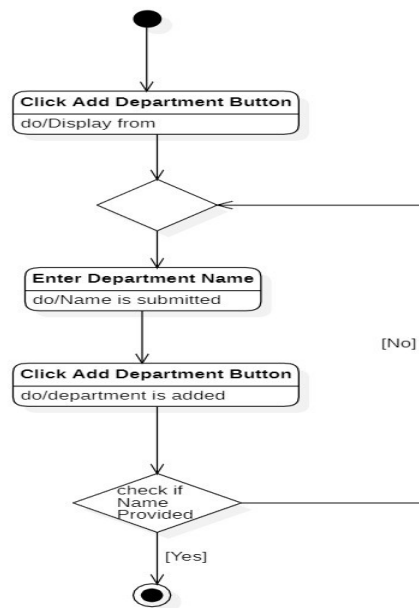
Librarian - view all books report:



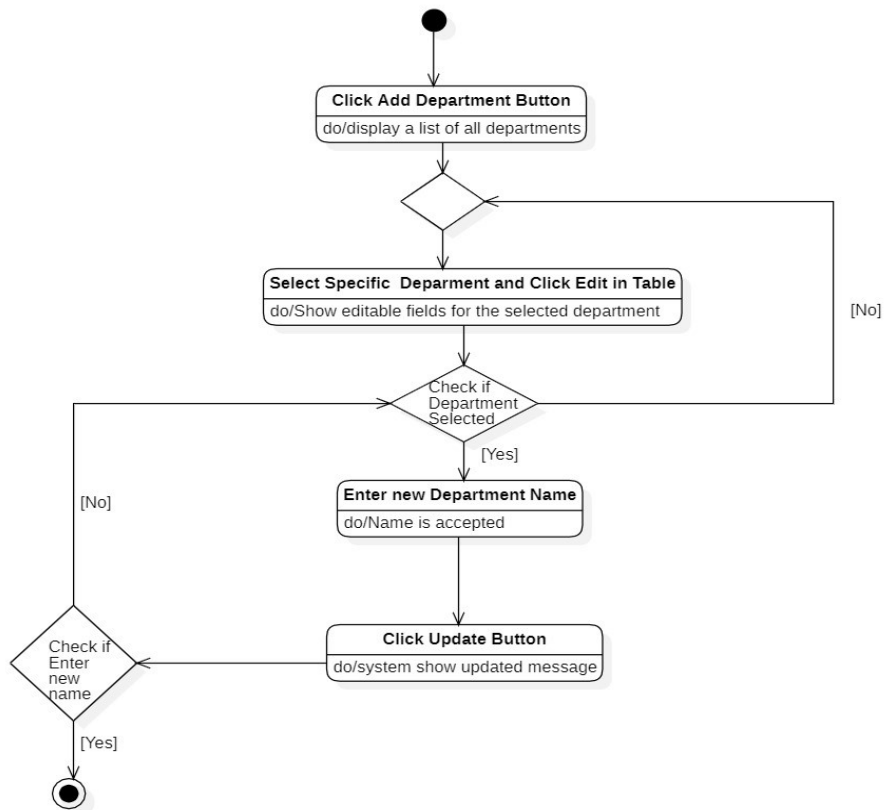
Librarian - Book Details:



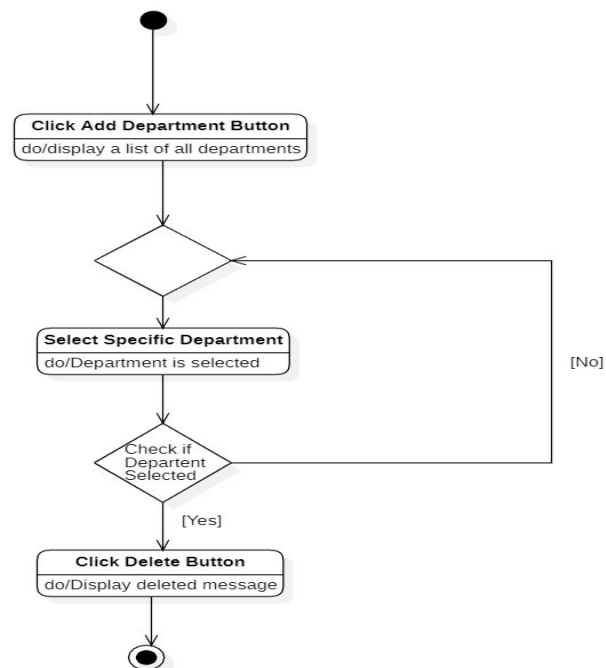
Add department:



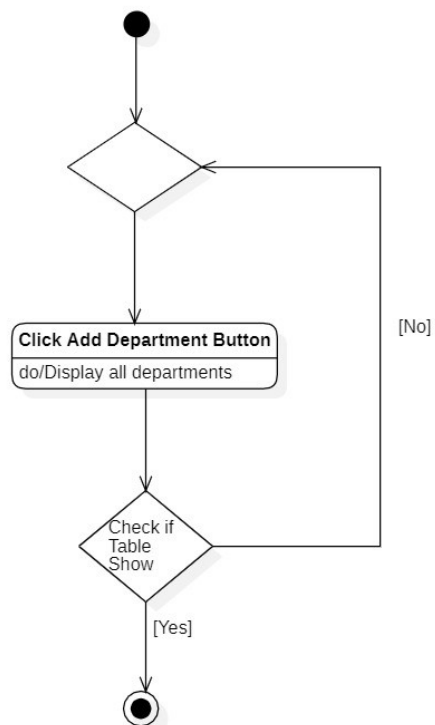
Edit department:



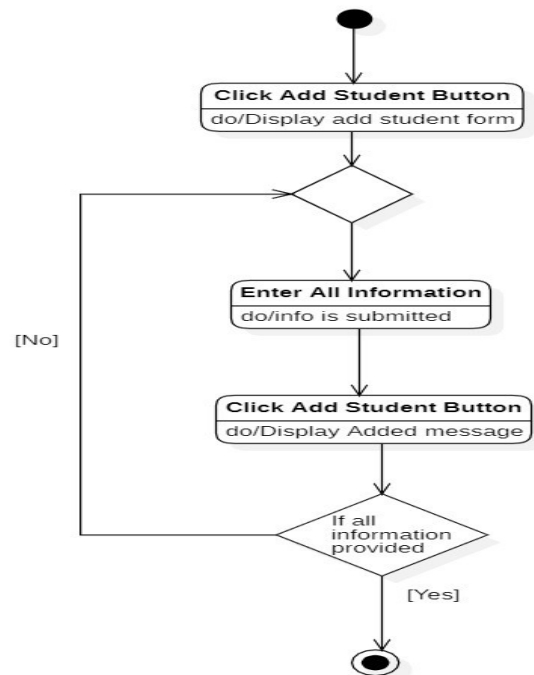
Delete department:



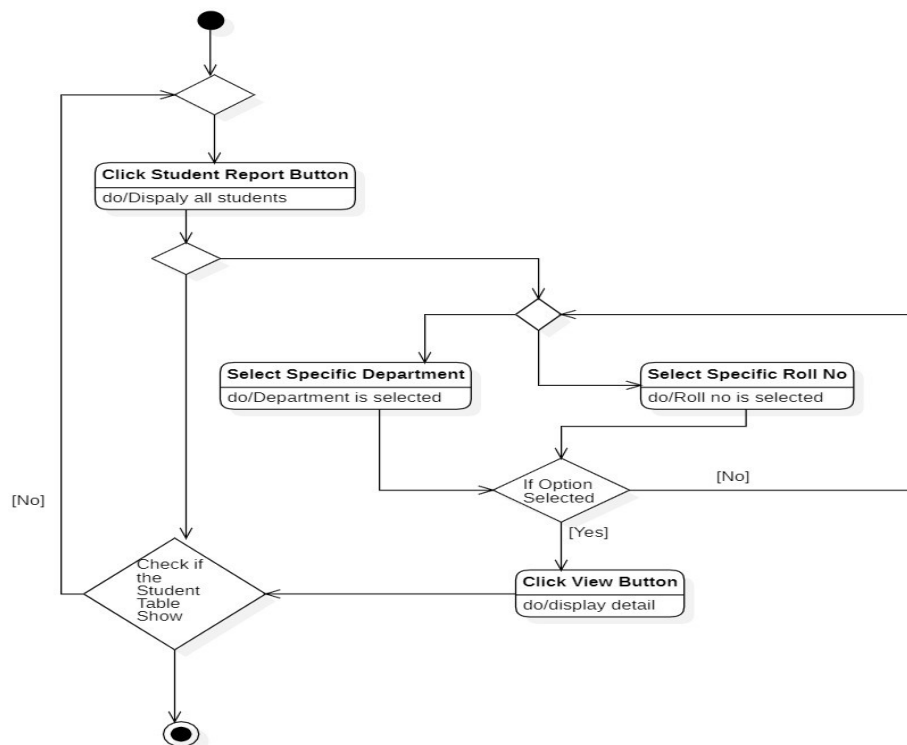
All departments:



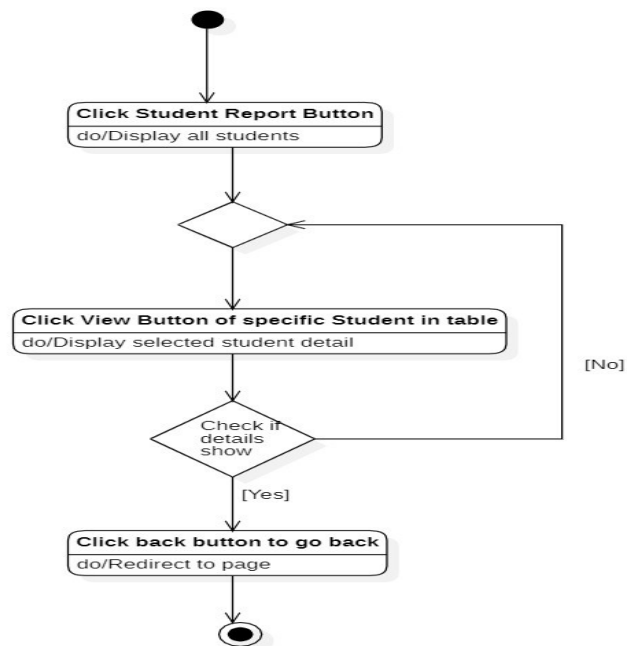
Add students:



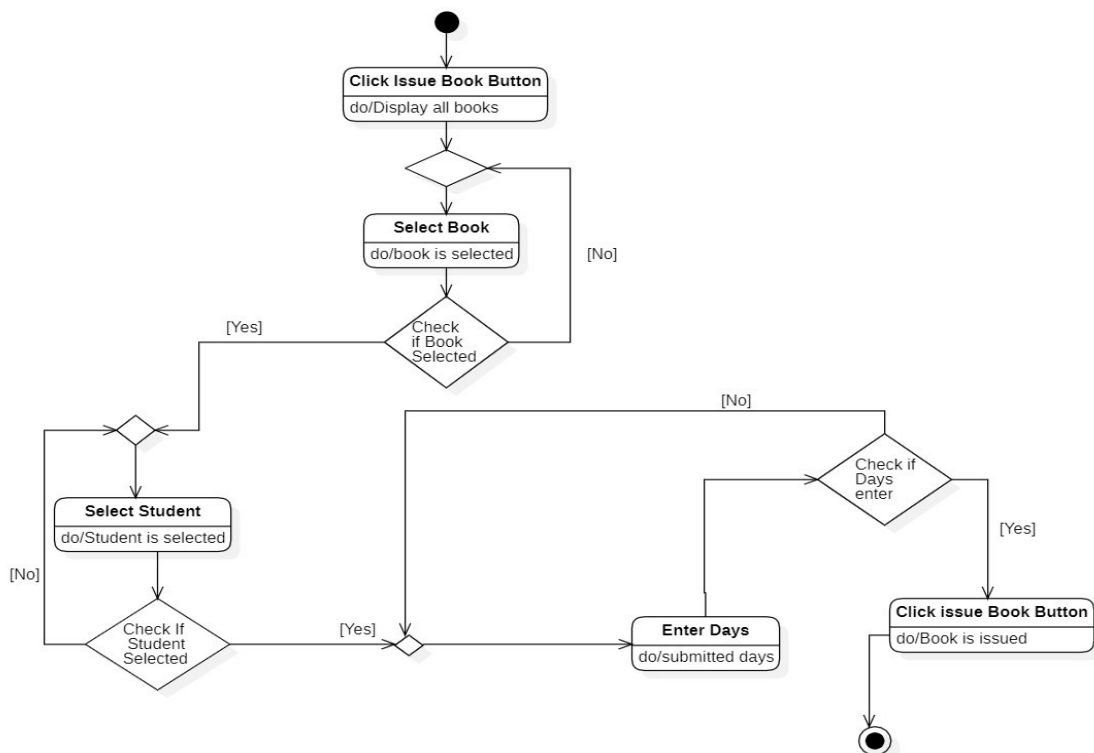
View students report:



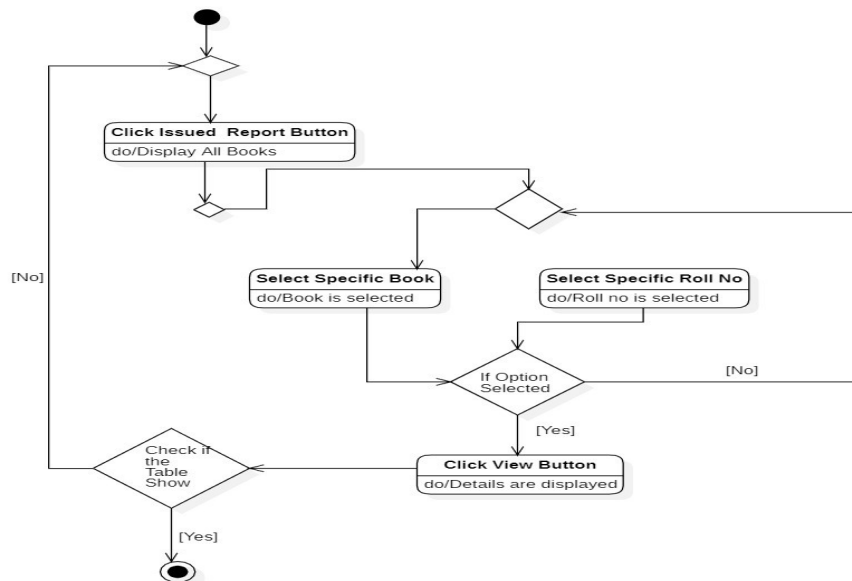
Student Detail:



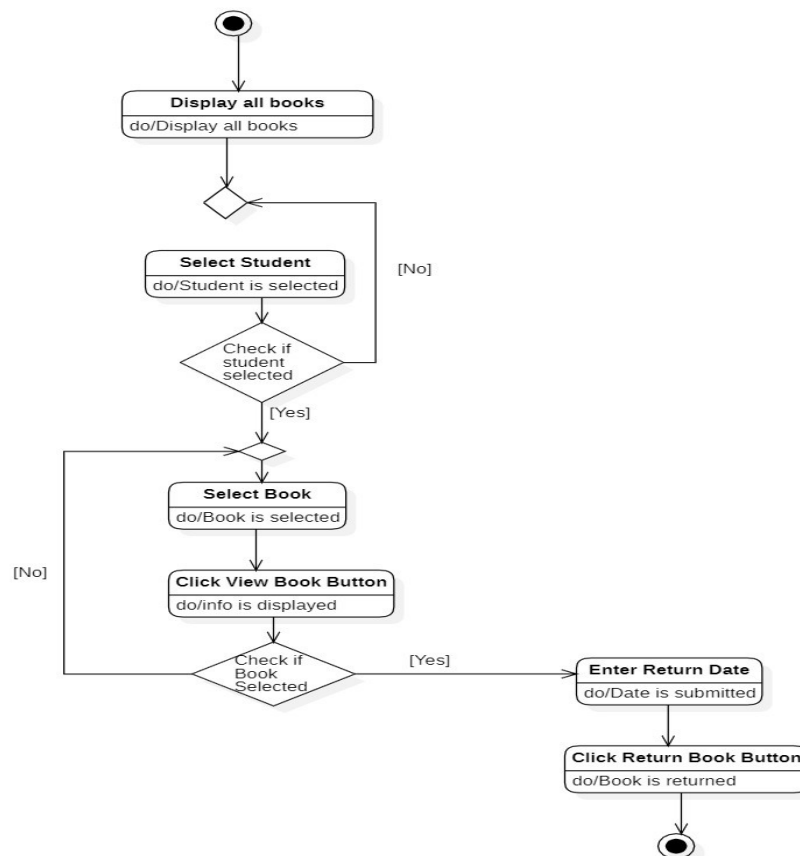
Issue Book:



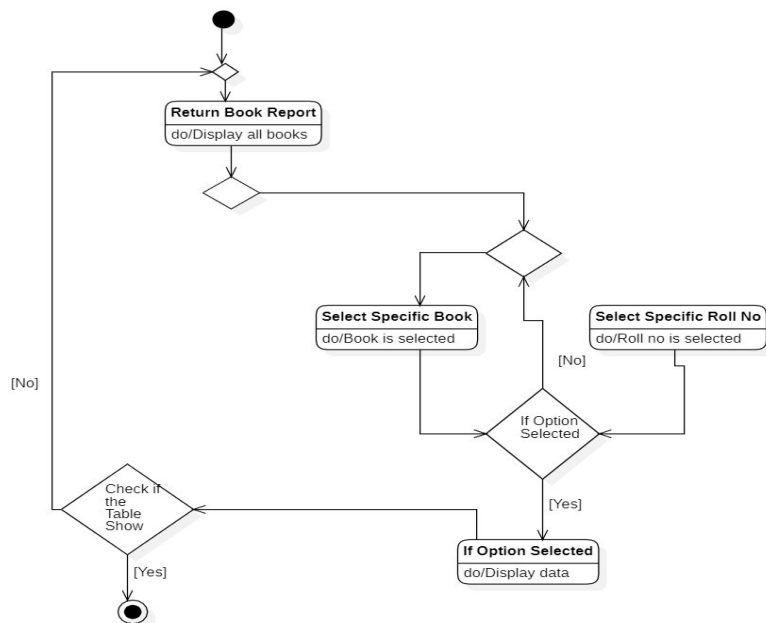
Librarian - Issued Books Report:



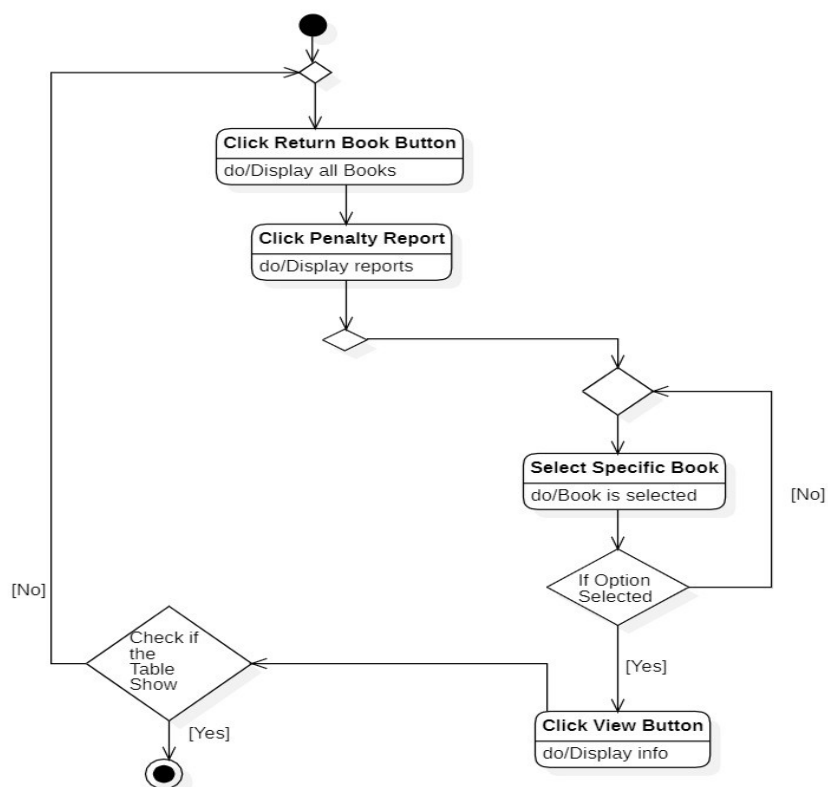
Return Borrow Book:



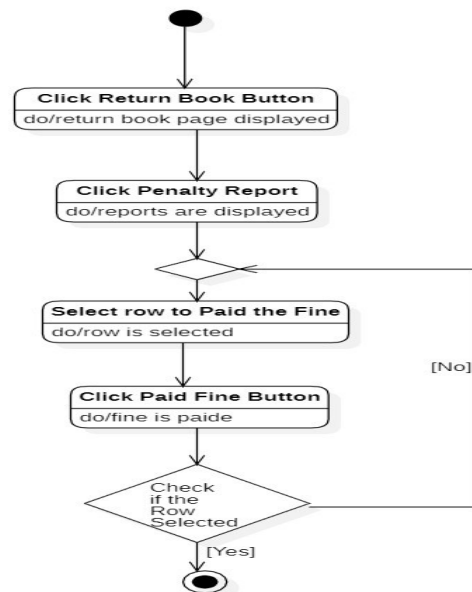
Librarian - Returned Books Report:



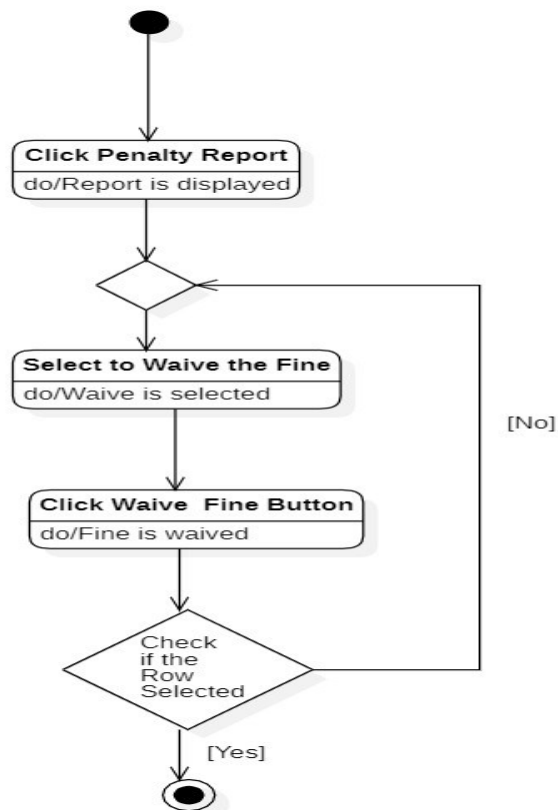
Librarian view Penalty Report:



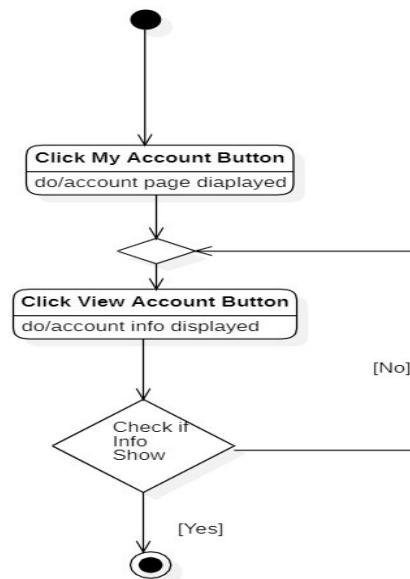
Paid the Penalty Fine:



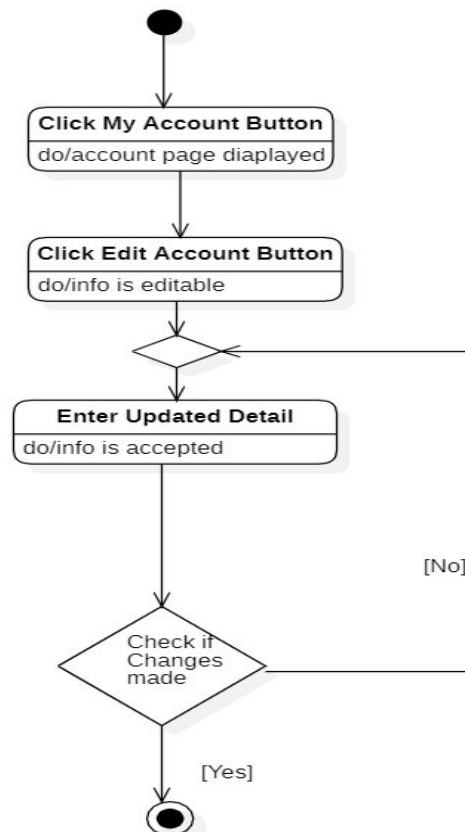
Waive the Penalty Fine:



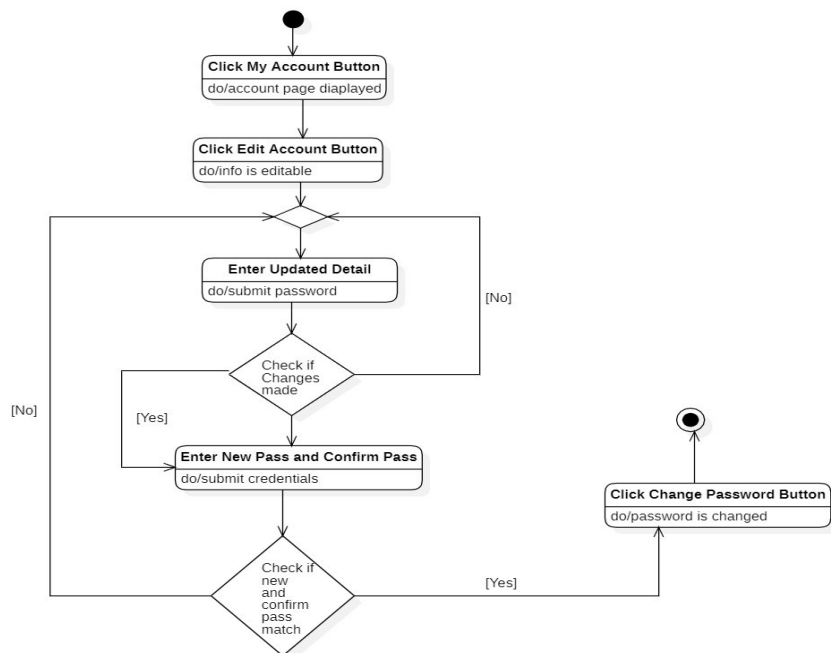
Student - view his Account:



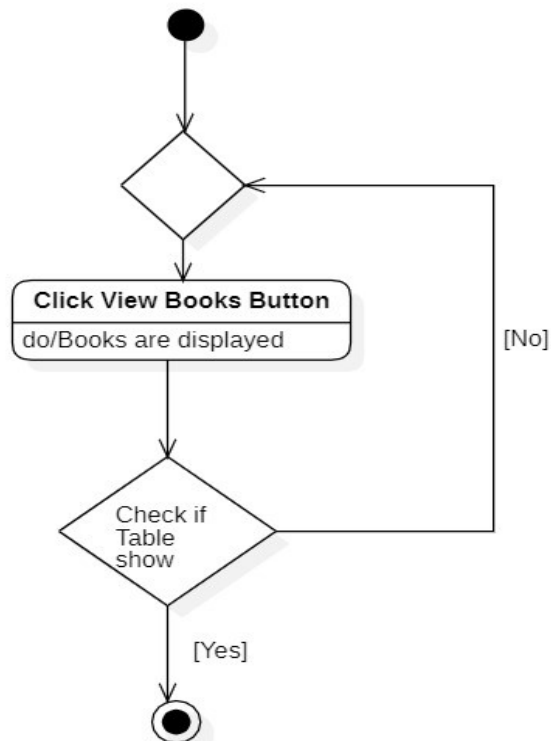
Student - Edit his Account Detail:



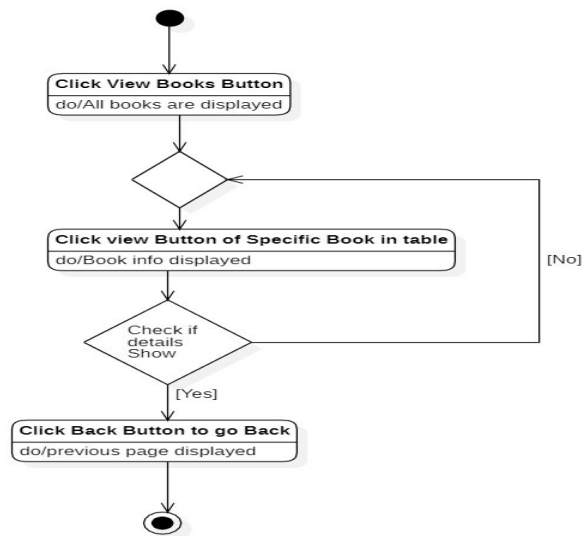
Student - Change his Password:



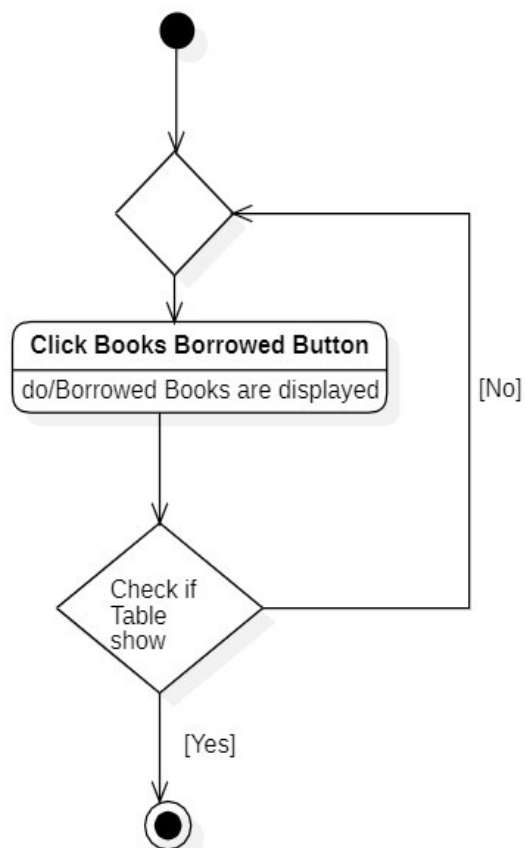
Student - View All Books:



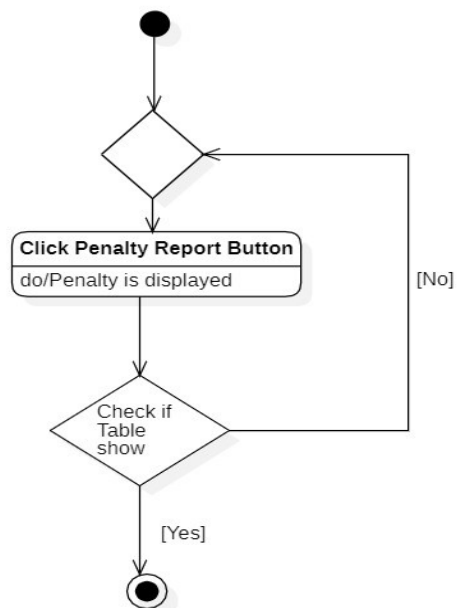
Student - view detail of specific Book:



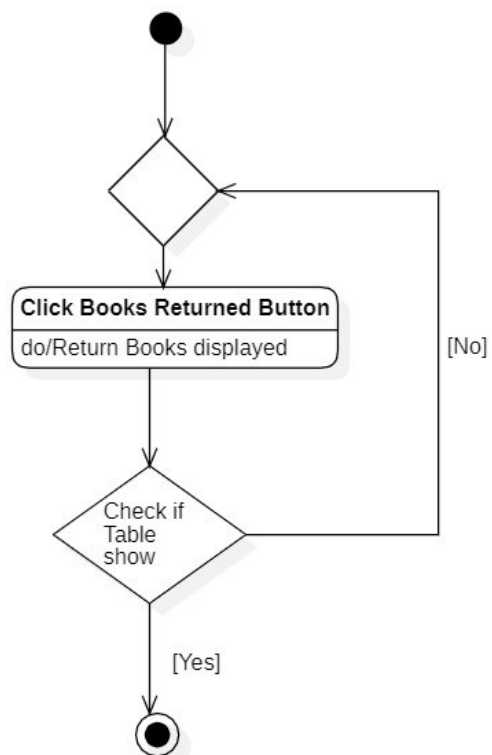
Student - view Borrow Books:



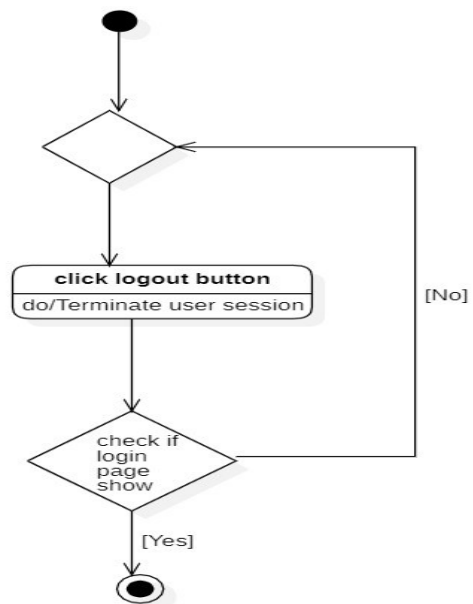
Student - View Penalty Reports:



Student - view Return Books:

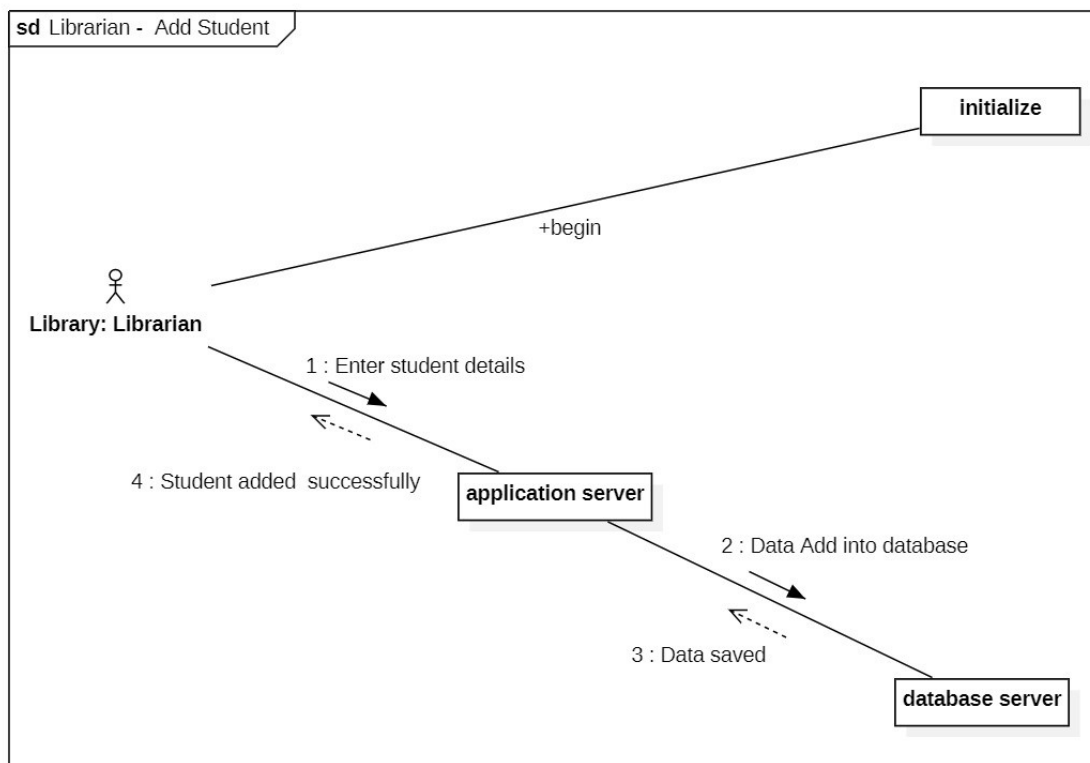


Logout:

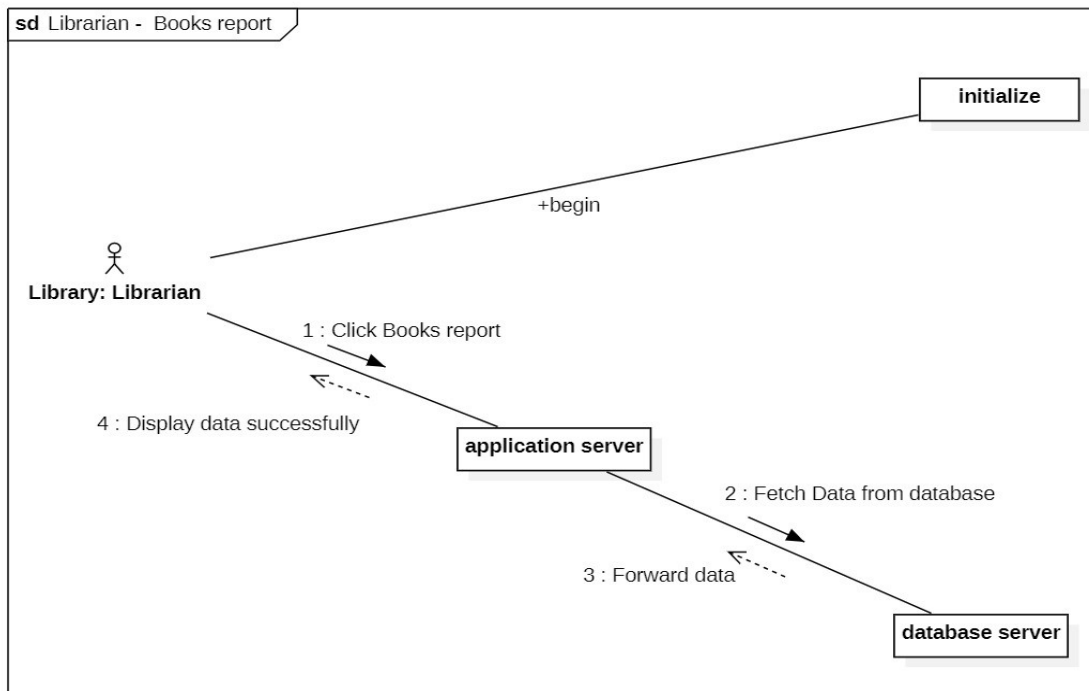


Collaborative Diagram:

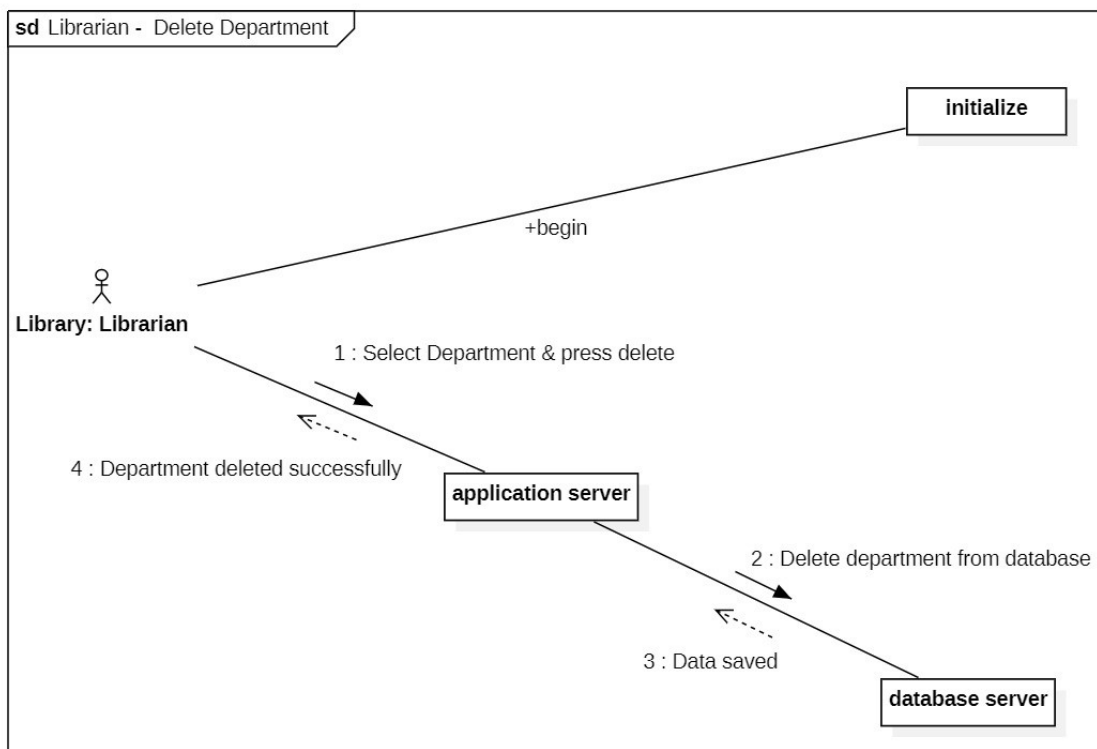
Add students:



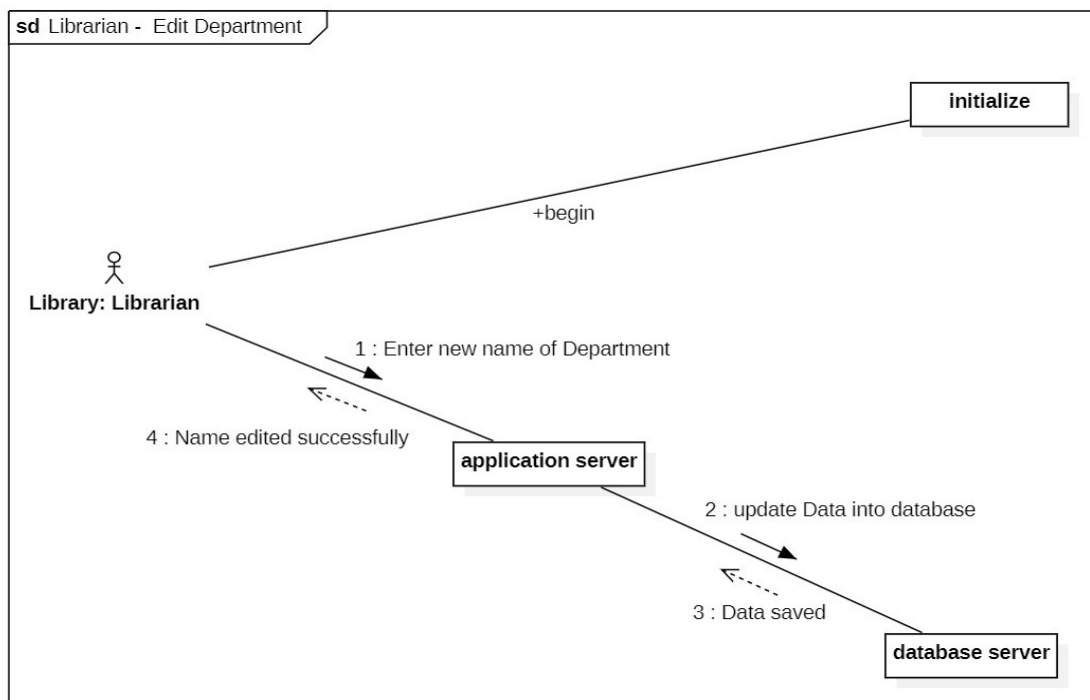
Librarian - view all books report:



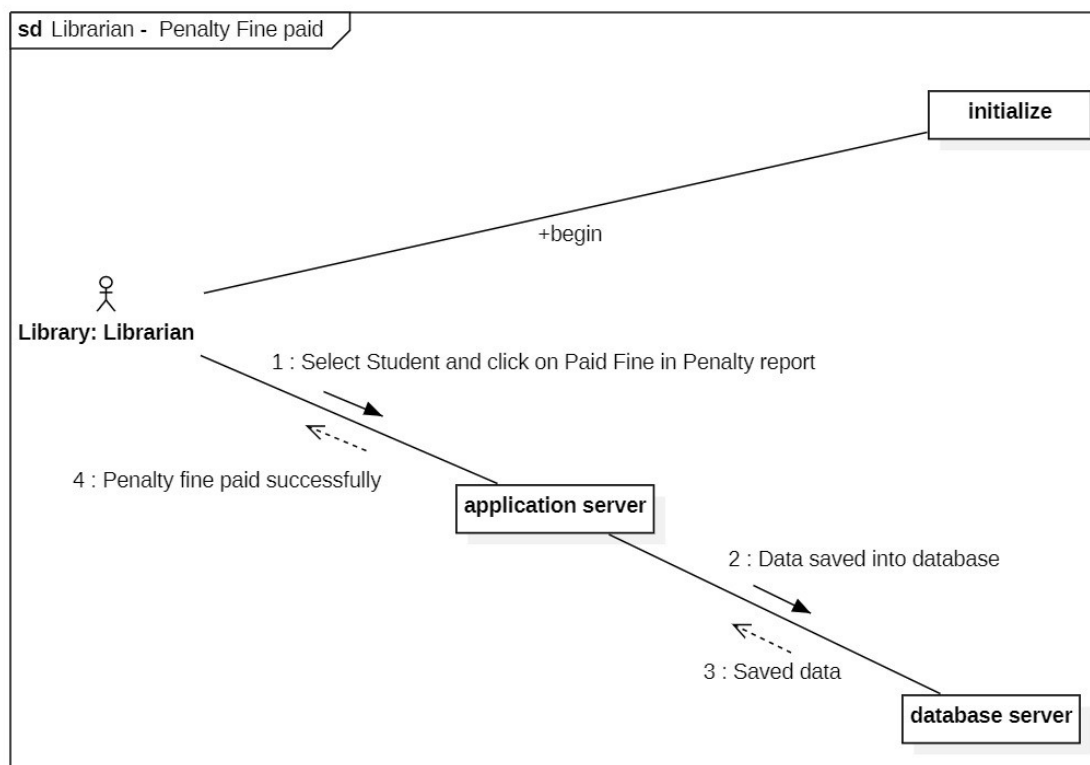
Delete department:



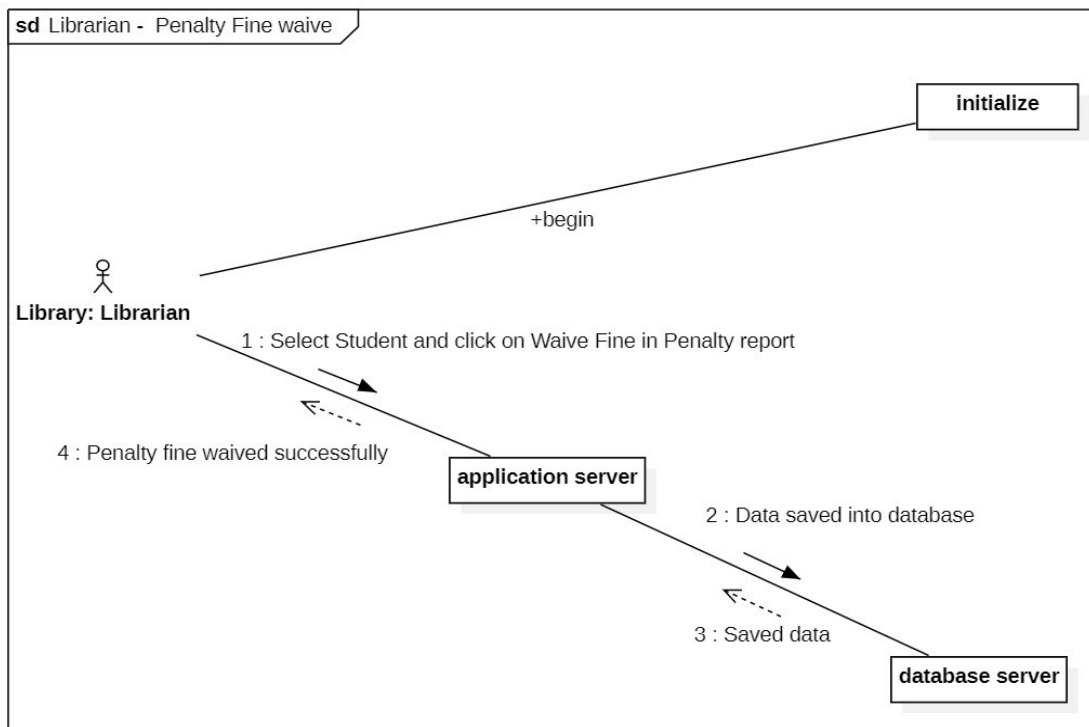
Edit department:



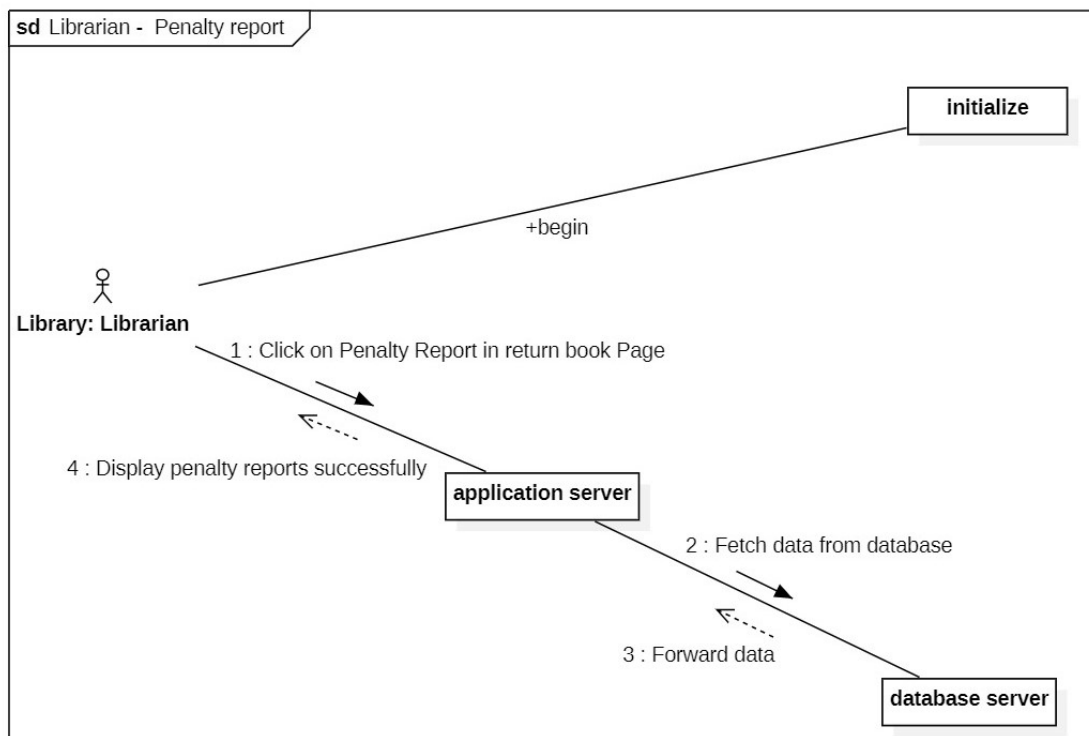
Paid the Penalty Fine:



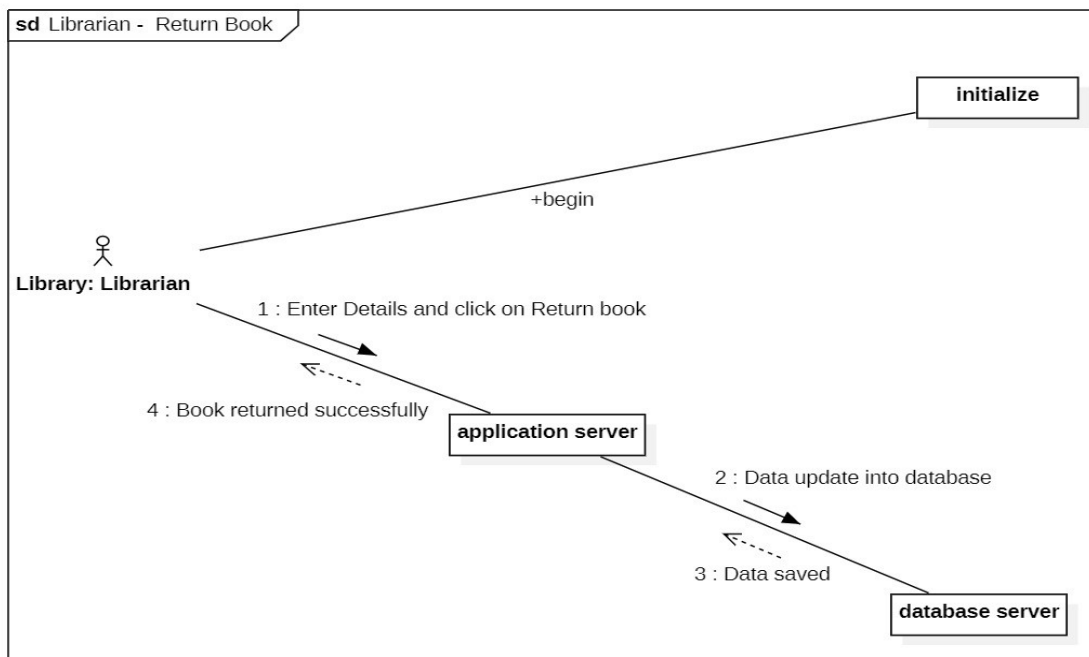
Waive the Penalty Fine:



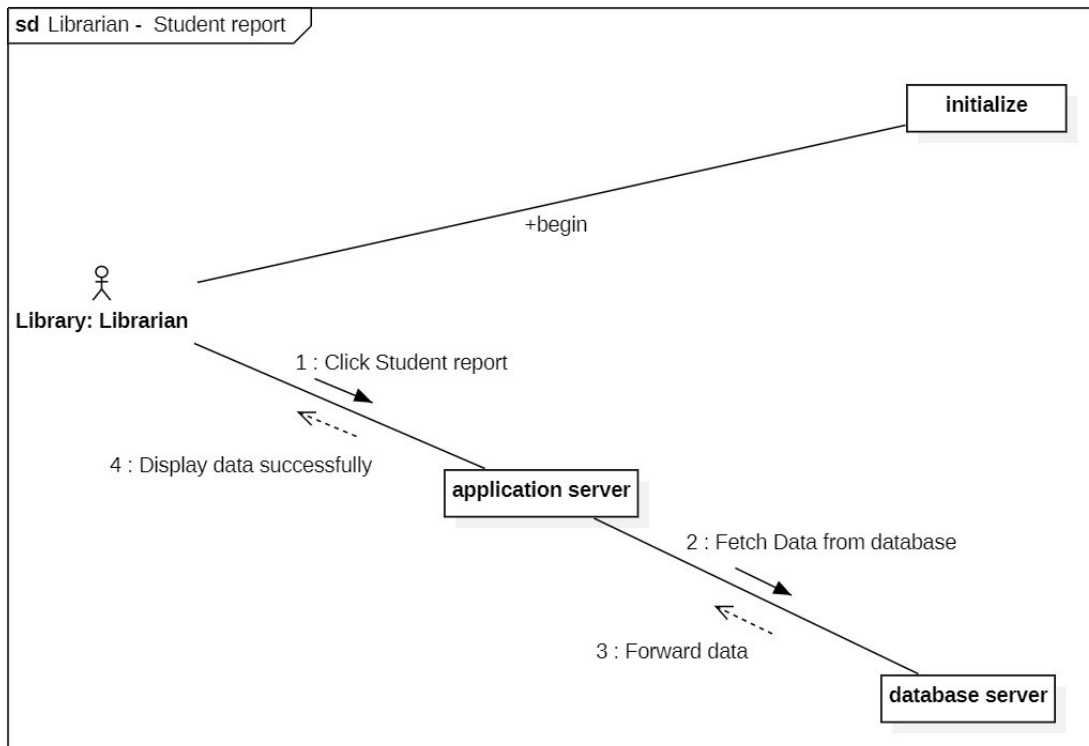
Penalty Report:



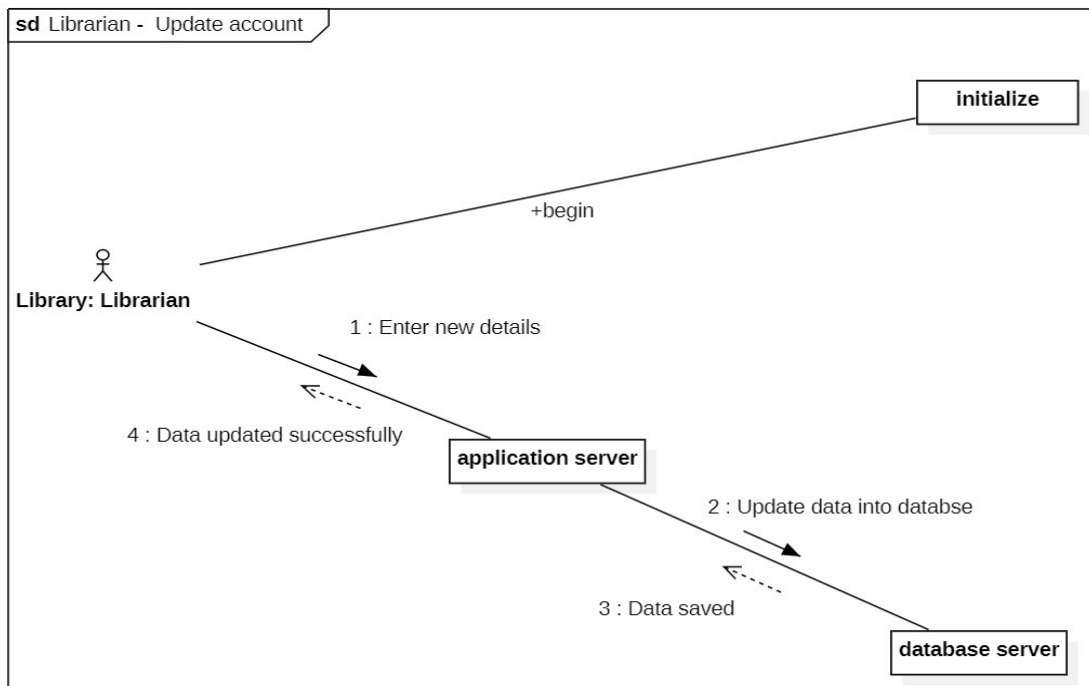
Return Borrow Book:



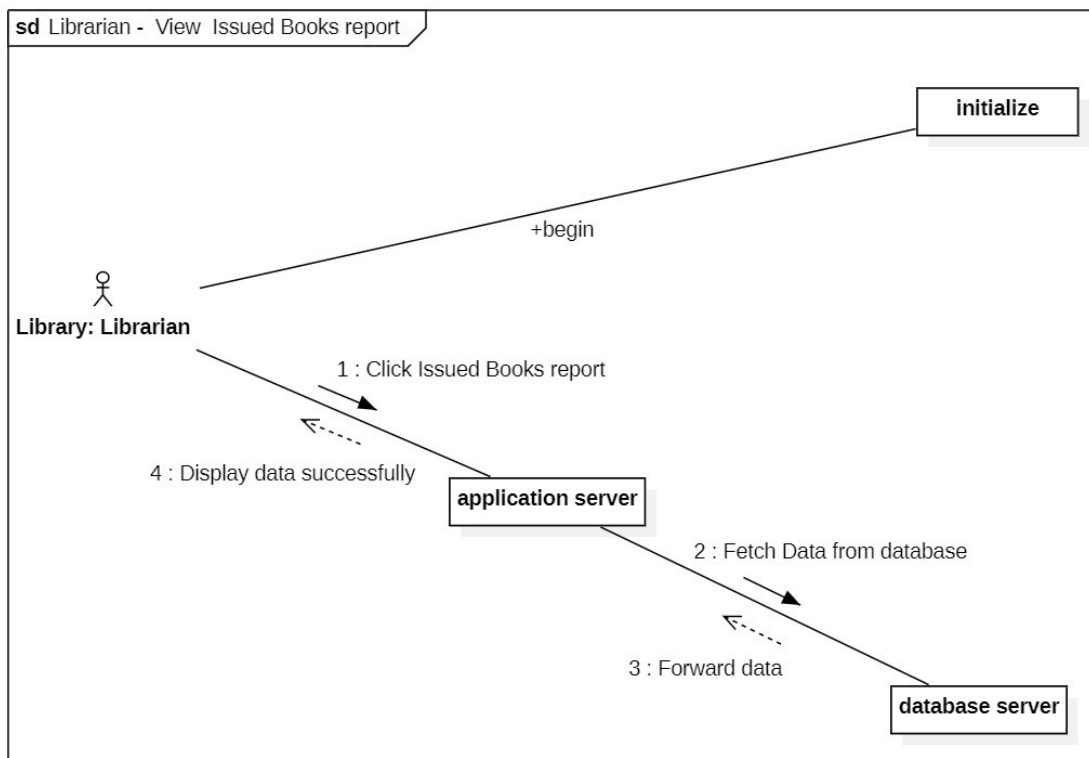
Student Report:



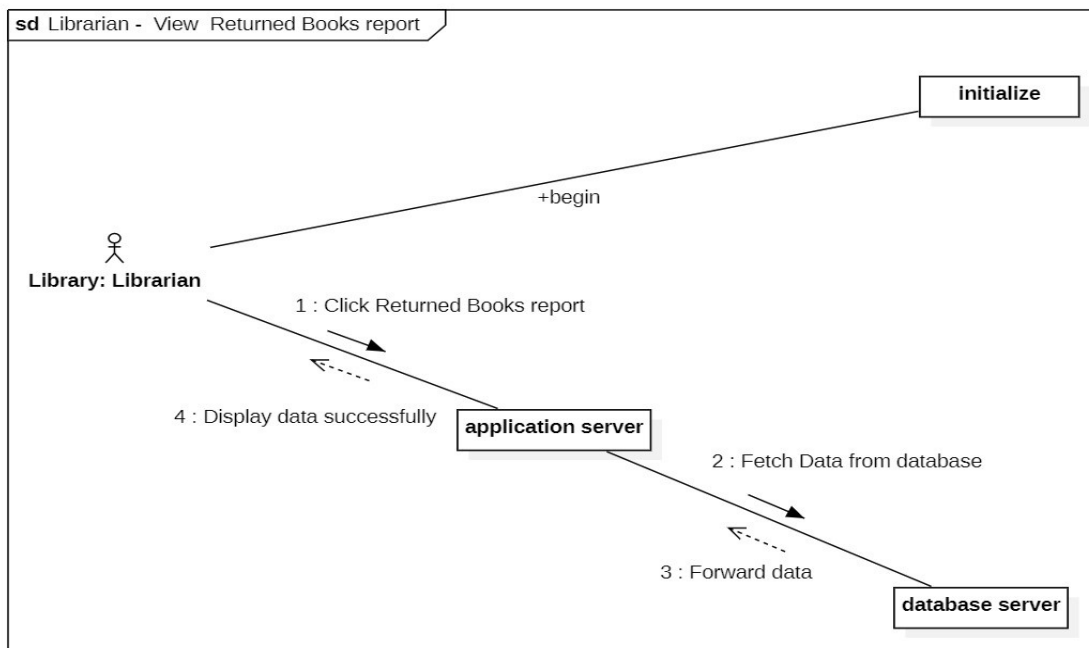
Librarian - update his information:



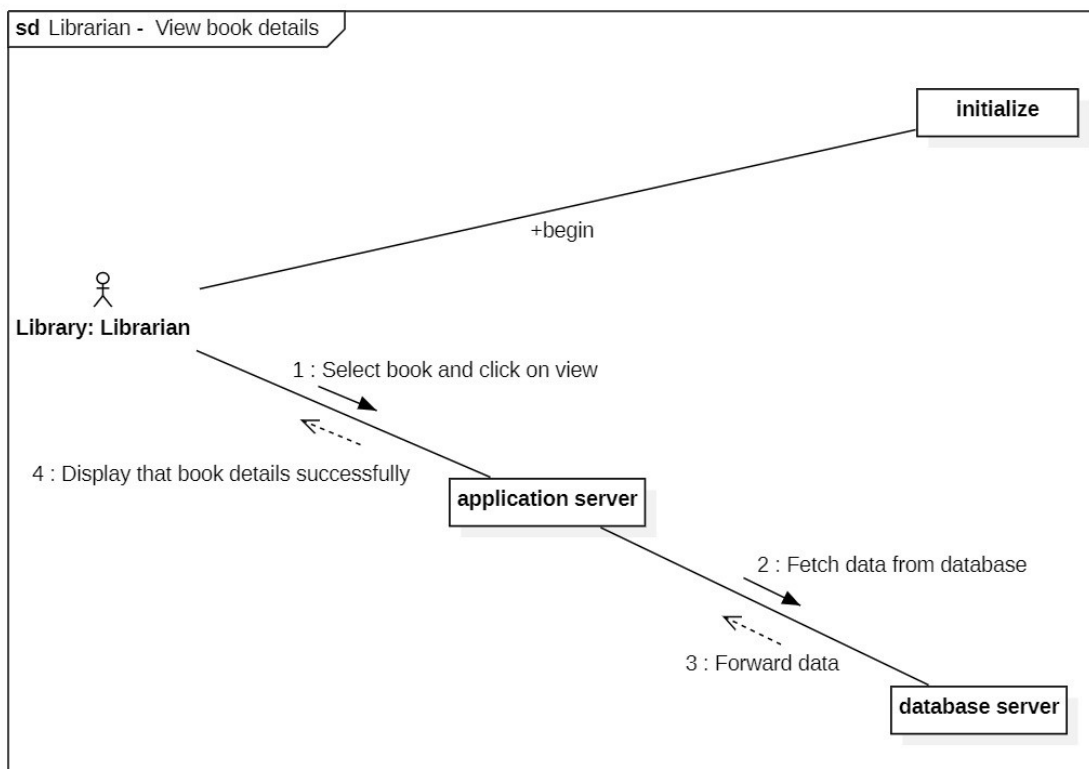
Librarian - Issued Books Report:



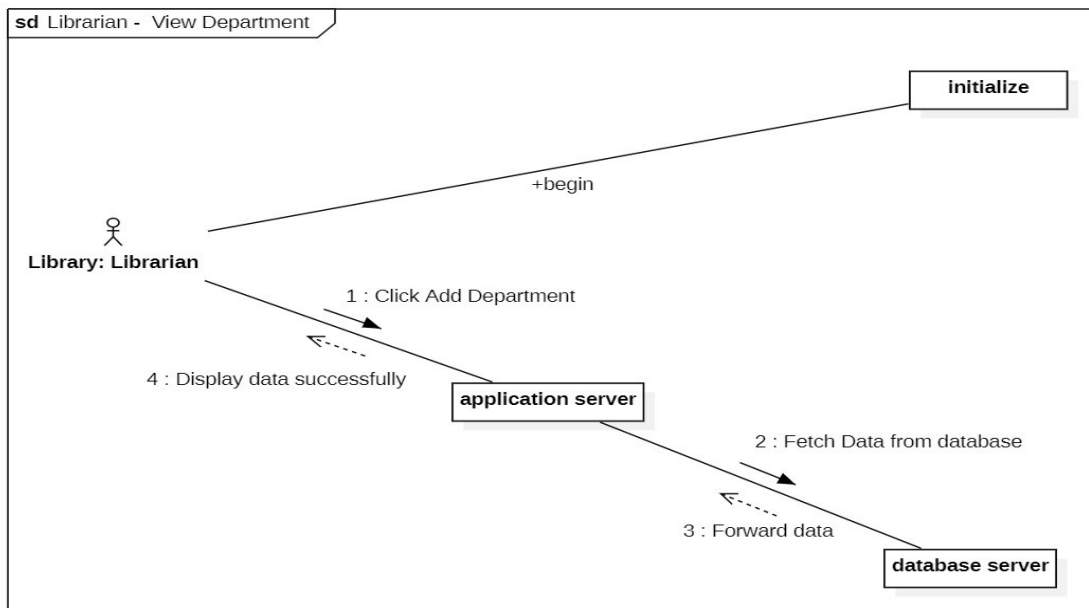
Librarian - Returned Books Report:



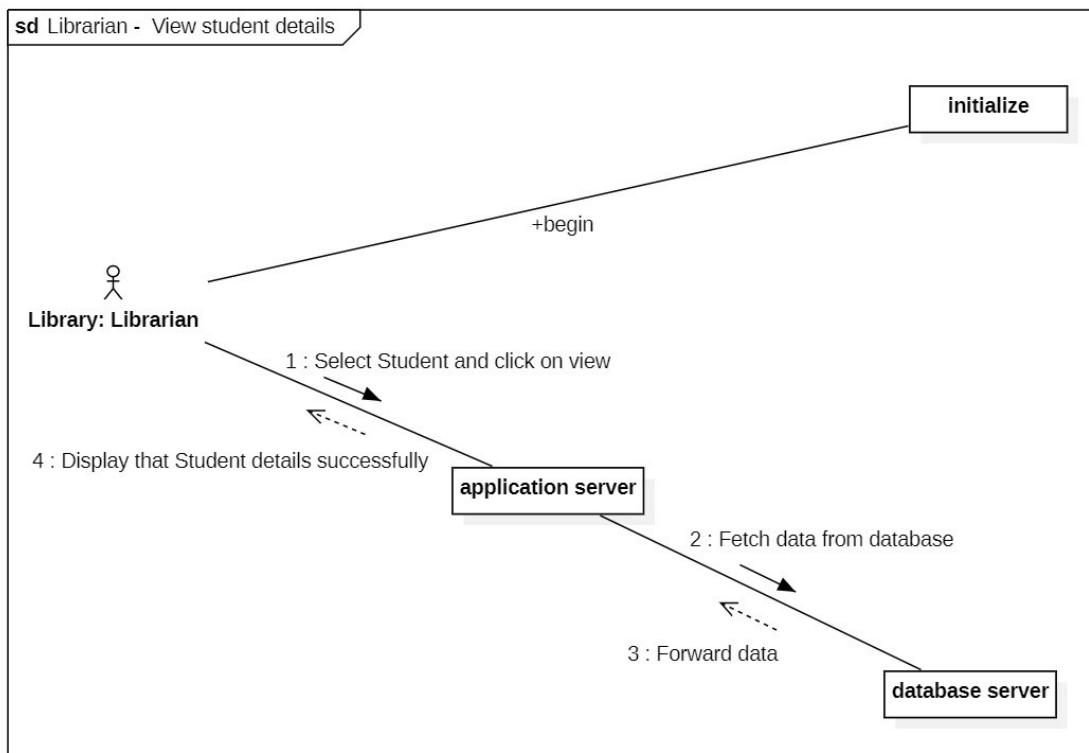
Librarian - Book Details:



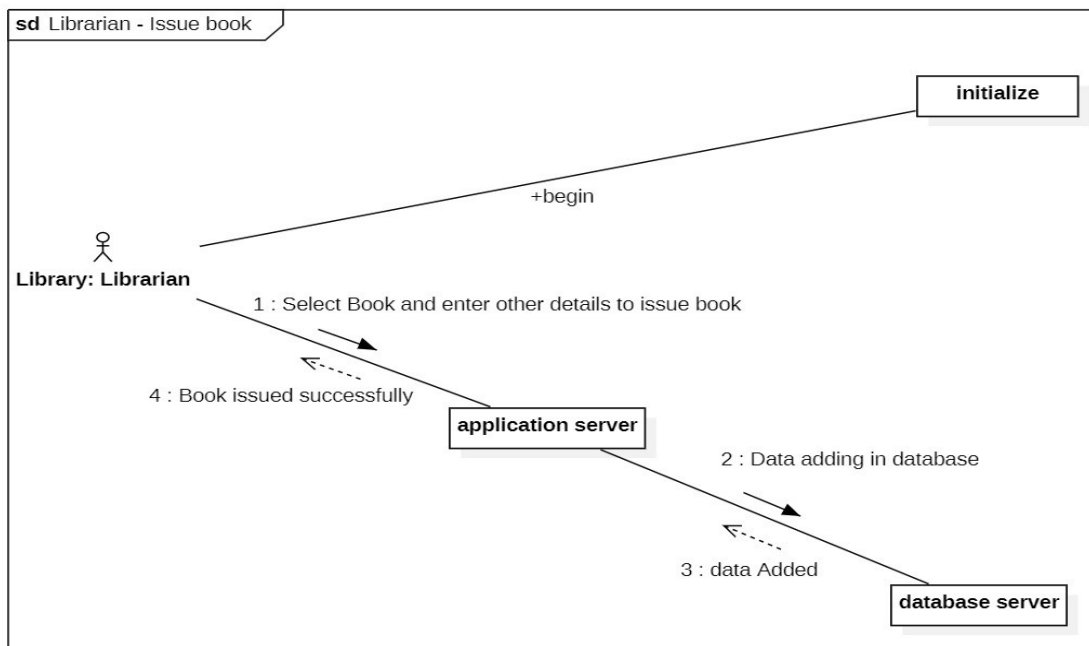
All departments:



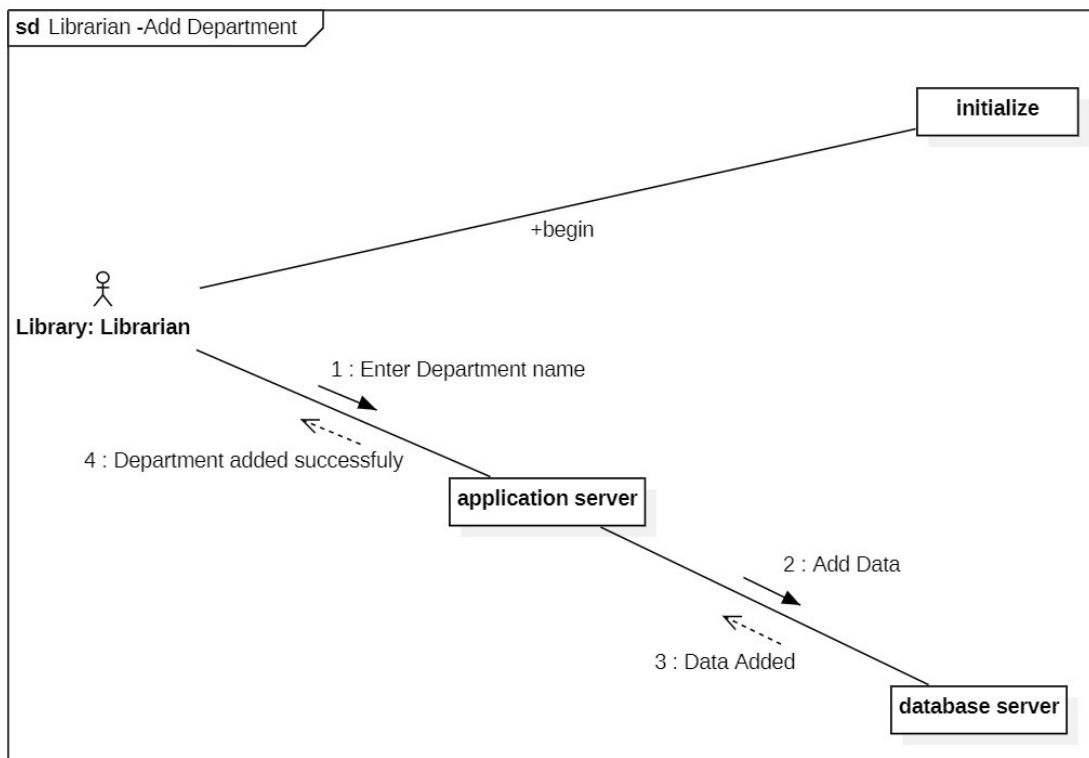
Student Detail:



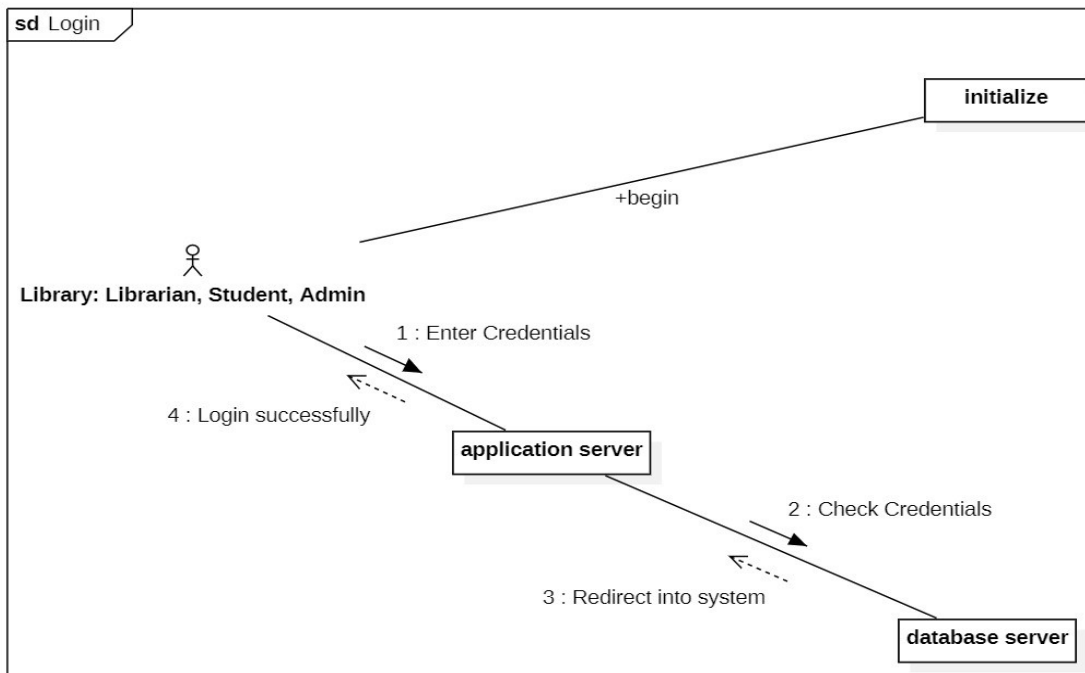
Issue Book:



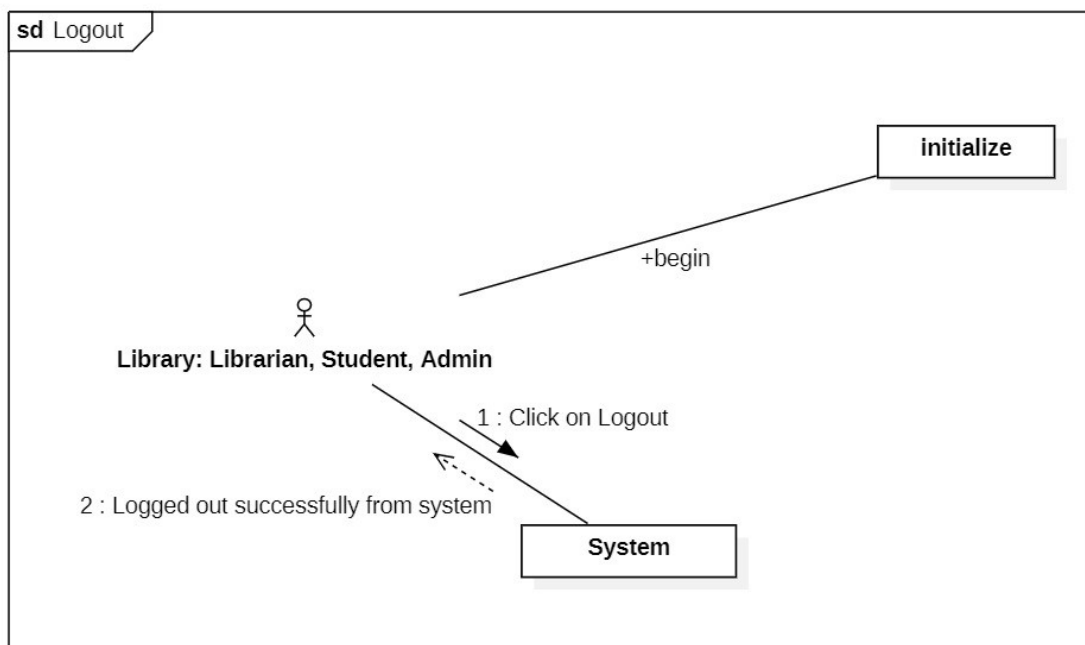
Add Department:



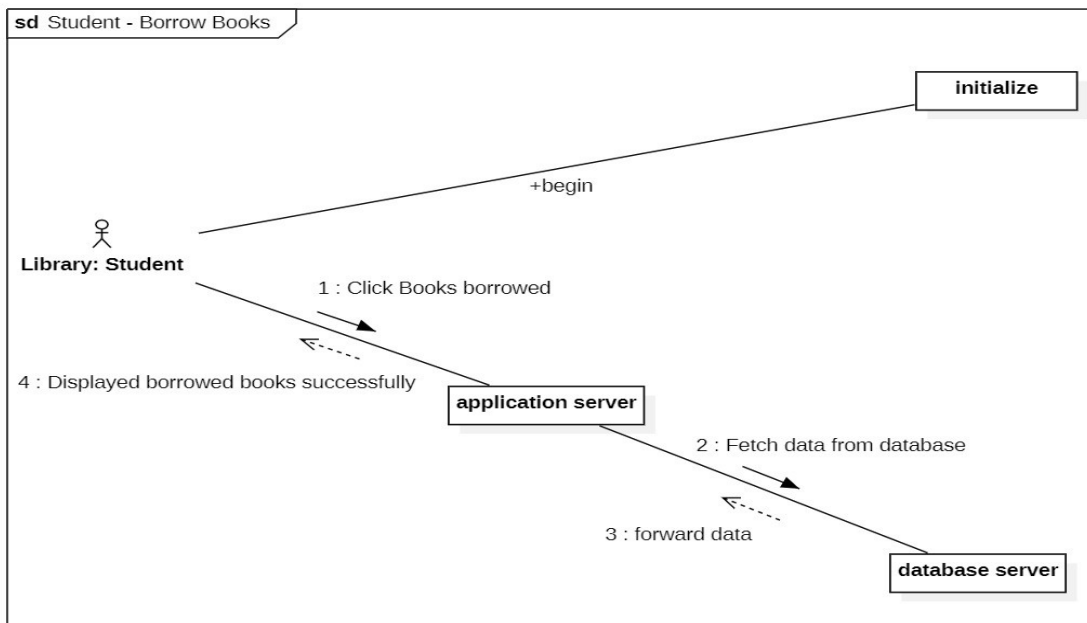
Login:



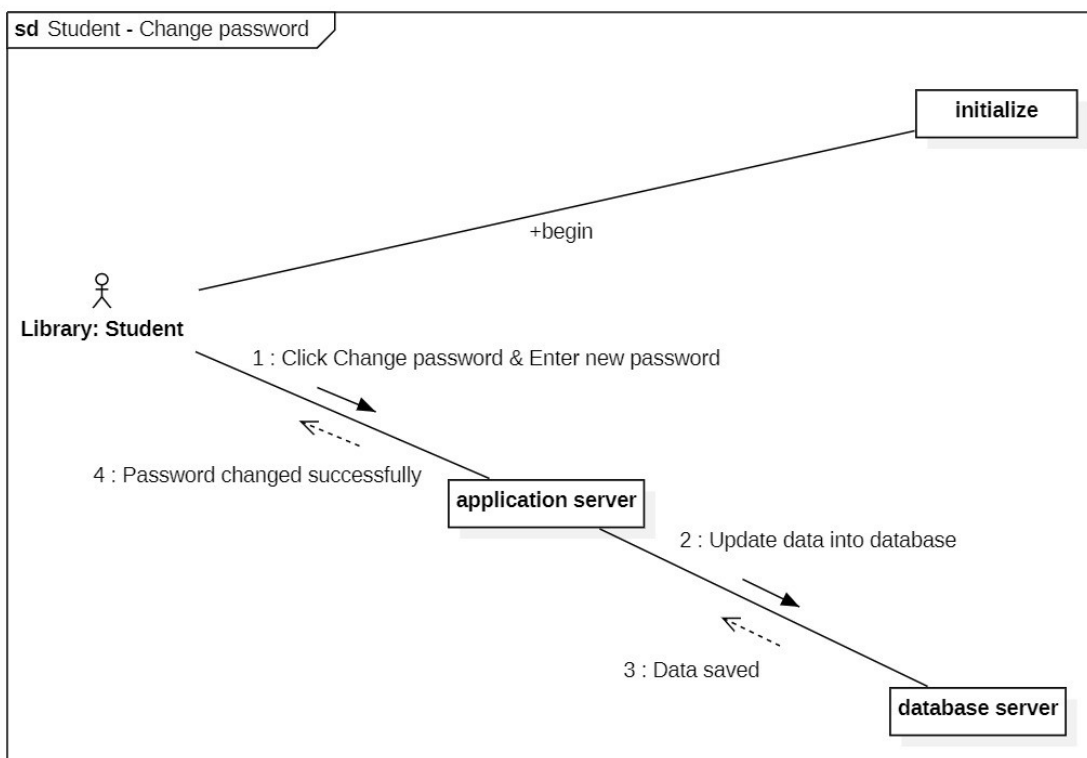
Logout:



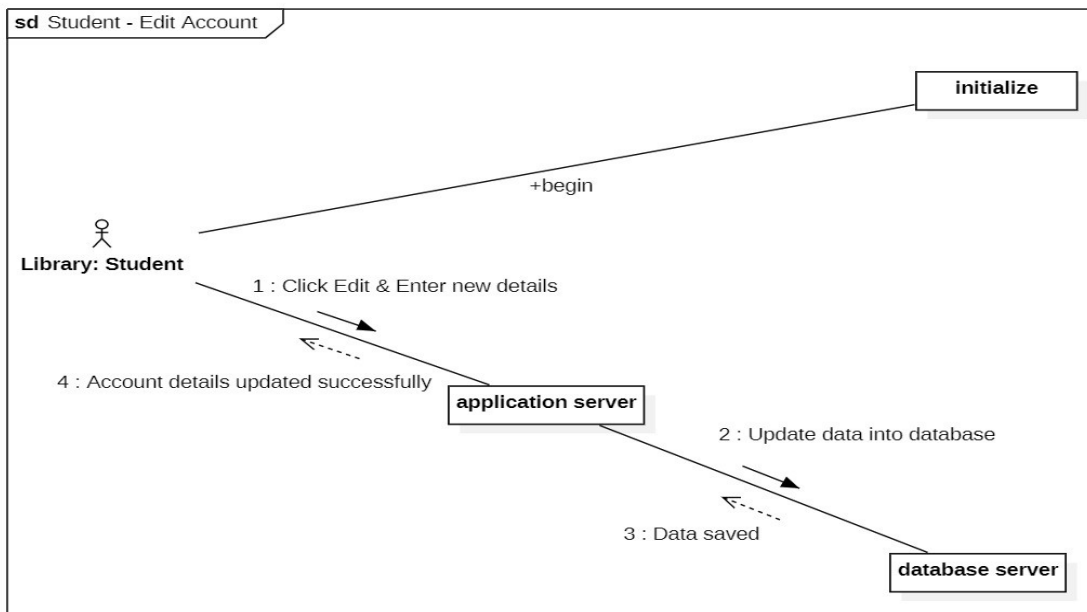
Student - view Borrow Books:



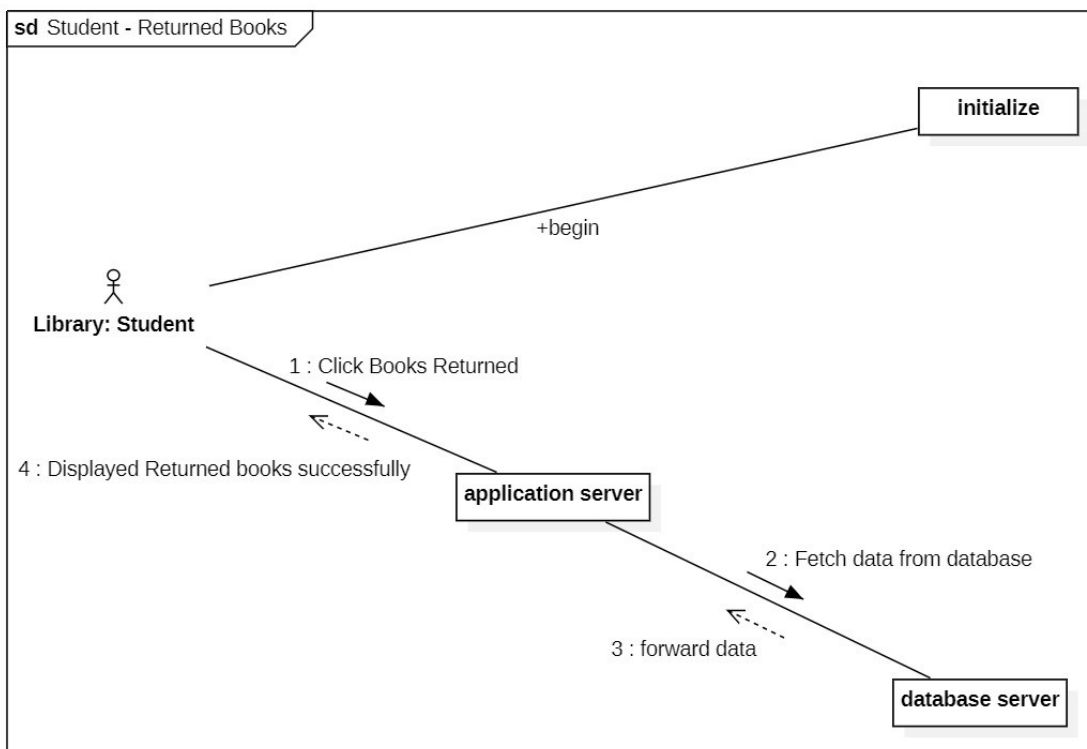
Student - Change his Password:



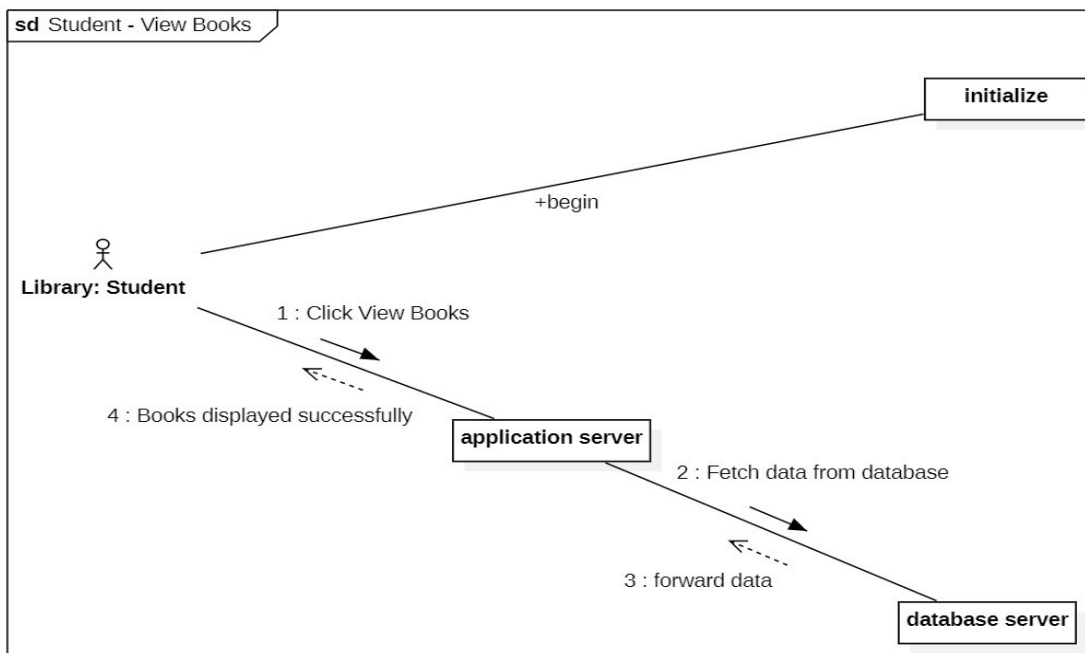
Student - Edit his Account Detail:



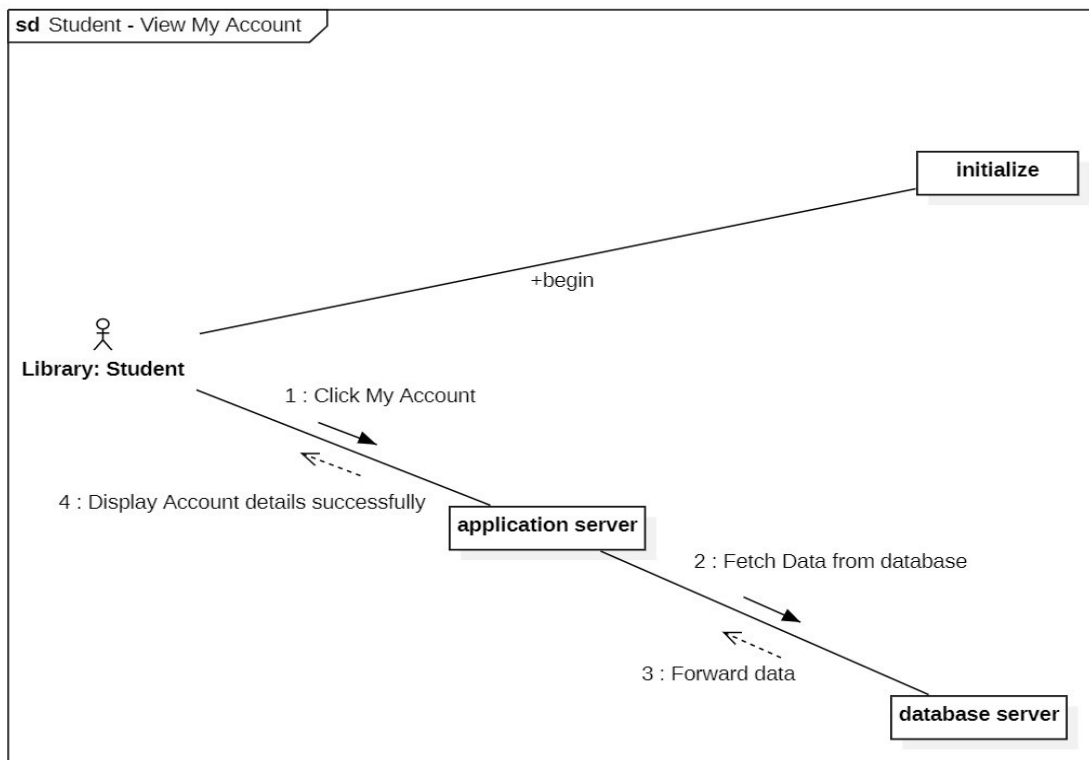
Student - view Return Books:



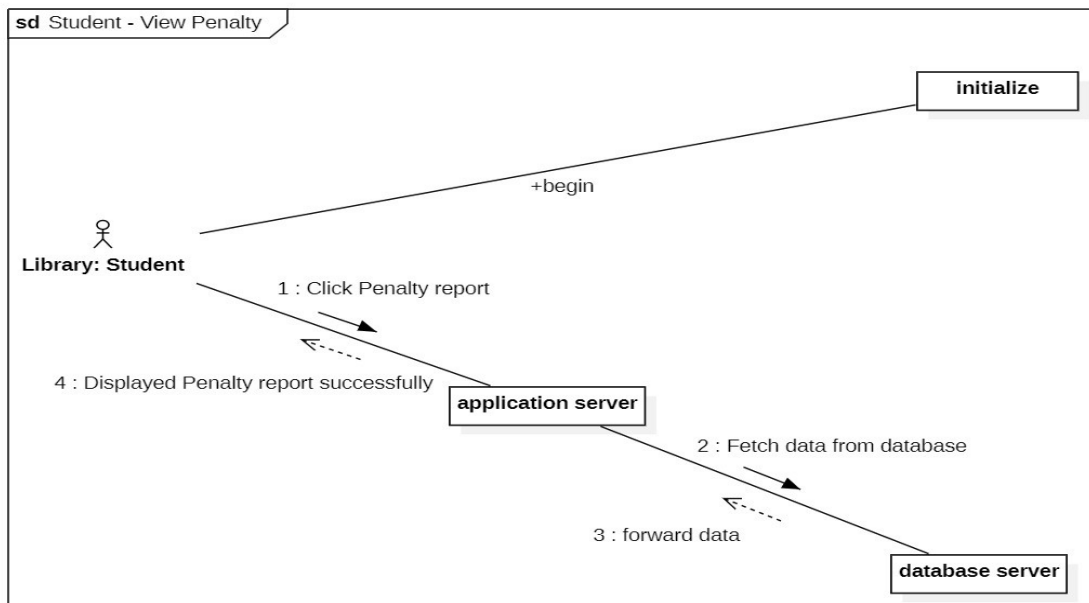
Student - View All Books:



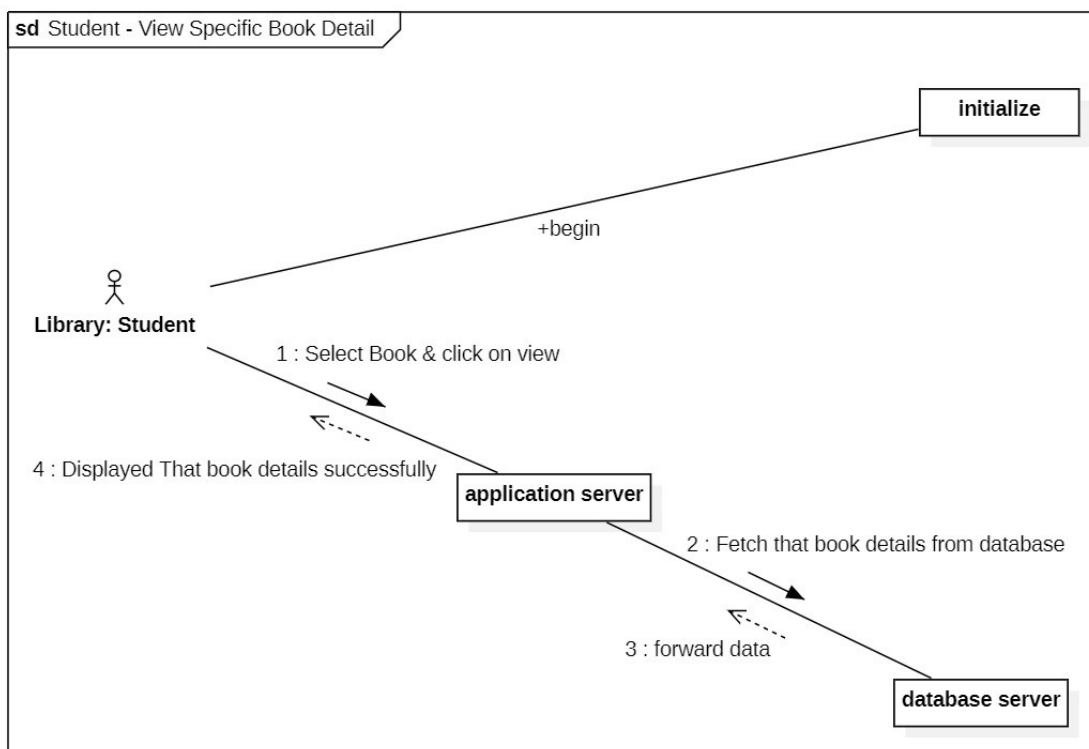
Student - view his Account:



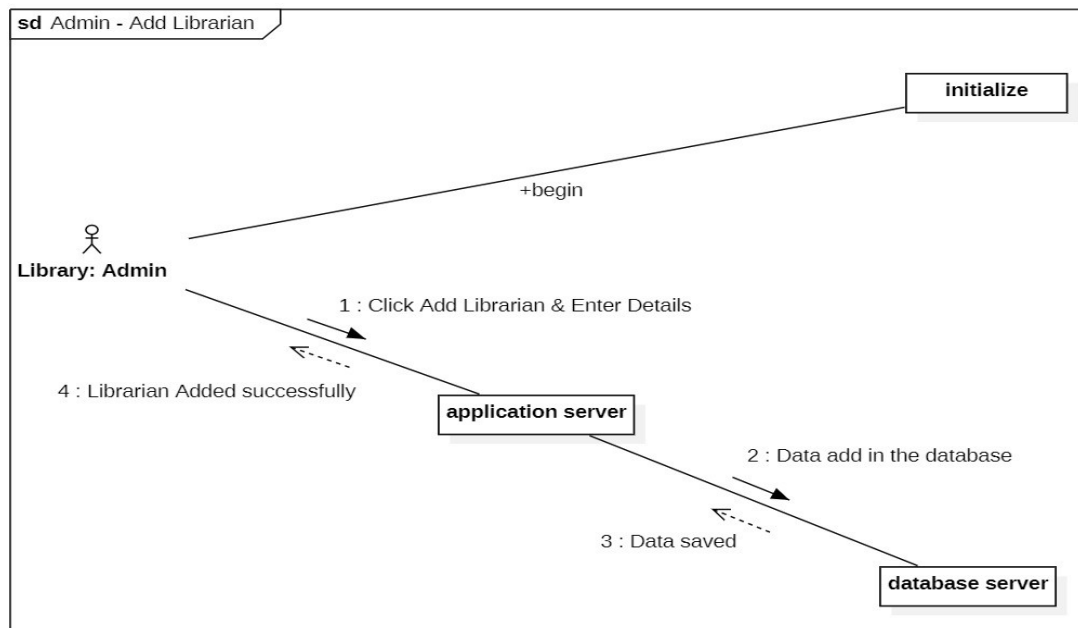
Student - View Penalty Reports:



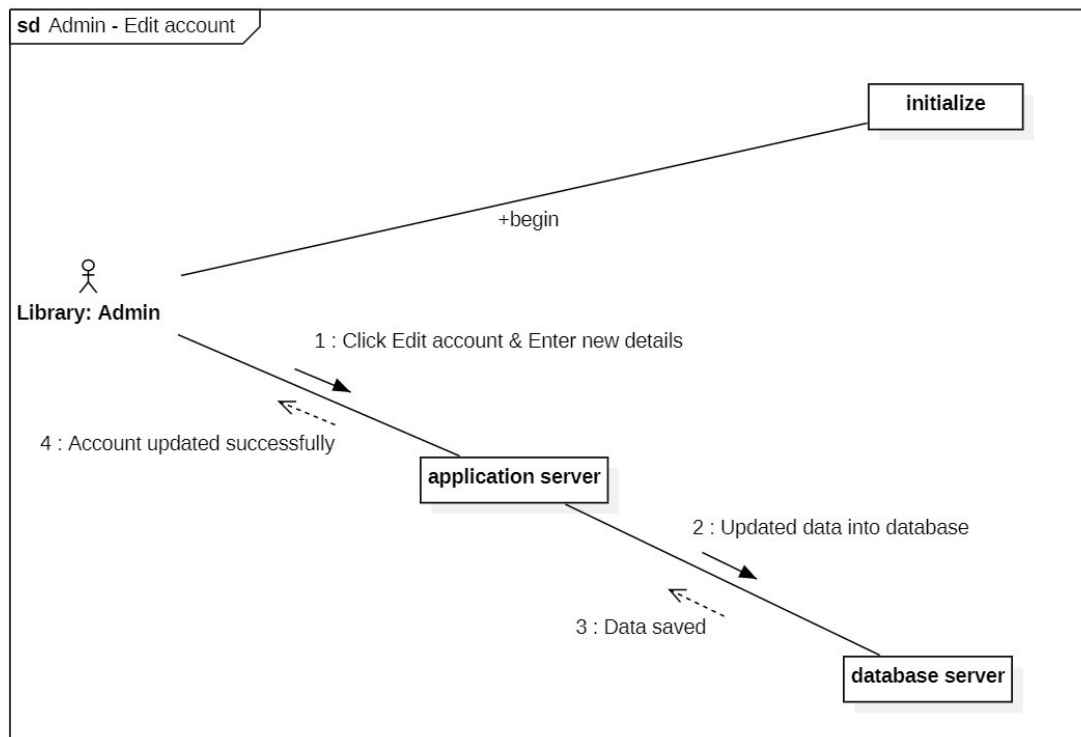
Student - view detail of specific Book:



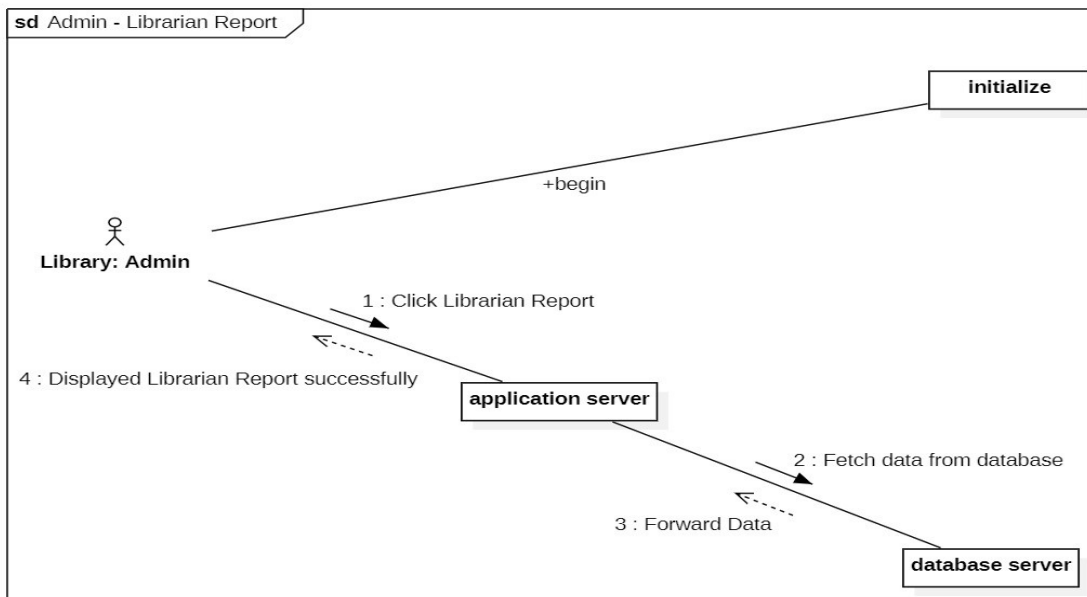
Add Librarian:



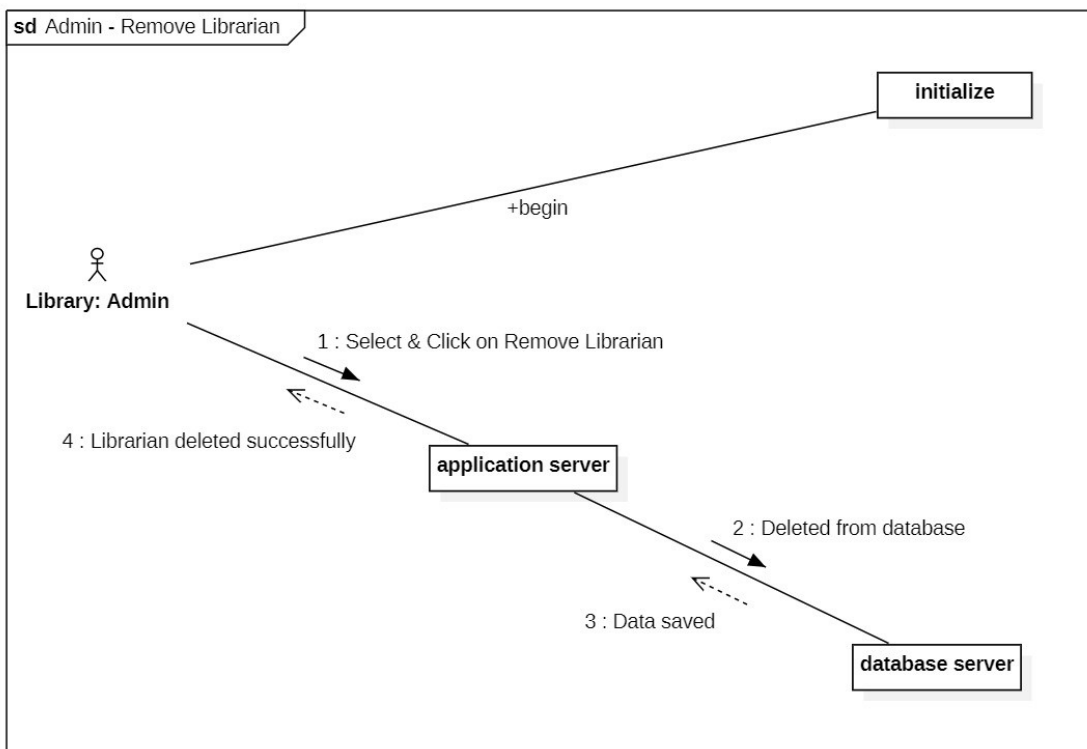
Admin - Update his Information:



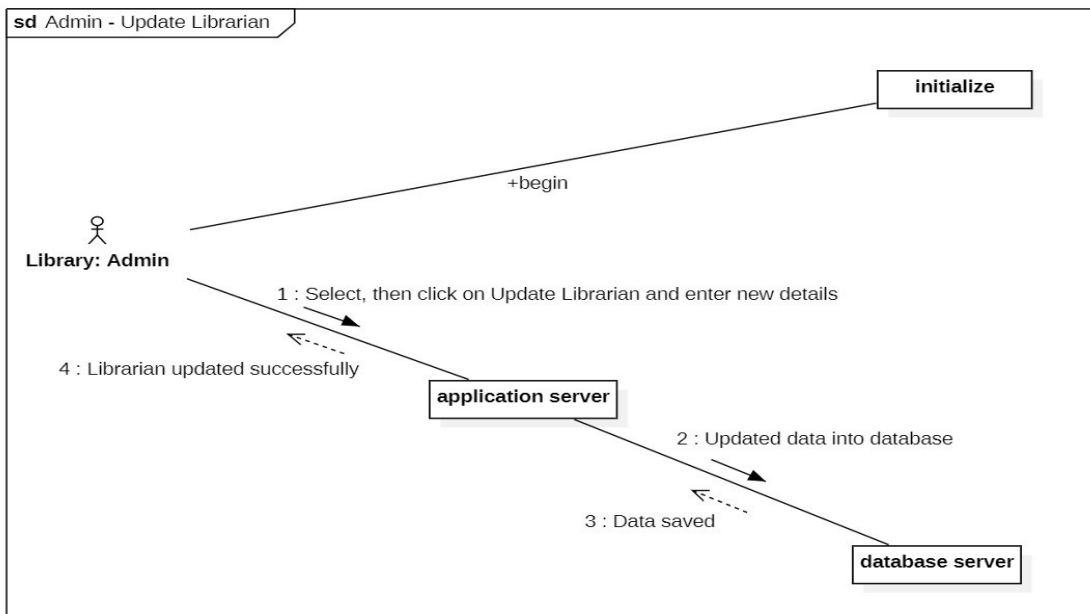
View Librarian Report:



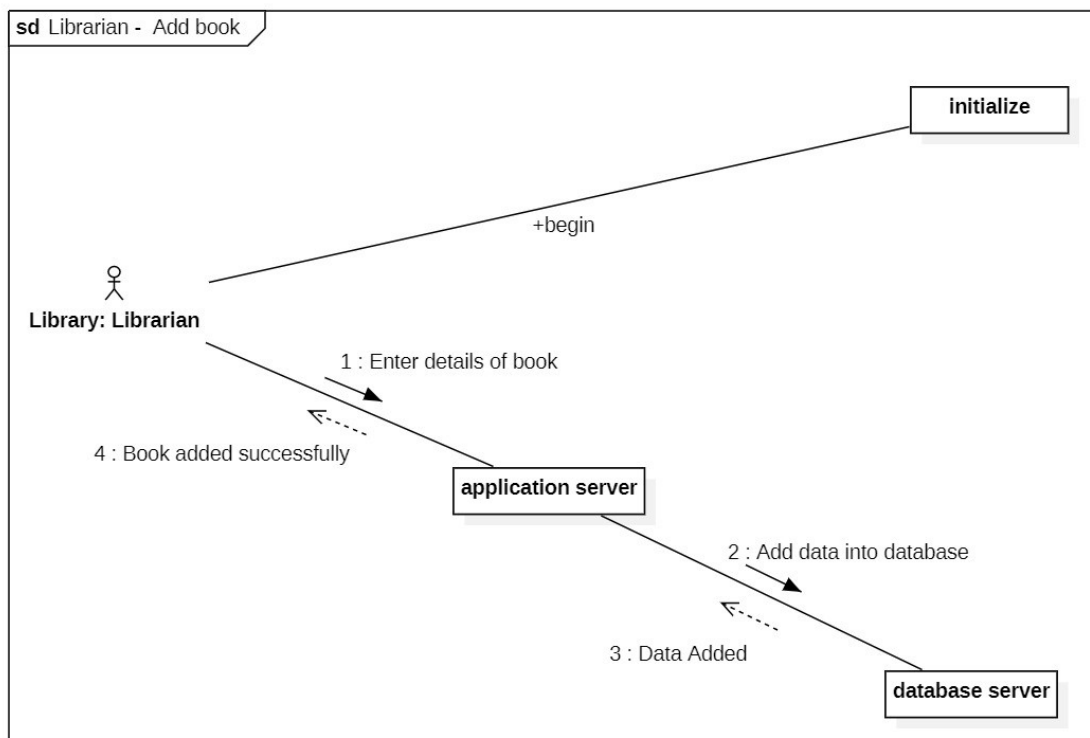
Remove Librarian:



Update Librarian Information:



Add Books:



TESTING

Unit Testing

Login Page

Rules:

1. Username must be alphabets only.
2. A user role (Student, Librarian, or Admin) must be selected before login.
- 3 Password must be at least 8 characters long and include alphabets, numbers, and special characters.
4. Username should not be empty.
5. Password should not be empty.
6. The login button must not be clickable if any input field is empty.

1. Username V 1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Hammad123	Should be Rejected	Has been rejected	Pass
Hammad12@	Should be Rejected	Has been accepted	Fail
1234	Should be Rejected	Has been rejected	Pass
ShayanArshad	Should be Accepted	Has been accepted	Pass

2. Username V 1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Hammad123	Should be Rejected	Has been rejected	Pass
Hammad12@	Should be Rejected	Has been rejected	Pass
1234	Should be Rejected	Has been rejected	Pass
ShayanArshad	Should be Accepted	Has been accepted	Pass

3. Password V 1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Hammad123@	Should be Accepted	Has Been Accepted	Pass
Hammad@	Should be Rejected	Has Been Accepted	Fail
H123@	Should be Rejected	Has Been Accepted	Fail
@@@@@@	Should be Rejected	Has Been Accepted	Fail
123	Should be Rejected	Has Been Accepted	Fail
Hai123@!xyz	Should be Accepted	Has Been Accepted	Pass

4. Password V 1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Hammad123@	Should be Accepted	Has Been Accepted	Pass
Hammad@	Should be Rejected	Has Been Rejected	Pass
H123@	Should be Rejected	Has Been Rejected	Pass
@@@@@@	Should be Rejected	Has Been Rejected	Pass
123	Should be Rejected	Has Been Rejected	Pass
Hai123@!xyz	Should be Accepted	Has Been Accepted	Pass

5. Role V 1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Student	Should be Accepted	Has Been Accepted	Pass
Empty	Should be Rejected	Has Been Rejected	Pass
Admin	Should be Accepted	Has Been Accepted	Pass

6. Login Button V 1.0

Test Data	Expected Result	Actual Results	Pass/Fail
The User clicks on login button	Page should be jumped to Home page	Page has been jumped to Home	Pass

Librarian Home

Rules:

Buttons like Home, Add Book, Book Report, Add Department, etc., must be responsive and functional.

1. Add Book V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "Add Book" button.	The form to add a new book should be displayed.	Data form has been displayed.	Pass

2. Book Report V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "Book Report" button	A report of all books should be displayed	Report displayed	Pass

3. Add department V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "Add Department" button	The form to add a new department should be displayed.	Has been displayed.	Pass

4. Add Student V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "Add Student" button	The form to add a new student should be displayed	Has been displayed successfully	Pass

5. Student Report V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "Student Report" button	A report of all students should be displayed	Report displayed correctly	Pass

6. Issue Book V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "Issue Book" button	The form to issue a book should be displayed	Has been displayed successfully	Pass

7. Issue Report V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "Issue Report" button	A report of all issued books should be displayed	Report displayed correctly	Pass

8. Return Book V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "Return Book" button	The form to return a book should be displayed	Has been displayed successfully	Pass

9. Logout V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "Logout" button	The user should be logged out and redirected to the login page	Redirected successfully	Pass

10. Change Your info V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "Change Your Info"	The form to change the user's personal information should be displayed	Has been displayed successfully	Pass

Update Librarian/Add Librarian

Rules:

1. Librarian Id must be a unique identifier for the librarian.
2. Name must only contain alphabets.
3. Gender must be selected.
4. Mobile no. Should not be empty.
5. The address should be of a reasonable length (e.g., not more than 200 characters).
6. Username should not be empty.
7. The password should not match the previous password used by the librarian and should not be empty.
8. The button should be enabled only when all the fields are filled out correctly and pass validation.

1. Librarian ID V 1.0

Test Data	Expected Result	Actual Results	Pass/Fail
1234	Should be accepted	Has been accepted	Pass
0001	Should be accepted	Has been accepted	Pass
1234	Should be rejected	Has been accepted	Fail
5678	Should be accepted	Has been accepted	Pass

2. Librarian ID V 1.1

Test Data	Expected Result	Actual Results	Pass/Fail
1234	Should be accepted	Has been accepted	Pass
0001	Should be accepted	Has been accepted	Pass
1234	Should be rejected	Has been rejected	Pass
5678	Should be accepted	Has been accepted	Pass

3. Name V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Ali	Should be accepted	Has been accepted	Pass
M Ali	Should be accepted	Has been accepted	Pass
Ali123	Should be rejected	Has been rejected	Pass
@Ali	Should be rejected	Has been accepted	Fail

4. Name V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Ali	Should be accepted	Has been accepted	Pass
M Ali	Should be accepted	Has been accepted	Pass
Ali123	Should be rejected	Has been rejected	Pass
@Ali	Should be rejected	Has been rejected	Fail

5. Gender V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Male	Should be Accepted	Has Been Accepted	Pass
Empty	Should be Rejected	Has Been Rejected	Pass
Female	Should be Accepted	Has Been Accepted	Pass

6. Mobile No V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
03001234567	Should be accepted	Has been accepted	Pass
123456789	Should be rejected	Has been rejected	Pass
0300ABC4567	Should be rejected	Has been rejected	Pass
@3001234567	Should be rejected	Has been accepted	Fail

7. Mobile No V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
03001234567	Should be accepted	Has been accepted	Pass
123456789	Should be rejected	Has been rejected	Pass
0300ABC4567	Should be rejected	Has been rejected	Pass
@3001234567	Should be rejected	Has been rejected	Pass

8. Address V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
House 123, Street 45	Should be accepted	Has been accepted	Pass
Flat #10, Building ABC	Should be accepted	Has been accepted	Pass
123456	Should be rejected	Has been rejected	Pass
@@@Street Name	Should be rejected	Has been rejected	Pass

9. Username V 1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Hammad123	Should be Rejected	Has been rejected	Pass
Hammad12@	Should be Rejected	Has been accepted	Fail
1234	Should be Rejected	Has been rejected	Pass
ShayanArshad	Should be Accepted	Has been accepted	Pass

10. Username V 1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Hammad123	Should be Rejected	Has been rejected	Pass
Hammad12@	Should be Rejected	Has been rejected	Pass
1234	Should be Rejected	Has been rejected	Pass
ShayanArshad	Should be Accepted	Has been accepted	Pass

11. Password V 1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Hammad123@	Should be Accepted	Has Been Accepted	Pass
Hammad@	Should be Rejected	Has Been Accepted	Fail
H123@	Should be Rejected	Has Been Accepted	Fail
@@@@@@	Should be Rejected	Has Been Accepted	Fail

12. Password V 1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Hammad123@	Should be Accepted	Has Been Accepted	Pass
Hammad@	Should be Rejected	Has Been Rejected	Pass
H123@	Should be Rejected	Has Been Rejected	Pass
@@@@@@	Should be Rejected	Has Been Rejected	Pass

13. Login Button V 1.0

Test Data	Expected Result	Actual Results	Pass/Fail
The User clicks on Update Librarian button	info should be updated	Info has been updated	Pass

Change librarian password

Rules:

1. Username should not be empty.
2. Username must be alphabets only.
3. Old password must be correct.
4. New Password should not match the previous password.
5. A new Password should contain at least one special character, number, and letter.
6. Confirm password must be match with new password.

1. Username V 1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Hammad123	Should be Rejected	Has been rejected	Pass
Hammad12@	Should be Rejected	Has been accepted	Fail
1234	Should be Rejected	Has been rejected	Pass
ShayanArshad	Should be Accepted	Has been accepted	Pass

2. Username V 1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Hammad123	Should be Rejected	Has been rejected	Pass
Hammad12@	Should be Rejected	Has been rejected	Pass
1234	Should be Rejected	Has been rejected	Pass
ShayanArshad	Should be Accepted	Has been accepted	Pass

3. Old Password V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian123	Should be accepted	Has been accepted	Pass
WrongPass123	Should be rejected	Has been rejected	Pass
Admin@1234	Should be rejected	Has been rejected	Pass
Admin@199	Should be rejected	Has been accepted	Fail

4. Old Password V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian123	Should be accepted	Has been accepted	Pass
WrongPass123	Should be rejected	Has been rejected	Pass
Admin@1234	Should be rejected	Has been rejected	Pass
(Empty)	Should be rejected	Has been rejected	Pass

5. New Password V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
NewPass@123	Should be accepted.	Has been accepted	Pass
12345678	Should be rejected	Has been rejected	Pass
Password123!	Should be accepted	Has been accepted	Pass
OldPassword123!	Should be rejected	Has been rejected	Pass

6. Confirm Password V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
NewPass@123	Should be accepted.	Has been accepted	Pass
12345678	Should be rejected	Has been rejected	Pass
Password123!	Should be accepted	Has been accepted	Pass

Student Home

Rules:

- Buttons like My Account, Books Borrowed, Books Report, Penalty Report, etc., must be responsive and functional.

1. My Account V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Student clicks on the "My Account" button	The form to view and update personal account details should be displayed.	form detail has been displayed	Pass

2. Books Borrowed V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Student clicks on the "Books Borrowed" button	A report of borrowed books should be displayed.	Borrowed report has been displayed	Pass

3. Books Returned V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Student clicks on the "Books Returned" button	A report of Returned books should be displayed.	Returned report has been	Pass

4. View Books V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Student clicks on the "View Books" button	A list of available books should be displayed.	Available Books have been displayed	Pass

5. Penalty Report V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Student clicks on the "Penalty Report" button	A report of any penalties or fines should be displayed.	Penalty report has been displayed	Pass

6. Logout V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Student clicks on the "Logout" button	The user should be logged out and redirected to the login page	Redirected successfully	Pass

Student (Edit Account Detail)

Rules:

1. Name should only contain alphabets and spaces.
2. The name must not be empty.
3. Roll numbers must be unique. It must be a mixture of alphabets and numbers.
4. Gender must not be empty.
5. The department must not be empty, and it should only contain alphabets and spaces.
6. The address must be a string containing letters, numbers, and spaces.
7. Username should not be empty.
8. Username must be alphabets only.
9. Password should not be empty.

1. Name V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
John Doe	Should be accepted	Has been accepted	Pass
Jane123	Should be rejected	Has been rejected	Pass
Ali	Should be accepted	Has been accepted	Pass
@John	Should be rejected	Has been accepted	Fail

2. Name V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
John Doe	Should be accepted	Has been accepted	Pass
Jane123	Should be rejected	Has been rejected	Pass
Ali	Should be accepted	Has been accepted	Pass
@John	Should be rejected	Has been rejected	Pass

3. Roll No V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
AB123	Should be accepted	Has been accepted	Pass
A1B2C3	Should be accepted	Has been accepted	Pass
12345	Should be rejected	Has been rejected	Pass
ABC123	Should be rejected	Has been accepted	Fail

4. Roll No V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
AB123	Should be accepted	Has been accepted	Pass
A1B2C3	Should be accepted	Has been accepted	Pass
12345	Should be rejected	Has been rejected	Pass
ABC123	Should be rejected	Has been rejected	Pass

5. Gender V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Male	Should be Accepted	Has Been Accepted	Pass
Empty	Should be Rejected	Has Been Rejected	Pass
Female	Should be Accepted	Has Been Accepted	Pass

6. Department V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Computer Science	Should be accepted	Has been accepted	Pass
Electrical Engg	Should be accepted	Has been accepted	Pass
12345	Should be rejected	Has been rejected	Pass
!@#Department	Should be rejected	Has been rejected	Pass
Mathematics	Should be accepted	Has been accepted	Pass

7. Address V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
House 123, Street 45	Should be accepted	Has been accepted	Pass
Flat #10, Building ABC	Should be accepted	Has been accepted	Pass
123456	Should be rejected	Has been rejected	Pass
@@@Street Name	Should be rejected	Has been rejected	Pass

8. Username V 1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Hammad123	Should be Rejected	Has been rejected	Pass
Hammad12@	Should be Rejected	Has been accepted	Fail
1234	Should be Rejected	Has been rejected	Pass
ShayanArshad	Should be Accepted	Has been accepted	Pass

9. Username V 1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Hammad123	Should be Rejected	Has been rejected	Pass
Hammad12@	Should be Rejected	Has been rejected	Pass
1234	Should be Rejected	Has been rejected	Pass
ShayanArshad	Should be Accepted	Has been accepted	Pass

10. Password V 1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Hammad123@	Should be Accepted	Has Been Accepted	Pass
Hammad@	Should be Rejected	Has Been Accepted	Fail
H123@	Should be Rejected	Has Been Accepted	Fail
@@@@@@	Should be Rejected	Has Been Accepted	Fail

Password V 1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Hammad123@	Should be Accepted	Has Been Accepted	Pass
Hammad@	Should be Rejected	Has Been Rejected	Pass
H123@	Should be Rejected	Has Been Rejected	Pass
@@@@@@	Should be Rejected	Has Been Rejected	Pass

Student Change Password

Rules:

1. Old password must be correct.
2. New Password should not match the previous password.
3. A new Password should contain at least one special character, number, and letter.
4. Confirm password must be match with new password.

1. Old Password V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian123	Should be accepted	Has been accepted	Pass
WrongPass123	Should be rejected	Has been rejected	Pass
Admin@1234	Should be rejected	Has been rejected	Pass
Admin@908	Should be rejected	Has been accepted	Fail

2. Old Password V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian123	Should be accepted	Has been accepted	Pass
WrongPass123	Should be rejected	Has been rejected	Pass
Admin@1234	Should be rejected	Has been rejected	Pass
Admin@908	Should be rejected	Has been rejected	Pass

3. New Password V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
NewPass@123	Should be accepted.	Has been accepted	Pass
12345678	Should be rejected	Has been rejected	Pass
Password123!	Should be accepted	Has been accepted	Pass
OldPassword123!	Should be rejected	Has been rejected	Pass

4. Confirm Password V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
NewPass@123	Should be accepted.	Has been accepted	Pass
12345678	Should be rejected	Has been rejected	Pass
Password123!	Should be accepted	Has been accepted	Pass

Admin Home

Rules:

- Buttons like Add Librarian, Librarian Report, Edit Account and logout must be responsive and functional.

1. Add Librarian V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Admin clicks on the "Add Librarian" button	The form to add a new librarian should be displayed.	Form to add a new librarian has been displayed.	Pass

2. Librarian Report V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Admin clicks on the "Librarian Report" button	A report of all librarians should be displayed.	Librarian report has been displayed.	Pass

3. Edit Account V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Admin clicks on the "Edit Account" button	The form to view and update admin account details should be displayed.	Admin account form has been displayed.	Pass

4. Logout V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Admin clicks on the "Logout" button	The admin should be logged out and redirected to the login page	Redirected successfully	Pass

View Librarian Report

Rules:

1. A librarian must be selected.

1. Remove Librarian V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Admin clicks on the "Add Remove Librarian" button.	Selected librarian should be removed.	Selected librarian has been removed.	Pass

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2. Update Librarian V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Admin clicks on the "Update Librarian" button.	The form to update librarian info should be displayed.	Form has been displayed.	Pass

All Books (Student)

Rules:

1. The View button must be present next to each book.
2. Clicking the View button for a specific book should display the detailed information of that book.

1. View V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Student clicks the View button for Book 1	The details of Book 1 should be displayed correctly.	Book 1 details have been displayed.	Pass
Student clicks the View button for Book 2	The details of Book 2 should be displayed correctly.	Book 2 details have been displayed.	Pass

Add Book

Rules:

1. Book Title must not be empty and allow only alphabets, spaces, and punctuation.
2. Description must be 10-1000 characters long and not empty.
3. Author must not be empty and allow only alphabets and spaces.
4. Edition must not be empty and can be numeric or alphanumeric.
5. Price must be a positive number and not empty.
6. Quantity must be a whole positive number and not empty.
7. Book Image must be uploaded in .jpg, .png, or .jpeg format.
8. Blank spaces alone are not valid input.
9. All fields must pass validation to enable the "Add Book" button.

1. Book Title V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Software Engineering	Should be accepted	Has been accepted	Pass
To Kill a Mockingbird!	Should be accepted	Has been accepted	Pass
12345	Should be rejected	Has been rejected	Pass
War & Peace	Should be accepted	Has been accepted	Pass
!@#\$\$%	Should be rejected	Has been accepted	Fail

Book Title V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Software Engineering	Should be accepted	Has been accepted	Pass
To Kill a Mockingbird!	Should be accepted	Has been accepted	Pass
12345	Should be rejected	Has been rejected	Pass
War & Peace	Should be accepted	Has been accepted	Pass
!@#\$\$%	Should be rejected	Has been rejected	Pass

1. Description V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
An academic book that provides a detailed overview of software development methodologies and practices.	Should be accepted	Has been accepted	Pass
A classic novel that explores racial injustice and moral growth in the Deep South during the 1930s.	Should be accepted	Has been accepted	Pass
Short.	Should be rejected	Has been rejected	Pass
(Empty)	Should be rejected	Has been rejected	Pass

2. Author V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
John Doe	Should be accepted	Has been accepted	Pass
Harper Lee	Should be accepted	Has been accepted	Pass
12345	Should be rejected	Has been rejected	Pass
Leo Tolstoy	Should be accepted	Has been accepted	Pass

3. Edition V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
1st	Should be accepted	Has been accepted	Pass
2023	Should be accepted	Has been accepted	Pass
3rd Edition	Should be accepted	Has been accepted	Pass
(Empty)	Should be rejected (empty field)	Has been accepted	Fail
2	Should be accepted	Has been accepted	Pass

4. Edition V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
1st	Should be accepted	Has been accepted	Pass
2023	Should be accepted	Has been accepted	Pass
3rd Edition	Should be accepted	Has been accepted	Pass
(Empty)	Should be rejected	Has been rejected	Pass
2	Should be accepted	Has been accepted	Pass

5. Price V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
500	Should be accepted	Has been accepted	Pass
1000.50	Should be accepted	Has been accepted	Pass
-250	Should be rejected (negative)	Has been rejected	Pass
0	Should be rejected (not positive)	Has been rejected	Pass

6. Quantity V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
10	Should be accepted	Has been accepted	Pass
1	Should be accepted	Has been accepted	Pass
0	Should be rejected	Has been rejected	Pass
-5	Should be rejected	Has been rejected	Pass

7. Image V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
book_cover.jpg	Should be accepted	Has been accepted	Pass
cover_image.png	Should be accepted	Has been accepted	Pass
book_photo.jpeg	Should be accepted	Has been accepted	Pass
document.pdf	Should be rejected	Has been rejected	Pass

Add department

Rules:

1. Department names must be alphabets only.
2. Edit button must edit respective department name.
3. Delete button must delete the respective department.

1. Department Name V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Computer Science	Should be accepted	Has been accepted	Pass
Mechanical 123	Should be rejected	Has been rejected	Pass
Chemical Engineering	Should be accepted	Has been accepted	Pass
@Software Engineering	Should be rejected	Has been accepted	Fail

2. Department Name V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Computer Science	Should be accepted	Has been accepted	Pass
Mechanical 123	Should be rejected	Has been rejected	Pass
Chemical Engineering	Should be accepted	Has been accepted	Pass
@Software Engineering	Should be rejected	Has been rejected	Pass

3. Edit Department V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "edit" button.	Selected department should be editable.	Selected librarian has been editable.	Pass

4. Delete Department V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "delete" button.	Selected department should be deleted.	Selected librarian has been deleted.	Pass

View Student Report

Rules:

1. Department dropdown must not be empty if filtering by department.
2. Roll Number field must not be empty if filtering by roll number.
3. Department filters must display only the students of the selected department.
4. Roll Number filter must display the exact student details for the entered roll number.
5. The "Reset Filter" button must clear all filters and display the full list.
6. The "View" button in the table must open detailed student information.

1. Department V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Computer Science	Should be accepted	Has been accepted	Pass
Mechanical 123	Should be rejected	Has been rejected	Pass
Chemical Engineering	Should be accepted	Has been accepted	Pass
@Software Engineering	Should be rejected	Has been accepted	Fail

2. Department V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Computer Science	Should be accepted	Has been accepted	Pass
Mechanical 123	Should be rejected	Has been rejected	Pass
Chemical Engineering	Should be accepted	Has been accepted	Pass
@Software Engineering	Should be rejected	Has been rejected	Pass

3. Roll No V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
2023-CS-625	Should be accepted	Has been selected	Pass
2022-IT-234	Should be accepted	Has been selected	Pass
2024-ENG-678	Should be accepted	Has been selected	Pass
(Empty)	Should be rejected	Has been accepted	Pass

4. Roll No V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
2023-CS-625	Should be accepted	Has been selected	Pass
2022-IT-234	Should be accepted	Has been selected	Pass
2024-ENG-678	Should be accepted	Has been selected	Pass
(Empty)	Should be rejected	Has been rejected	Pass

5. Reset Filter V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian applies filters and clicks "Reset Filter"	All filters should be cleared and the full list displayed	Filters has been cleared and full list displayed	Pass

6. View V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "view" button.	Details of respective student should be displayed.	Details of respective student has been displayed.	Pass

Book issue form

Rules:

1. The book must be selected from the available options in the dropdown.
2. The selection must not be empty.
3. A roll number must be selected from the available options in the dropdown.
4. The number of days should be a positive integer.
5. The button must only be clickable if all fields (Book, Roll Number, and Days) are filled correctly.

1. Book V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Book A	Should be accepted and selected	Has been accepted and selected	Pass
Book B	Should be accepted and selected	Has been accepted and selected	Pass
Book C	Should be accepted and selected	Has been accepted and selected	Pass
(Empty)	Should be rejected	Has been be accepted	Fail

2. Book V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Book A	Should be accepted and selected	Has been accepted and selected	Pass
Book B	Should be accepted and selected	Has been accepted and selected	Pass
Book C	Should be accepted and selected	Has been accepted and selected	Pass
(Empty)	Should be rejected	Has been be rejected	Fail

3. Roll No V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
2023-CS-625	Should be accepted	2023-CS-625 has been selected	Pass
2022-IT-234	Should be accepted	2022-IT-234 has been selected	Pass
2024-ENG-678	Should be accepted	2024-ENG-678 has been selected	Pass
(Empty)	Should be rejected	Has been accepted	Pass

4. Roll No V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
2023-CS-625	Should be accepted	2023-CS-625 has been selected	Pass
2022-IT-234	Should be accepted	2022-IT-234 has been selected	Pass
2024-ENG-678	Should be accepted	2024-ENG-678 has been selected	Pass
(Empty)	Should be rejected	Has been rejected	Pass

5. No of Days V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
7	Should be accepted	has been accepted	Pass
15	Should be accepted	has been accepted	Pass
0	Should be rejected	has been rejected	Pass
-5	Should be rejected	has been accepted	Fail

6. No of Days V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
7	Should be accepted	has been accepted	Pass
15	Should be accepted	has been accepted	Pass
0	Should be rejected	has been reject	Fail
-5	Should be rejected	has been rejected	Pass

7. Issue Book V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "Issue Book" button.	The book should be issued successfully.	The book has been issued successfully.	Pass

Issued Books report/Return Book Report

Rules:

1. A roll number must be selected from the available options in the dropdown.
2. Roll Number must follow a valid format (e.g., "2023-CS-625").
3. The book must be selected from the available options in the dropdown.
4. The "Reset Filter" button must clear all filters and display the full list.
5. The "View" button must display selected student information.

1. Roll No V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
2023-CS-625	Should be accepted	Has been selected	Pass
2022-IT-234	Should be accepted	Has been selected	Pass
2024-ENG-678	Should be accepted	Has been selected	Pass
(Empty)	Should be rejected	Has been accepted	Pass

2. Roll No V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
2023-CS-625	Should be accepted	Has been selected	Pass
2022-IT-234	Should be accepted	Has been selected	Pass
2024-ENG-678	Should be accepted	Has been selected	Pass
(Empty)	Should be rejected	Has been rejected	Pass

3. Book V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Book A	Should be accepted and selected	Has been accepted and selected	Pass
Book B	Should be accepted and selected	Has been accepted and selected	Pass
Book C	Should be accepted and selected	Has been accepted and selected	Pass
(Empty)	Should be rejected	Has been be accepted	Fail

4. Book V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Book A	Should be accepted and selected	Has been accepted and selected	Pass
Book B	Should be accepted and selected	Has been accepted and selected	Pass
Book C	Should be accepted and selected	Has been accepted and selected	Pass
(Empty)	Should be rejected	Has been be rejected	Fail

5. Reset Filter V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian applies filters and clicks "Reset Filter"	All filters should be cleared and the full list displayed	Filters has been cleared and full list displayed	Pass

6. View V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "view" button.	Issued Books report of respective student should be displayed.	Issued Books report of respective student should be displayed.	Pass

Book Return Form

Rules:

7. A roll number must be selected from the available options in the dropdown.
8. Roll Number must follow a valid format (e.g., "2023-CS-625").
9. The book must be selected from the available options in the dropdown.
10. The "View Book" button must view selected student and books that he/she borrowed.
11. The "Penalty Report" button must open selected student information.
12. The "Report Book report" button must open return book page.
13. The librarian must select a return date from the calendar. The date must be after the book issue date.

1. Roll No V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
2023-CS-625	Should be accepted	Has been selected	Pass
2022-IT-234	Should be accepted	Has been selected	Pass
2024-ENG-678	Should be accepted	Has been selected	Pass
(Empty)	Should be rejected	Has been accepted	Pass

2. Roll No V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
2023-CS-625	Should be accepted	Has been selected	Pass
2022-IT-234	Should be accepted	Has been selected	Pass
2024-ENG-678	Should be accepted	Has been selected	Pass
(Empty)	Should be rejected	Has been rejected	Pass

3. Book V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Book A	Should be accepted and selected	Has been accepted and selected	Pass
Book B	Should be accepted and selected	Has been accepted and selected	Pass
Book C	Should be accepted and selected	Has been accepted and selected	Pass
(Empty)	Should be rejected	Has been be accepted	Fail

4. Book V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
Book A	Should be accepted and selected	Has been accepted and selected	Pass
Book B	Should be accepted and selected	Has been accepted and selected	Pass
Book C	Should be accepted and selected	Has been accepted and selected	Pass
(Empty)	Should be rejected	Has been be rejected	Fail

5. Return date selection V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian selects a valid future return date.	The book should be marked as returned.	The book has been returned successfully.	Pass

6. Return Book V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "Return book" button.	Selected book should be returned.	Selected book has been returned.	Pass

Penalty Report

Rules:

1. A roll number must be selected from the available options in the dropdown.
2. Roll Number must follow a valid format (e.g., "2023-CS-625").
3. The "Reset Filter" button must clear all filters and display the full list.
4. Clicking the "Waive Fine" button should remove the fine associated with the selected student.
5. Clicking the "Paid Fine" button should mark the fine as paid.

1. Roll No V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
2023-CS-625	Should be accepted	Has been selected	Pass
2022-IT-234	Should be accepted	Has been selected	Pass
2024-ENG-678	Should be accepted	Has been selected	Pass
(Empty)	Should be rejected	Has been accepted	Pass

2. Roll No V1.1

Test Data	Expected Result	Actual Results	Pass/Fail
2023-CS-625	Should be accepted	Has been selected	Pass
2022-IT-234	Should be accepted	Has been selected	Pass
2024-ENG-678	Should be accepted	Has been selected	Pass
(Empty)	Should be rejected	Has been rejected	Pass

3. Reset Filter V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian applies filters and clicks "Reset Filter"	All filters should be cleared and the full list displayed	Filters has been cleared and full list displayed	Pass

4. View V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "view" button.	Penalty report of selected student should be displayed.	Issued Books report of respective student should be displayed.	Pass

5. Waive Fine V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "Waive Fine" button.	The fine should be waived, and the penalty report should be updated.	The fine has been waived, and the report has been updated.	Pass

Paid Fine V1.0

Test Data	Expected Result	Actual Results	Pass/Fail
Librarian clicks on the "Paid Fine" button.	The fine should be marked as paid, and the penalty report should be updated.	The fine has been marked as paid, and the penalty report has been updated.	Pass

Integration Level Testing**1. User Authentication**

Test Case ID	Test Case Description	Expected Result	Actual Result	Pass/Fail
INT_1	User signs in successfully	User is redirected to the dashboard	As Expected	Pass
INT_2	User fails to sign in (invalid credentials)	Error message "Invalid Credentials" displayed	As Expected	Pass

2. Book Management

Test Case ID	Test Case Description	Expected Result	Actual Result	Pass/Fail
INT_3	Add a new book to the library	Book is added successfully	As Expected	Pass
INT_4	Issue a book to a student	Book status updates to "Issued"	As Expected	Pass
INT_5	Return a borrowed book	Book status updates to "Available"	As Expected	Pass

3. Department Management

Test Case ID	Test Case Description	Expected Result	Actual Result	Pass/Fail
INT_6	Add a new Department	Department is added successfully	As Expected	Pass
INT_7	Edit Department	Information is updated successfully	As Expected	Pass
INT_8	Delete Department	Deleted successfully	As Expected	Pass

4. User Account Management

Test Case ID	Test Case Description	Expected Result	Actual Result	Pass/Fail
INT_9	View user profile	Profile details are displayed correctly	As Expected	Pass
INT_10	Edit user profile	Changes are saved successfully	As Expected	Pass
INT_11	Update user password	Password is updated	As Expected	Pass

5. Reports

Test Case ID	Test Case Description	Expected Result	Actual Result	Pass/Fail
INT_12	Generate librarian report	Report is generated and displayed	As Expected	Pass
INT_13	Generate Student report	Report is generated and displayed	As Expected	Pass
INT_14	Generate issued books report	Report lists all issued books	As Expected	Pass
INT_15	Generate penalty report	Penalty details are displayed	As Expected	Pass

6. Penalties

Test Case ID	Test Case Description	Expected Result	Actual Result	Pass/Fail
INT_16	Waive penalty	Penalty is removed successfully	As Expected	Pass
INT_17	Pay penalty	Penalty is Paid successfully	As Expected	Pass

7. Administrative Tasks

Test Case ID	Test Case Description	Expected Result	Actual Result	Pass/Fail
INT_18	Add a new librarian	Librarian account created successfully	As Expected	Pass
INT_19	Remove a librarian	Librarian account removed successfully	As Expected	Pass
INT_20	Update librarian information	Changes are saved successfully	As Expected	Pass

8. Session Management

Test Case ID	Test Case Description	Expected Result	Actual Result	Pass/Fail
INT_22	User logout	Session ends and login page displayed	As Expected	Pass
INT_23	Session timeout	When the user closes the system session will be end.	As Expected	Pass

System Level Testing

Test the Login Functionality

Test Case ID		LMS_001		Test Case Description		Test the Login Functionality
Created By	Syed Shayan	Reviewed By	Hammad Khalid	Version	1.0.1	

QA Testers Log		Review comments from XYZ incorporate in version 1.0.1				
Tester Name	Adeel Ul Rehman	Date Tested	20-12-2024	Test Case (Pass/Fail/Not Executed)	Pass	

S #	Prerequisites:
1.	Access to System of library management.
2.	Valid user credentials for Admin, Librarian, and Student.

S #	Test Data
3.	AdminID = admin Password = 12345
4.	LibrarianID = Shabir Password = 12345
5.	StudentID = ali123 Password = ali123

Test Scenario		Verify user able to Be Logged into System.			
S #	Step Details	Expected Results	Actual Results	Pass/Fail/Not executed/Suspended	
6.	Navigate to the login page.	The login page should open.	As Expected	Pass	
7.	Enter valid Admin credentials and click login.	Admin dashboard is displayed.	As Expected	Pass	
8.	Enter valid Librarian credentials and click login.	Librarian dashboard is displayed.	As Expected	Pass	
9.	Enter valid Student credentials and click login.	Student dashboard is displayed.	As Expected	Pass	

Post Condition	User will be able to use his dashboard functionalities.
-----------------------	---

Test the Add Librarian Functionality

Test Case ID		LMS_002	Test Case Description		Test the Add Librarian Functionality
Created By	Hammad Khalid	Reviewed By	Adeel Ul Rehman	Version	1.0.1

QA Testers Log		Review comments from XYZ incorporate in version 1.0.1			
Tester Name	Syed Shayan	Date Tested	20-12-2024	Test Case (Pass/Fail/Not Executed)	Pass

S #	Prerequisites:		S #	Test Data	
1.	Admin account access.		1.	Librarian ID = “Lib123”; Name = “Nouman”; Gender = “M”; Mobile No = “1223”; Address = “XYZ”; Username = “Nouman”; Password = 12345	
2.	Valid librarian details.				
Test Scenario		Verify that Admin can add a new Librarian successfully.			
S #	Step Details	Expected Results	Actual Results	Pass/Fail/Not executed/Suspended	
1.	Login as Admin and navigate to "Add Librarian".	"Add Librarian" form is displayed.	As Expected	Pass	
2.	Enter Librarian details and click "Add Librarian".	Librarian is added and confirmation is shown.	As Expected	Pass	

Post Condition	After adding a librarian, the admin can add more librarians or update the existing ones.
-----------------------	--

Test the View Librarian Report Functionality

Test Case ID	LMS_003	Test Case Description	Test the View Librarian Report Functionality
Created By	Adeel Ur Rehman	Reviewed By	Syed Shayan
		Version	1.0.1

QA Testers Log	Review comments from XYZ incorporate in version 1.0.1				
Tester Name	Hammad Khalid	Date Tested	20-12-2024	Test Case (Pass/Fail/Not Executed)	Pass

S #	Prerequisites:
1.	Admin account access.
2.	At least one Librarian exists.

S #	Test Data
1.	N/A

Test Scenario	Verify that Admin can view the list of Librarians and their details.			
S #	Step Details	Expected Results	Actual Results	Pass/Fail/Not executed/Suspended
1.	Login as Admin and navigate to "Librarian Report".	Librarian report page is displayed.	As Expected	Pass
2.	Verify that all Librarians are listed.	All existing Librarians are displayed.	As Expected	Pass

Post Condition	After displaying Librarians report, admin can remove or update any librarian.
-----------------------	---

Test the Add Book or Student Functionality

Test Case ID	LMS_004	Test Case Description	Test the Add Book or Student Functionality
Created By	Syed Shayan	Reviewed By	Hammad Khalid
		Version	1.0.1

QA Testers Log	Review comments from XYZ incorporate in version 1.0.1				
Tester Name	Adeel Ur Rehman	Date Tested	20-12-2024	Test Case (Pass/Fail/Not Executed)	Pass

S #	Prerequisites:
3.	Librarian account access.

Test Scenario	Verify that the Librarian can add a new book or Student to the library.
----------------------	---

S #	Test Data
1.	Book Title = "Data Structures"; Description = "ABC"; Author = "Mark Allen"; Edition = "12 th "; Price = 2000. Quantity = 1; Book Image = "IMG-jpg"
2.	Student Name = "Tanzeel" Roll No = 567 Gender = "M" Department = "Electrical" Mobile = 890 Username = Tanzeel Password =09876

S #	Step Details	Expected Results	Actual Results	Pass/Fail/Not executed/Suspended
1.	Login as Librarian and navigate to "Add Book" or "Add Student".	"Add Book" form is displayed.	As Expected	Pass
2.	Enter Book details and click "Save".	Book is added and confirmation is shown.	As Expected	Pass
3.	Enter Student details and click "Save".	Student is added and confirmation is shown.	As Expected	Pass

Post Condition	After clicking save a book or a student will be added to the system, and librarian can navigate to other pages.
-----------------------	---

Test the Issue Book Functionality

Test Case ID	LMS_005	Test Case Description	Test the Issue Book Functionality
Created By	Syed Shayan	Reviewed By	Adeel Ul Rehman
		Version	1.0.1

QA Testers Log	Review comments from XYZ incorporate in version 1.0.1				
Tester Name	Hammad Khalid	Date Tested	20-12-2024	Test Case (Pass/Fail/Not Executed)	Pass

S #	Prerequisites:
1.	Librarian account access.
2.	At least one book exists.
3.	At least one student exists.

S #	Test Data
1.	Book = "Data Structures" StudentID = ali123 Days = 5

Test Scenario	Verify that Librarian can issue a book to a student successfully.			
S #	Step Details	Expected Results	Actual Results	Pass/Fail/Not executed/Suspended
1.	Login as Librarian and navigate to "Issue Book".	"Issue Book" form is displayed.	As Expected	Pass
2.	Select Book, Student and number of days, then click "issue".	Book is issued to the student.	As Expected	Pass

Post Condition	After issuing the book to the student, the Librarian can see it in the Reports.
-----------------------	---

Test the Book or Student Report Functionality

Test Case ID	LMS_006	Test Case Description	Test the Book or Student Report Functionality
Created By	Syed Shayan	Reviewed By	Hammad Khalid
		Version	1.0.1

QA Testers Log	Review comments from XYZ incorporate in version 1.0.1				
Tester Name	Adeel Ur Rehman	Date Tested	20-12-2024	Test Case (Pass/Fail/Not Executed)	Pass

S #	Prerequisites:
1.	Librarian account access.

S #	Test Data
1.	N/A

Test Scenario	Verify that the Librarian can see all the Books or Students Reports successfully.
----------------------	---

S #	Step Details	Expected Results	Actual Results	Pass/Fail/Not executed/Suspended
1.	Login as Librarian and navigate to View "Books Report" or "Students Report".	Lists of Books or Students Report will be displayed	As Expected	Pass

Post Condition	After the Reports are displayed, the Librarian can add the filters to the list to view the list according to that filter.
-----------------------	---

Test Add Department Functionality

Test Case ID	LMS_007	Test Case Description	Test the Issue Book Functionality
Created By	Hammad Khalid	Reviewed By	Adeel Ul Rehman
		Version	1.0.1

QA Testers Log	Review comments from XYZ incorporate in version 1.0.1				
Tester Name	Syed Shayan	Date Tested	20-12-2024	Test Case (Pass/Fail/Not Executed)	Pass

S #	Prerequisites:
2.	Librarian account access.

S #	Test Data
3.	Department Name = "Architecture"

Test Scenario	Verify that Librarian can add Department to the system successfully.			
S #	Step Details	Expected Results	Actual Results	Pass/Fail/Not executed/Suspended
4.	Login as Librarian and navigate to "Add Department".	"Add Department" form is displayed.	As Expected	Pass
5.	Enter the Name of Department and click "Add Department".	Department will be added to the system	As Expected	Pass

Post Condition	After Adding the department librarian can Edit, Delete or view all Departments.
-----------------------	---

Test the Return Book Functionality

Test Case ID		LMS_008	Test Case Description	Test the Return Book Functionality	
Created By	Adeel Ul Rehman	Reviewed By	Syed Shayan	Version	1.0.1

QA Testers Log		Review comments from XYZ incorporate in version 1.0.1			
Tester Name	Hammad Khalid	Date Tested	20-12-2024	Test Case (Pass/Fail/Not Executed)	Pass

S #	Prerequisites:
1.	Librarian account access.
2.	A book has been issued to a student.

S #	Test Data
1.	Book = "Data Structures" StudentID = ali123

Test Scenario		Verify that the Librarian can mark a book as returned.			
S #	Step Details	Expected Results	Actual Results	Pass/Fail/Not executed/Suspended	
1.	Login as Librarian and navigate to "Return Book".	"Return Book" form is displayed.	As Expected	Pass	
2.	Select Book and Student, then click "Return Book".	Book is marked as returned.	As Expected	Pass	

Post Condition	After returning the book, Librarian can see it in the Reports.
-----------------------	--

Test the View Book Functionality

Test Case ID	LMS_009	Test Case Description	Test the View Book Functionality
Created By	Syed Shayan	Reviewed By	Adeel Ul Rehman
		Version	1.0.1

QA Testers Log	Review comments from XYZ incorporate in version 1.0.1				
Tester Name	Hammad Khalid	Date Tested	20-12-2024	Test Case (Pass/Fail/Not Executed)	Pass

S #	Prerequisites:
3.	Librarian or Student account access.
4.	At least one book must be exists.

S #	Test Data
5.	N/A

book as returned.

Test Scenario

Verify that the Librarian can mark a

S #	Step Details	Expected Results	Actual Results	Pass/Fail/Not executed/Suspended
6.	Login as Librarian or Student and navigate to "View Books".	All the books will be displayed	As Expected	Pass

Post Condition	After the books displayed Student or Librarian can add filters on the list of books.
-----------------------	--

Test the View Penalty Report Functionality

Test Case ID		LMS_0010	Test Case Description	Test the View Penalty Report Functionality	
Created By	Hammad Khalid	Reviewed By	Adeel Ul Rehman	Version	1.0.1

QA Testers Log		Review comments from XYZ incorporate in version 1.0.1			
Tester Name	Syed Shayan	Date Tested	20-12-2024	Test Case (Pass/Fail/Not Executed)	Pass

S #	Prerequisites:
1.	Librarian account access.
2.	At least one penalty exists.

S #	Test Data
1.	N/A

Test Scenario		Verify that the Librarian can view the penalty report for all overdue books.			
S #	Step Details	Expected Results	Actual Results	Pass/Fail/Not executed/Suspended	
2.	Login as Librarian and navigate to "Penalty Report".	"Penalty Report" page is displayed.	As Expected	Pass	
3.	Verify that all overdue penalties are listed.	All penalties are displayed.	As Expected	Pass	

Post Condition	After Penalty Reports are displayed, Librarian can change the status of fine to Waived Fine or Paid Fine.
-----------------------	---

Test Account Update Functionality

Test Case ID	LMS_0011	Test Case Description	Test the Student Account Update Functionality		
Created By	Adeel Ul Rehman	Reviewed By	Hammad Khalid	Version	1.0.1

QA Testers Log	Review comments from XYZ incorporate in version 1.0.1				
Tester Name	Syed Shayan	Date Tested	20-12-2024	Test Case (Pass/Fail/Not Executed)	Pass

S #	Prerequisites:
1.	Admin, Librarian or Student account access.

Test Scenario	Verify that Student Account details are updated.
----------------------	--

S #	Test Data
1.	Username = "Admin1"; Old Password = 12345; New Password = 12389; confirm Password = 12389
2.	Librarian ID = 102 Name = Shakir; Gender = "M" Mobile No: "6789" Address = "Multan" Username = "Shakir" Password = 12345
3.	Name = "Ali Raza" Roll No = 1234; Gender = "M" Department = "Computer Science"; Mobile No = 123 Address = "XYZ"; Username = "aliraza123";

S #	Step Details	Expected Results	Actual Results	Pass/Fail/Not executed/Suspended
1.	Login as Student to "Update Account"	"Update Account" form is displayed.	As Expected	Pass
2.	Enter new details and click "Save".	Account details are updated.	As Expected	Pass

Post Condition	After updating Student will be able to be logged into the system with new username.
-----------------------	---

Performance Testing

1. Load Testing

S #	Functionality	Estimated Load Time (seconds)
1.	Login	0.2 to 0.5
2.	View Account	0.2 to 0.5
3.	Update Account	3 to 5
4.	See Books List	0.2 to 0.5
5.	View Books Report	0.2 to 0.5
6.	View Student Report	0.2 to 0.5
7.	View Penalty Report	0.2 to 0.5
8.	View Librarian Report	0.2 to 0.5
9.	Add Book	3 to 5
10.	Add Student	3 to 5
11.	Add Department	1 to 2
12.	Add Librarian	3 to 5
13.	Issue Book	1 to 3
14.	Return Book	1 to 2
15.	Penalty Fine waive or pay	1 to 2
16.	Edit or delete Department	0.5 to 1
17.	Remove or Update Librarian	0.2 to 0.5
18.	Logout	0.2 to 0.5

2. Stress Testing

S #	Functionality	Estimated Load Time (seconds)
1.	Login	1 to 3
2.	View Account	1 to 3
3.	Update Account	4 to 7
4.	See Books List	1 to 3
5.	View Books Report	1 to 3
6.	View Student Report	1 to 3
7.	View Penalty Report	1 to 3
8.	View Librarian Report	1 to 3
9.	Add Book	4 to 7
10.	Add Student	4 to 7

11.	Add Department	1 to 2
12.	Add Librarian	4 to 7
13.	Issue Book	3 to 6
14.	Return Book	3 to 5
15.	Penalty Fine waive or pay	3 to 5
16.	Edit or delete Department	2 to 4
17.	Remove or Update Librarian	1 to 3
18.	Logout	1 to 3

3. Endurance Testing

S #	Endurance Test
1.	Add many Data so the system not able to handle the coming so much data.
2.	Issue or return many books so the system sustained to issue or return more.
3.	View many Reports quickly so the system is not able to load data properly.
4.	Edit or Delete many Data so the system not able to track the changes.

4. Spike Testing

S #	Spike Test
1.	Enter many wrong IDs and passwords again and again.
2.	Enter too much Data with wrong details again and again.
3.	Issue many books to same student.
4.	Unable to handle the multiple user logins.

5. Usability Testing

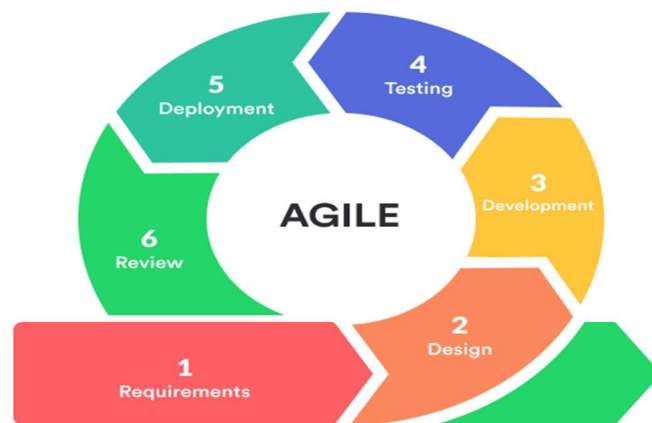
S #	System Flow
1.	After login the user will show his dashboard and he able to use his functionalities
2.	Admin can add librarian by entering his/her data and add to the system.
3.	Admin see the reports of librarian and update or remove him/her.
4.	Admin can change his password by entering old and new password.
5.	Librarian can update his data by entering new data and save it.
6.	Librarian can add books, department or student by entering respective data.
7.	Librarian can view books, Penalty and Student reports.
8.	Librarian can issue to student or return book from student.
9.	Librarian can waive or pay the fine of student penalty by changing its status.

10.	Student can view his account and edit it.
11.	Student can see all books list and their specific details.
12.	Student can see the books he borrowed, returned and penalty reports.

Proposed Methodology/System:

The development of the Digital Library Management System (DLMS) will follow **Scrum Agile Approach**. This methodology is selected for its ability to foster collaboration, flexibility, and iterative development. The key reasons for choosing Scrum include:

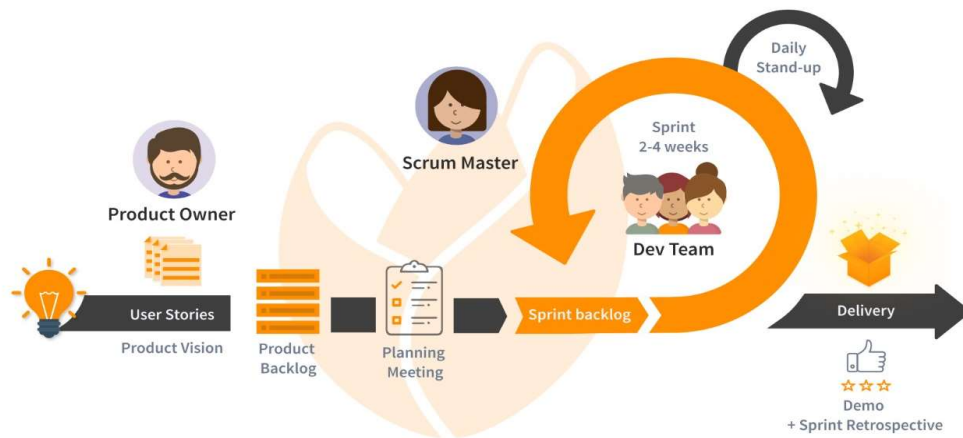
- **Collaboration:** It enhances teamwork and communication among all team members.
- **Short Sprints:** Quick development cycles (sprints) allow rapid adjustments based on continuous feedback.
- **Daily Scrums:** Each team member's progress is reviewed and discussed in brief daily meetings to ensure alignment.
- **Continuous Testing:** Development and testing occur simultaneously during each sprint, enabling immediate feedback and adjustments.
- **Clear Visibility:** Regular scrum meetings promote transparency and ensure everyone is on the same page.



Roles in the Project:

- **Product Owner:** Sir Zeeshan
- **Scrum Master:** Shayan
- **Development Team:** Shayan, Hammad, Adeel

Scrum Process:



Scrum Ceremonies:

- **Product Backlog:** Requirements are gathered from the client to form a comprehensive product backlog.
- **Sprint Planning:** Functional requirements are broken down into manageable tasks and divided across multiple sprints.
- **Daily Standups:** Short daily meetings to review progress, discuss challenges, and ensure efficient teamwork.
- **Sprint Review:** At the end of each sprint, the team evaluates progress, gathers feedback, and identifies areas for improvement to ensure continuous development.

This approach provides flexibility, encourages collaboration, and ensures the timely delivery of a high-quality digital library management system.

Related Work:

When exploring existing library management systems, several key options emerge as significant players in the field. Each of these systems has garnered attention for their unique features and capabilities, yet they also come with distinct strengths and weaknesses. This is particularly true when comparing them to the needs of smaller and medium-sized libraries, such as the one we are developing for. Here, we will look into these three related systems i.e., **Koha [1]**, **Evergreen [2]**, and **LibraryWorld [3]**.

Koha [1]:

Koha is one of the most recognized open-source library management systems available today. Having been around for many years, it has become popular among various institutions for its extensive array of features, including cataloguing, circulation tracking, and reporting capabilities. One of Koha's primary advantages is its high degree of customization, enabled by its open-source

nature, which allows libraries to tailor the system to their specific needs without incurring hefty licensing fees.

Strengths:

Koha's flexibility is a standout feature, as it allows libraries to modify the system according to their unique operational requirements. The vibrant user community surrounding Koha contributes to its ongoing development, providing regular updates, bug fixes, and enhancements. This ensures that libraries can benefit from continuous improvements. Furthermore, Koha is a comprehensive solution, covering everything from managing book inventories to generating detailed analytical reports. This makes it particularly suitable for large institutions with diverse and complex needs.

Weaknesses:

However, Koha is not without its challenges. Setting up the system can be a difficult task, especially for smaller libraries that may lack the necessary IT expertise. While the vast range of features can be advantageous for large libraries, it can also feel excessive for smaller ones, leading to potential overwhelm. Moreover, as databases grow over time, users have reported slower performance, especially during peak usage periods. This slowdown can lead to frustration for both staff and patrons, undermining the system's overall effectiveness.

Evergreen [2]:

Another notable open-source library management system is Evergreen, which is particularly acclaimed for its ability to facilitate collaboration among libraries that share resources across multiple locations. This functionality makes Evergreen an excellent choice for networks of libraries working together. However, it is also suitable for individual libraries seeking a robust management solution.

Strengths:

Evergreen excels in promoting resource sharing, making it an ideal choice for libraries that operate within collaborative environments. Its capacity to handle large databases further enhances its suitability for larger institutions with extensive collections. Like Koha, Evergreen's open-source framework allows libraries to customize the system to meet their specific operational needs, providing a level of flexibility that is highly valued in the library community.

Weaknesses:

On the downside, Evergreen can present a steep learning curve, requiring significant IT expertise for effective implementation and maintenance. This complexity can be overwhelming for smaller libraries, which may find themselves burdened by features they do not need. Although the system allows for customization, this can lead to technical challenges, including performance issues if not implemented carefully. Without adequate technical knowledge, managing Evergreen can become a daunting task for many library staff.

LibraryWorld [3]:

In contrast to Koha and Evergreen, LibraryWorld is a cloud-based library management system specifically designed for smaller libraries. Its feature set includes essential functions such as catalogues, circulation management, and patron management. One of its most appealing aspects is its cloud-based nature, which allows for easy setup without the need for extensive infrastructure.

Strengths:

LibraryWorld stands out for its user-friendly design, which simplifies navigation for library staff and minimizes the training time required. This aspect is crucial for smaller institutions that may have limited resources for staff development. Additionally, LibraryWorld is budget-friendly, tailored specifically for smaller libraries, and does not necessitate costly hardware or software investments. The cloud-based format also enables quick implementation, allowing libraries to begin operations with minimal technical expertise.

Weaknesses:

Despite its advantages, LibraryWorld does have limitations. While it serves smaller libraries well, it lacks the advanced features that larger institutions might require for more complex operations. Additionally, its scalability is limited; as libraries grow and expand their collections, LibraryWorld may struggle to meet their evolving needs. Unlike its open-source counterparts, LibraryWorld offers limited customization options, which can hinder libraries from tailoring the system to better suit their specific requirements.

How Our System Stands Out:

Our Digital Library Management System (DLMS) is purposefully designed to provide a straightforward and efficient solution, particularly for smaller or medium-sized educational libraries, such as those found in schools or colleges. We believe that simplicity and usability are paramount in creating an effective library management experience.

Simplicity:

Unlike Koha and Evergreen, which can often feel complex and overwhelming for smaller institutions, our system emphasizes a clean and intuitive interface. This focus on usability significantly reduces the need for extensive technical support and lengthy training sessions for both librarians and students, allowing them to spend more time on what truly matters serving their community.

Cost-Effectiveness:

Built using C# and Microsoft SQL Server, our system is designed to be more affordable than alternatives like Koha, which require specialized IT expertise for setup and ongoing maintenance. Furthermore, our system does not carry the recurring costs associated with cloud services, such as those offered by LibraryWorld. This cost-effective approach ensures that more resources can be allocated toward improving library services rather than managing software expenses.

Performance:

We have prioritized performance in our design, offering real-time book tracking and a streamlined process for issuing and returning books. This is in stark contrast to the performance issues that can arise in larger systems like Koha and Evergreen, particularly as their databases grow and during busy periods. Our system is engineered to maintain efficiency, ensuring that library staff and users enjoy a seamless experience.

Customization Without Complexity:

While LibraryWorld excels in simplicity, it often falls short in terms of flexibility. Our system strikes a balance by providing easy-to-use features that can be customized to meet the specific needs of each library without introducing unnecessary complexity. Additionally, we have designed our system to avoid the performance bottlenecks that can plague more intricate systems like Koha, ensuring that libraries can effectively manage their operations without frustration.

In **conclusion**, our Digital Library Management System is crafted to take the best aspects of existing systems such as ease of use, reliability, and the capacity for customization and tailor them specifically to the needs of educational libraries. By focusing on simplicity, cost-effectiveness, performance, and practical customization, we offer a solution that enhances day-to-day library operations for both staff and students. Our goal is to empower libraries to thrive in an increasingly digital world, without the unnecessary complexity or high costs that often accompany larger, one-size-fits-all solutions.

Data Gathering Approach

For this project, I began by conducting thorough research and analysis of existing library management systems, including well-known platforms such as **Koha [1]**, **Evergreen [2]**, and **LibraryWorld [3]**. This comparative analysis helped me gain a deeper understanding of how these systems function, their core features, and where they fall short in addressing the needs of smaller educational libraries. By critically examining these systems, I could identify common features such as cataloguing, circulation management, and report generation, as well as pinpoint the areas that often prove challenging for smaller institutions, like system complexity and scalability.

Through this process, I also explored how these systems handle user interactions, data security, and integration with other educational tools. Furthermore, I examined how these platforms adapt to evolving technologies, such as cloud computing, to assess their long-term viability.

Given that my project focuses on creating a solution for school and college libraries, I paid special attention to how these existing systems meet (or fail to meet) the specific needs of such environments. My goal was to determine the essential features that would benefit smaller libraries, particularly in educational settings, without overburdening users with unnecessary complexity. This step-by-step analysis helped me understand which features were crucial such as easy user management, real-time book tracking, and simplified reporting and what aspects could be streamlined to improve the overall user experience.

Instead of conducting formal surveys or interviews, I relied on insights gained from this research and my personal observations as a student. Having interacted with library systems firsthand, I had a clear perspective on what would work best in an educational context. I reflected on how both students and librarians use library resources daily, which allowed me to design a system that aligns with their practical needs. For instance, I prioritized a user-friendly interface that requires minimal training, while ensuring that key functionalities such as book issuance, returns, and cataloguing was easily accessible.

In addition to studying the technical aspects of existing systems, I considered the operational challenges faced by librarians and students alike. This helped me shape a system that not only addresses the common technical gaps but also enhances day-to-day library workflows. By integrating these insights, I aimed to create a Digital Library Management System (DLMS) that is both efficient and adaptable, focusing on ease of use, affordability, and customization for smaller educational institutions.

This research-based approach, combined with my practical understanding of educational library needs, allowed me to develop a system which will be helpful specifically to smaller libraries, where simplicity and efficiency are paramount. The final product is designed to reduce administrative burdens while ensuring that library staff can manage collections and assist students with ease. By concentrating on these real-world requirements, I ensured that this library management system would be a valuable tool in any educational library setting, improving overall efficiency without adding unnecessary complexity.

Team Members Individual Tasks:

Syed Shayan Arshad:

- Related Work & Data Gathering
- References
- Proposed Methodology
- Pages on C#
- Designing
- Requirements

Adeel Ur Rehman:

- Use Cases
- Story Board
- Testing
- Project Overview

M. Hammad Khalid:

- User Stories
- Testing
- Requirements
- Project Overview
- Wireframes

Timeline/Gantt Chart:

Date	10-Sep to 20 Sep	21-Sep to 30 Sep	1-Oct to 10-Oct	11-Oct to 20-Oct
Project Overview				
Requirements				
User Stories				
Use Case				
Pages on C#				
Story Board				
Related Work & Data Gathering				
Wireframes				
Proposed Methodology				

References:

- [1] “Official Website of Koha Library Software.” Accessed: Oct. 19, 2024. [Online]. Available: <https://koha-community.org/>
- [2] “Evergreen ILS.” Accessed: Oct. 19, 2024. [Online]. Available: <https://evergreen-ils.org/>
- [3] “LibraryWorld | Cloud Library | Library Automation | Online Library Service.” Accessed: Oct. 19, 2024. [Online]. Available: <https://www.libraryworld.com/>